Social Ties and Residential Choice: Micro Evidence and Aggregate Implications*

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Abstract

Why don't more people move to places where they can earn more? Using individual-level data from Facebook, we show that social ties play a crucial role in explaining this fact. Exploiting variation in the timing of friends' moves around individuals' college graduation, we show that friends shape both whether people migrate and where they move to. Since social ties are geographically concentrated, the estimated per-friend effects translate into a large aggregate pull, helping to explain why people often remain in poorer places, and why less-educated individuals are less responsive to local economic shocks.

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1 Introduction

Why don't more people move to places where they can earn more? Regional wage gaps are large—average wages in Boston are about 50% higher than in Cleveland[1]—yet migration toward higher-wage places is limited: around 70% of people growing up in areas with below median wages remain there as adults, and even those who do move often relocate to places with similarly low wages (Bartik, 2009; Sprung-Keyser et al., 2022). This disconnect between economic opportunity and migration is especially pronounced for less-educated individuals. To explain these patterns, standard spatial equilibrium models in the tradition of Rosen (1979) and Roback (1982) often appeal to very high moving costs, strong attachment to one's birthplace, or sizable unobserved local amenities. These forces can rationalize low net migration toward high-wage places in principle, but often serve as residual explanations for otherwise unexplained behavior and leave open why much of the migration that does occur is not more strongly directed toward places with better economic conditions.

In this paper, we show that social networks—ties to friends and family—provide a concrete and measurable force that helps explain these patterns. We first document that networks are geographically concentrated and estimate the causal effect of an individual friend's location on migration. We then embed these estimates in a spatial equilibrium model and show that the per-friend effects, combined with the concentration of people's networks, generate a large aggregate pull. This force helps explain both why people often remain in low-wage places and why migration is not more frequently directed toward higher-wage destinations. Because less-educated individuals tend to have more geographically concentrated networks, the same mechanism also provides a natural explanation for why they are less responsive to wage differences across places.

To study how networks shape location choices, we leverage individual-level panel data from Facebook that track both users' locations and social connections over time. We focus on users born between 1970 and 2000 living in the United States for whom we observe quarterly locations at the commuting zone (CZ) level from 2012 to 2025. Crucially, unlike much of the existing literature, our data lets us measure networks directly for a large part of people's networks. This unique combination of data not only lets us estimate the causal effect of an

[1] A long literature examines regional differences in economic performance, showing that a substantial part of these differences reflects causal effects of places, that they cannot entirely be explained by differences in prices, and that they persist over time (e.g., Topel, 1986; Blackaby and Manning, 1990; Barro and Sala-i Martin, 1991; Blanchard and Katz, 1992; Chetty et al., 2026; Deutscher and Mazumder, 2020; Bilal, 2023; Alesina et al., 2021; Asher et al., 2024; Card et al., 2025; Sprung-Keyser and Porter, 2026; Chetty and Hendren, 2018a; Handbury and Weinstein, 2015; Handbury, 2021; Diamond and Moretti, 2021). The average wage numbers discussed above can be found in QCEW (2024).

individual friend but also enables us to quantify how those per-friend effects translate into a large overall force in residential choice.

We begin with descriptive evidence. For the average individual in our data, around 60% of their friends live in the same CZ in 2012. Yet, this average masks important heterogeneity that turns out to be a powerful predictor of subsequent migration behavior. Only around 22% of individuals with at most one-third of their friends outside their 2012 CZ move away by 2025; in contrast, 58% of those with at least two-thirds of their friends elsewhere relocate. Relatedly, those with more friends in CZs that exhibit high wage growth are much more likely to migrate to these destinations by 2025, suggesting that the geography of one's network may shape access to places with stronger economic prospects. However, these correlations between networks and migration need not be causal, since the geography of one's network is likely correlated with other determinants of location choice.

To isolate the causal effect of social networks, we exploit variation in the timing of friends' moves around college graduation—a clear transition point when individuals choose where to live and begin their careers. Consider two observably very similar college graduates from the same home CZ and graduation cohort. Around graduation, each has a friend who leaves that home CZ. Critically, for one graduate, the friend leaves shortly before that graduate's graduation date; for the other, the friend leaves shortly after. As a result, when the two graduates decide where to live upon graduation, the first has one more friend outside the home CZ than the second. Our empirical approach leverages such differences in the timing of friends' moves around graduation to test whether having a larger network outside one's home CZ at this transition point increases the likelihood of leaving. More formally, leveraging the quarterly location measures in the Facebook data and restricting attention to friendships formed well before graduation, we regress whether an individual leaves their home CZ upon graduation on their friends' departures from that CZ at different points before versus after that date.

We show analytically that friends leaving before graduation—in the "pre-period"—identify a combination of the causal effect of networks and a potential bias. This bias can arise because of unobservables—for instance, shared preferences or "common shocks"—that simultaneously shape the location decisions of graduates and their friends. In contrast, friends leaving after graduation—in the "post-period"—only capture a bias, since future friends' relocations cannot affect earlier location choices. Our key identifying assumption is that these biases are the same for the pre- and post-period, so that the difference between the corresponding estimates recovers the causal effect of social networks. Thus, we only assume that whether friends move before or after the graduate's own graduation is quasi-random, not whether

those friends move.[2]

Empirically, we find that friends' moves in the "pre-period" predict graduates' migration decisions more strongly than the same moves in the "post-period." Graduates whose friends leave their home CZ before graduation are more likely to themselves leave upon graduation than graduates whose friends leave the same CZ shortly after. Interpreting the difference through our identifying assumption, a one-percentage-point increase in the share of one's network outside the home CZ raises the probability of leaving that CZ at graduation by 0.34 percentage points. Equivalently, if 10% of a graduate's friends currently living in the home CZ left that CZ, the graduate's probability of leaving would rise by 2.1 percentage points, a 17% increase relative to the average out-migration rate. This magnitude is comparable to the migration response to a roughly $7,300 decrease in average annual wages in one's home CZ when benchmarked against evidence from Sprung-Keyser et al. (2022).

Social networks shape not only whether graduates leave their home CZ, but also where they move to. To study network effects on directed migration, we adapt our design slightly: for a given CZ we ask whether graduates are more likely to move there if they have a friend moving there before graduation rather than after. We implement this approach for the 50 largest CZs and find positive network effects for every destination. For the average CZ, a one-percentage-point increase in the share of friends in that destination raises the probability of moving there by 0.1 percentage points. Strikingly, these effects are similar even for lower-wage destinations—such as Detroit—highlighting that social ties can exert a pull even toward places with limited economic opportunities.

Interpreting our results as reflecting network effects crucially depends on the assumption that whether friends move before or after the graduate's own graduation is quasi-random. Importantly, unobserved factors such as shared preferences or common shocks would, by themselves, *not* pose a threat to this assumption: such forces would bias our estimates only if friends who move before graduation are systematically more similar to the graduate in their unobserved location preferences, or more exposed to shocks that also affect the graduate, than friends who move afterward. We present several pieces of evidence to demonstrate that this is unlikely to be the case.

First, we find that friends moving earlier versus later are not systematically more or less similar to the graduates along observable characteristics. Our estimates are also stable when

[2]This timing-based approach is conceptually related to studies that exploit parents' firm transitions relative to children's labor-market entry to identify job-referral effects (e.g., Kramarz and Skans, 2014; Staiger, 2026), as well as to mover designs that use variation in the timing of moves to separate exposure effects from selection (e.g., Chetty and Hendren, 2018a).

we add very fine controls for remaining differences across graduates and when we exclude friends who are most likely to share college- or field-specific migration pipelines with the graduate. Second, we examine whether friends' moves to "sister CZs"—regions with industry compositions similar to that of the focal CZ (for instance, Cleveland and Detroit)—affect an individual's decision to live in the focal CZ. Following the logic of shift-share designs, the similar industry compositions of sister CZs suggest that they should be exposed to similar local economic shocks. Thus, if our estimates were driven by local economic shocks affecting both graduates and their friends, then friends' moves to a sister CZ should also predict whether graduates live in the focal CZ. They do not. Finally, we ask whether individuals' location choices are predicted by moves among people they eventually befriend, but with whom they were not yet friends by graduation. Because these future friends cannot causally affect earlier choices yet are likely to share similar location preferences and respond to similar location-specific shocks, an explanation based on shared unobservables would attribute similar "effects" to the moves of these future friends. Reassuringly, we find no such effects.

To shed light on the mechanisms behind the network effects, we examine evidence on informational channels, which have long been emphasized in the network literature (e.g., Granovetter, 1973). The robustness exercises above already suggest that job-related information is unlikely to be the primary driver: the estimates remain large even when we exclude friends who attended the same college or studied a similar major. We then study the effects of friends who return to one's home CZ as well as of friends who move away from a potential destination CZ. If information were the main channel, and if friends can continue to provide it after leaving a place, then these moves should have limited effects; instead, we find large negative effects, pointing to benefits tied to contemporaneous presence. This makes a purely information-based interpretation less plausible.

The emphasis on contemporaneous presence also clarifies what our design adds relative to standard peer-composition approaches. Standard peer-effects designs exploit variation in exposure to peers with particular backgrounds or experiences. Applied to migration, this approach naturally shifts attention to exposure to peers with prior experience in a place—for example, because they previously lived there or have other place-specific knowledge. Recent work provides evidence that such exposure can shape later mobility (e.g., Green, 2024). Our approach complements this evidence by focusing on the current locations of individuals' actual friends at the moment residential choices are made. This distinction is important in our setting: the results above suggest that a large part of the network effect reflects the value of friends' contemporaneous presence in the destination itself.

So far, our causal analysis has focused on college graduates, but our primary interest lies in understanding migration decisions for the broader population. We therefore adapt our timing-based design to a broader sample of individuals by measuring people's locations at the beginning of 2012 and again at the end of 2017, and compare observably similar individuals who differ in whether their friends relocate before versus after 2017. Relative to the graduation setting, this design requires a stronger identifying condition: if unobservables shape the location choices of these individuals and their friends, they must be sufficiently stable over time for our design to identify causal network effects. With this caveat in mind, we obtain estimates that are comparable in magnitude to those for college graduates and remain positive across a range of subgroups, providing suggestive evidence that similar network effects extend beyond the college graduation setting.

In the second part of the paper, we quantify how important social ties are for explaining the observed aggregate migration patterns that motivate this study. To do so, we incorporate our results into a Rosen-Roback-style spatial equilibrium model: people derive utility not only from local wages, rents, and amenities, but also from living near their social network. Importantly, we calibrate this preference using our quasi-experimental estimate from the college graduation design and, taking individuals' networks as given, feed in the directly observed distribution of where people's friends live in the broader population to capture the concentration of networks, as well as any variation between individuals and across regions. We then compare the performance of this "network model" with that of a "basic model" that abstracts from network effects but is otherwise identical.

We find that the network model can explain the observed variation in CZ population size much more easily. As is common in Rosen-Roback-style models, we rationalize the variation in population size that cannot be explained by differences in wages and rents by the presence of unobserved local amenities (e.g., Albouy, 2008, 2016; Hsieh and Moretti, 2019). Crucially, in the network model, the standard deviation of these amenities—or residuals—is less than 40% of that of the basic model, a reduction that would require a very large conventional moving-cost wedge in a model without networks. Moreover, the residuals in the network model align much more closely with their interpretation as local amenities. To make this point, we correlate the CZ-specific residuals with various measures of highly valued place characteristics: air quality (Chay and Greenstone, 2005), weather (Rappaport, 2007) and retail environment (Glaeser et al., 2001). In the basic model, we find at most modest correlations, always below 0.25 and sometimes negative. In the network model, the correlations are substantially larger, ranging from 0.21 to 0.51.

There are three reasons why our network model can explain the data in a much more straight-

forward way. First, our quasi-experimental results highlight that network effects are substantial, so that every additional friend in a given CZ has a meaningful impact on where one chooses to live. Second, networks are spatially concentrated, so that the per-friend effects translate into a large aggregate pull; this can explain why people are reluctant to move even in the absence of high moving costs. Third, we find a negative correlation between a CZ's economic conditions and the extent to which people living there have geographically concentrated networks. This observation further helps to explain the low out-migration rates for regions characterized by economic hardship (e.g., Autor et al., 2013, 2025) without having to assign these areas particularly high amenity terms. For example, in the basic model, a relatively large positive residual is required to explain the high population of Detroit; in our framework, this fact can be explained by the observation that people in Detroit have particularly large social networks there.

Finally, we consider whether social networks can help explain differences in migration patterns between more- and less-educated individuals. We show that less-educated individuals have more geographically concentrated networks: individuals without listed college information have only 33% of their friends outside the CZ they live in, compared to nearly 55% among those who attended the most selective colleges. We then simulate local productivity shocks in our network model, holding fixed preferences and all other structural parameters across individuals. The model predicts substantially larger out-migration responses among individuals with more dispersed networks, and generates responses for individuals who attended the most selective colleges that are nearly 70% larger than those for individuals without college information. These findings show that social networks provide one concrete source of the heterogeneous moving costs or preferences to stay that are often used (e.g., Diamond, 2016) to explain why migration responses to local economic shocks differ across education groups.

Overall, this study shows that social networks play a first-order role in shaping residential choice at both the individual and aggregate level. Taking this force into account helps explain why many people remain in economically distressed areas and why this pattern is especially pronounced among less-educated individuals, while also offering a natural reason why migration flows are not always directed toward places with stronger economic conditions.

This paper contributes to several strands of literature. First, it relates to the literature on social networks and migration; see Munshi (2020) for a recent review. Much of this work uses aggregate measures of connectedness across places, or identifies the role of networks using variation in specific subsets of people's social environments such as co-ethnics, co-workers, military peers, or shared origin locations (e.g., Munshi, 2003; McKenzie and Rapoport, 2007;

Costa et al., 2018; Mahajan and Yang, 2020; Stuart and Taylor, 2021; Porcher, 2025; Munshi, 2020; Egger et al., 2022; Ilyés et al., 2023; Hansch et al., 2025; Green, 2024; Zou, 2025). While these approaches have provided important evidence that social ties influence migration decisions, they typically capture particular subsets of social connections or aggregate connectedness between places. Our contribution is to observe a broad set of each individual's *actual* social ties, which allows us to both estimate the causal effect of an individual's friends on migration and measure the overall geographic concentration of each person's network. This concentration is essential for translating per-friend effects into the aggregate force that networks exert on location choice. A small set of recent papers similarly leverages direct network measures, including Büchel et al. (2020); Sahai and Bailey (2022); Blumenstock et al. (2025).[3] We build on this literature by bringing together, in a developed-country setting, quasi-experimental identification of the causal effect of actual friends' locations with a quantitative assessment of how the observed geography of individual networks shapes aggregate residential choice. This combination allows us to connect micro-level network effects to the broader spatial patterns that motivate the paper.

Second, our paper relates to work on migration frictions that seeks to explain why migration responds weakly to local economic conditions (e.g., Sjaastad, 1962; Carrington et al., 1996; Chen and Rosenthal, 2008; Autor et al., 2013; Yagan, 2019; Wilson, 2022b; Sprung-Keyser et al., 2022). Recent studies emphasize frictions such as moving costs (e.g., Kennan and Walker, 2011; Diamond et al., 2019), search frictions (e.g., Bayer et al., 2016; Bergman et al., 2020), and home attachment (e.g., Diamond, 2016; Piyapromdee, 2021; Zerecero, 2021; Coate and Mangum, 2026; Wilson, 2026). We show that the size and spatial concentration of one's social network help explain why migration responds weakly to local economic conditions. Because social ties are place-specific but not limited to one's birthplace, they complement existing work on moving costs and preferences to stay by identifying a concrete source of attachment to places. This force can help rationalize both low net migration toward places with stronger labor markets and substantial gross migration across space: people may stay because many of their ties are local, but when they move they are pulled toward places where other ties already live.

Relatedly, our work connects to evidence that migration behavior differs sharply by education and other characteristics (e.g., Topel, 1986; Bound and Holzer, 2000; Borjas, 2001; Gregg et al., 2004; Wozniak, 2010; Malamud and Wozniak, 2012; Machin et al., 2012; Cadena and

[3]Particularly closely related in data and setting, Armona and Bailey (2025) exploit the staggered rollout of Facebook across U.S. colleges to estimate how Facebook access during college affects subsequent geographic mobility. While their analysis studies the effects of access to a social-networking technology, we study how friends' contemporaneous locations shape residential choice.

Kovak, 2016; Notowidigdo, 2020; Gaubert and Diamond, 2022; Sprung-Keyser et al., 2022). Existing explanations often emphasize differences in preferences and in the returns to and costs of migration (e.g., Moretti, 2011; Lkhagvasuren, 2014; Davis and Dingel, 2019; Gaubert and Diamond, 2022; Caldwell and Danieli, 2024; Amior, 2024). We show that differences in the local concentration of social networks can generate substantial differences in migration behavior, and may help microfound the heterogeneity in moving costs and home attachment pointed to in this literature.

Lastly, our findings are also linked to the macroeconomic literature on local economic shocks, where migration is a key adjustment margin (e.g., Blanchard and Katz, 1992; Monras, 2018; Hornbeck and Moretti, 2024; Foschi et al., 2025) and the effects of shocks can propagate through social networks (e.g., Diemer, 2026). At the same time, if social ties are geographically concentrated, leaving distressed areas may be less attractive to workers, potentially slowing adjustment relative to a world with more dispersed networks (Zabek, 2024). This mechanism also matters for policy. On the one hand, it may strengthen a potential rationale for place-based interventions: when moving severs valuable social ties, the welfare costs of leaving may be larger than standard models imply (e.g., Glaeser and Gottlieb, 2008; Kline and Moretti, 2014; Glaeser and Redlick, 2009). On the other hand, since networks are not inherently place-based, our results suggest that, in settings where mobility is a policy goal, policies that help individuals move with their ties—for example, subsidies that facilitate coordinated relocation to more productive areas—may differ substantially from policies targeting individuals in isolation.

The remainder of this paper is structured as follows. Section 2 describes the data in more detail and presents descriptive evidence on the role of social networks for residential choice. In Section 3 we outline the empirical framework to study the causal effect of social networks on residential choice, the results of which we show in Section 4. Lastly, Section 5 presents a model of spatial equilibrium which incorporates the role of social networks and highlights how this addition helps to explain observed migration patterns. Section 6 concludes.

2 Data & Descriptive Facts

2.1 Facebook Data

Prior studies of the relationship between social networks and migration have been limited by data availability, as few datasets combine panel data on individuals' locations and their social networks. In this paper, we leverage de-identified, individual-level data from Facebook that

combine both of these attributes, for a large sample. As of 2023, Facebook had 272 million monthly active users and 205 million daily active users in the U.S. and Canada (Facebook, 2023).

For each user, we observe a daily prediction of their CZ of residence from January 2012 through December 2025. This prediction does not rely on self-reported information and instead combines several sources of information, including an individual's IP address and their on-platform activity. We aggregate these daily predictions by quarter using the modal CZ. We validate the CZ-level migration rates from these data against those in Sprung-Keyser et al. (2022) to find a weighted correlation of 0.98 for inter-CZ migration rates and 0.91 for overall out-migration rates across CZs, indicating that our data capture migration patterns very well (see Appendix Figure A1 for details).

Crucially, the Facebook data let us measure people's social networks directly. For the entire user base, we observe all on-platform friends as well as changes in their connections over time.[4] These friendships are usually between individuals who know each other in person (Jones et al., 2013), and the same underlying Facebook user data have been used to study the relationship between social connections and various economic and social outcomes.[5]

We also observe basic demographic information such as a user's age, gender, the high school and college they attended, as well as their own and their parents' predicted socioeconomic status.[6,7] Using this dataset, we define three samples for the analyses below. Appendix B describes the construction of each sample in detail. All three samples draw on the same underlying de-identified Facebook data used in Chetty et al. (2022a,b).

General Sample. The General Sample constitutes our broadest sample of U.S. Facebook users. We use this sample for the descriptive results in Section 2.2 and for the external-validity exercises in Section 4.5. We restrict to users born between 1970 and 2000 because

[4]These connections are undirected in the sense that they represent reciprocal friendship links, rather than a one-sided follower relationship. An individual can have at most 5,000 connections at a given time.

[5]For instance, prior work has studied trade flows, patent citations, travel flows, disease transmission, bank lending, social capital, social program participation, investment decisions, product adoption decisions, housing choices, migration, immigrant integration, and beliefs and behaviors related to public health (Bailey et al., 2018a,b, 2019, 2020a,b, 2021, 2022, 2024; Chetty et al., 2022a,b; Kuchler et al., 2022a,b; Sahai and Bailey, 2022; Bailey et al., 2026; Rehbein and Rother, 2025; Wilson, 2022a).

[6]The construction and benchmarking of these variables are discussed in more detail in Chetty et al. (2022a) and Chetty et al. (2022b). In particular, Chetty et al. (2022a) describe the parent linkage and the construction of predicted socioeconomic status using characteristics such as a user's phone price and validate it against external measures. Chetty et al. (2022b) explain the assignment of users to high schools and colleges using self-reported information, network information, and administrative data sources such as NCES and IPEDS.

[7]Duggan et al. (2016) found that, among U.S. adults, usage rates were relatively constant across income groups, education levels, and race.

coverage is particularly high for those cohorts (Chetty et al., 2022a). We further limit to individuals who are active on the platform and have non-missing CZ locations in all quarters used in the analysis (i.e., Q1 of 2012, and Q4 of 2017 and 2025). For this sample, we measure networks using each individual's friends who were befriended before the start of 2012.

College Graduation Sample. The College Graduation Sample serves as the primary analysis sample for the causal analyses in Section 4. It consists of users who attended a four-year college in the United States and who are expected to graduate in the spring of 2017, 2018, or 2019 if following a "standard" timeline.[8] For brevity, we refer to this expected date as the individual's point of graduation. For the main analyses, we measure graduates' networks using "pre-graduation" friends: friends whom they had befriended by May of the year prior to expected graduation.[9]

Model Sample. The Model Sample is a subset of the General Sample, restricted to users who lived in one of the 50 CZs with the largest number of Facebook users residing there in Q1 of 2012. We refer to these throughout as the 50 largest CZs.[10] For this sample, we again measure networks using each individual's friends who were befriended before the start of 2012. We draw on this sample for the model results in Section 5.

All three samples are further restricted to individuals who have at least 50 friends on the platform and for whom we observe in which CZ they attended high school. All friend measures include only U.S.-based friends.

Panels (a)–(c) of Table 1 report summary statistics regarding the migration behavior, social networks, and demographics of individuals in the three samples.[11] We discuss various aspects of these statistics throughout the paper.

[8]We match self-reported colleges to administrative data from IPEDS, dropping online colleges as well as colleges that do not appear in the Carnegie Classification. Our approach follows the procedure described in the data appendix of Chetty et al. (2022b). We define an individual's expected year of college graduation as four years after their expected high-school graduation, with expected high-school graduation inferred from birth date and the relevant school-entry cutoff. Violations of the standard-timeline assumption would lead us to mis-measure an individual's time of graduation, which, in turn, would attenuate the estimates discussed in Section 4.

[9]We relax this restriction for the placebo exercise focused on "post-graduation" friends described in Section 4.2.

[10]Appendix Figure A2 presents a map of these CZs, and the corresponding figure note lists them by name. 54% of individuals in the General Sample and 58% of individuals in the College Graduation Sample live in one of these CZs in Q1 of 2012 and upon graduation, respectively. For comparison, 45 of the 50 CZs selected by Facebook user count also rank among the 50 largest CZs by 2012 ACS population, and the remaining five all fall within the top 62 by ACS population.

[11]We note that all three samples over-represent women: Across the samples, 62–63% of individuals are female. To address concerns that this over-representation affects our estimates, in Sections 4.4 and 4.5, we estimate network effects separately for men and women. Reassuringly, we find no meaningful difference between men and women.

2.2 Descriptive Facts

Drawing on the General Sample, we document several stylized facts about the spatial concentration of social networks and the relationships between migration decisions, social networks, and local economic opportunity.

First, we find that social networks are geographically concentrated, at least on average. Measuring friendships and locations at the start of our panel in 2012, Figure 1 shows that nearly 60% of the average individual's friends live in the same CZ as they do, and around 74% (87%) of friends live within 100 (500) miles.[12] Yet, as pointed out by Panel (a) of Table 1, this average masks substantial heterogeneity: on one end of the distribution, 5% of individuals have 7% or fewer of their friends living in a CZ other than their own, whereas on the other end, 5% have 92% or more of their friends elsewhere.

This heterogeneity in network concentration is a powerful predictor of people's migration behavior. While 32% of individuals overall have left their 2012 CZ by the end of the sample period in 2025, Panel (a) of Figure 2 shows that the likelihood of moving away strongly differs by the extent to which one has friends living in CZs other than one's own as of 2012. Those with at most one-third of their friends outside their initial CZ leave in only around 22% of cases; in contrast, 58% of individuals with at least two-thirds of their friends relocate. The predictive importance of network concentration also stands out relative to other commonly considered determinants of migration. In Panel (a) of Table 2 we report results from regressions of out-migration on the share of friends living outside the CZ, age, college background, and whether one grew up in the CZ. In the univariate specifications, the friend measure has the greatest explanatory power; when all measures are included jointly, its coefficient remains nearly unchanged, whereas those on college background and having grown up in the CZ fall sharply. This comparison suggests that the locations of people's social networks are a particularly important predictor of out-migration.

Where an individual's friends live predicts not only whether one stays in their initial location, but also where one moves to. In Panel (b) of Figure 2, we explore whether moving to a CZ with high predicted wage growth by 2025 is correlated with the extent to which people have pre-existing friends there, subsetting to individuals who did not live in such a CZ initially.[13]

[12] These patterns are consistent with prior evidence on the spatial concentration of social networks using similar data, see for instance Bailey et al. (2018b).

[13] High-wage-growth CZs are defined as the ten CZs (among the largest 50 CZs) with the highest predicted wage growth between 2012 and 2024. We construct predicted wage growth at the CZ level using a standard shift-share (Bartik-style) approach. We obtain employment and wage data by 6-digit NAICS industry at the county level from the Quarterly Census of Employment and Wages (QCEW) and aggregate these to the CZ level. For each CZ, we compute 2012 industry employment shares and interact them with national

Strikingly, people with relatively more friends in high-wage-growth CZs are substantially more likely to have moved to such locations by 2025: while only around 2% of those with no—or almost no—2012 friends in high-wage-growth CZs have moved there by 2025, those with more than 10% of friends there move to high-wage-growth CZs in almost 15% of cases. Panel (b) of Table 2 presents corresponding regressions of moving to a high-wage-growth CZ on the share of friends living in such CZs, age, college background, and distance from an individual's 2012 CZ to the nearest high-wage-growth CZ. Again, the network measure stands out as a much stronger predictor than any of the alternatives.

While the patterns in Figure 2 are descriptive in nature, they indicate that the location of an individual's social network may play an important role in determining where one lives, and hence the extent to which they harness the economic opportunities offered by different areas.

3 Estimation Strategy

Motivated by the above correlations, we now develop a research design to investigate whether networks causally affect migration decisions. While we examine network effects on both out-migration and directed migration, for brevity, we primarily describe our approach for the out-migration case before discussing how the design can be adapted to study directed migration later on.

Identification Challenge

In a hypothetical, "ideal" experiment designed to study how networks shape out-migration, an individual's friends would be randomly reallocated across geographies. In that case, the causal effect of having a larger network outside one's home CZ on out-migration could directly be estimated from:

$$Y_{i-j,t} = \beta_{-j} n_{i-j,t} + \varepsilon_{i-j,t} \tag{1}$$

where $Y_{i-j,t}$ is a binary indicator for whether individual i from "home CZ" j lives outside j at time t, and $n_{i-j,t}$ measures i's network outside j at that time, so that β_{-j} captures the

wage growth in the corresponding industries between 2012 and 2024, excluding the CZ itself. Predicted wage growth for a CZ is then given by the leave-out national industry wage growth rates, weighted by the CZ's initial industry shares. We define "high-wage-growth" CZs as the ten CZs with the highest predicted wage growth among the 50 largest CZs displayed in Appendix Figure A2. These CZs are Austin, Charlotte, Phoenix, Port St. Lucie, Portland, San Diego, San Francisco, San Jose, Seattle, Tampa.

causal effect of out-of-CZ ties on out-migration.

In reality, networks are not randomly assigned across geographies, and other, potentially unobserved factors $\theta_{i-j,t}$ may affect both $n_{i-j,t}$ and the likelihood of leaving j. We can represent this as:

$$Y_{i-j,t} = \beta_{-j} n_{i-j,t} + \theta_{i-j,t} + \varepsilon_{i-j,t} \tag{2}$$

Note that $\theta_{i-j,t}$ can capture both permanent factors—such as latent preferences or skills shared between individuals and their friends—and time-varying unobservables, including "common shocks" (e.g., industry- or occupation-specific local labor demand shocks) that affect both the individual and their friends (Manski, 1993).

Without observing $\theta_{i-j,t}$, a cross-sectional estimation of equation 2 would yield a biased estimate $b_{-j} = \beta_{-j} + \delta_{-j}$, combining the causal network effect (β_{-j}) with a bias term $\delta_{-j} = \frac{Cov(\theta_{i-j,t}, n_{i-j,t})}{Var(n_{i-j,t})}$. This bias arises because $n_{i-j,t}$ is correlated with $\theta_{i-j,t}$.

Studies of peer and network effects often address related concerns using quasi-random variation in peer composition. Such approaches are natural when the object of interest is exposure to peers with particular characteristics or experiences within a school or workplace (e.g., Hoxby, 2000; Sacerdote, 2001; Mas and Moretti, 2009). Applied to migration, a peer-composition design would typically vary exposure within the home CZ j to peers with prior experience elsewhere. For example, among students in Cleveland, such a design might compare those who are quasi-randomly exposed to classmates who previously lived in New York to those who are not, and then ask whether this exposure affects later moves to New York. Such variation is informative about how exposure to people with place-specific experience affects mobility. The object in equation 1, however, is different: $n_{i-j,t}$ measures the extent to which individual i's network is currently located outside the home CZ j. Even if the classmates in the preceding example become friends, the design does not isolate the value of having friends contemporaneously present in New York, since the relevant peers are encountered within Cleveland. As Section 4.3 shows, this distinction matters empirically because network effects in our setting are closely tied to friends' contemporaneous presence.

College Graduation Research Design

In light of these considerations, we require a design that both addresses unobserved determinants of migration and captures the contemporaneous-presence margin highlighted above. With these objectives in mind, we next develop a quasi-experimental strategy that exploits

variation in the timing of friends' moves around a clear transition point at which individuals are systematically prompted to choose where to live: college graduation. Appendix Figure A3 shows that the point of expected graduation coincides with substantial and persistent migration, underscoring its importance for location choices.[14] Appendix Figure A4 further highlights that the location choices made at the time of graduation are strongly predicted by where people's friends live.[15]

To build intuition for our design, we begin with a stylized example, visualized in Appendix Figure A5. Consider two college graduates, *Alice* and *Bob*, who grew up and attended college in the same CZ (e.g., Philadelphia) and graduate at the same time. Around their graduation, both decide whether to remain in Philadelphia or leave upon graduation. They begin college with the same network size in and outside Philadelphia, and each has one additional friend who leaves Philadelphia around *Alice*'s and *Bob*'s graduation. The key difference is *when* their friends leave: *Alice*'s friend leaves before *Alice* and *Bob* graduate—in the "pre-period"—while *Bob*'s friend leaves after *Alice*'s and *Bob*'s graduation—in the "post-period." Thus, when *Alice* and *Bob* make their location choice, *Alice* has one more friend outside Philadelphia than *Bob*. Our empirical design asks whether this difference results in *Alice* being more likely to leave Philadelphia upon graduation.[16]

How does this setup address unobservables? In principle, *Alice*'s and *Bob*'s friends' moves

[14]Panel (a) of Figure A3 plots the monthly share of college graduates moving to a CZ other than their home CZ (defined as the union of the CZs in which they attended high school or college). Around the expected graduation date, out-migration increases sharply. During the months between September and May of senior year, monthly out-migration rates are below 0.5%; they then rise to approximately 1.5–2.5% over several months, resulting in quarterly outflows of roughly 4% in both the third and fourth quarter of the graduation year. Thereafter, out-migration declines and stabilizes at just over 0.5% per month. Panel (b) shows that these location choices are persistent: nearly 80% of graduates who leave their home CZ at graduation remain outside their home CZ six years later, and almost half of them continue to reside in the same non-home CZ to which they initially moved.

[15]Panel (a) of Appendix Figure A4 shows that the likelihood of leaving one's home CZ upon graduation increases strongly with the share of friends residing outside the home CZ at the time of graduation. Panel (b) shows that graduates with a larger share of friends living in high-wage-growth CZs are substantially more likely to move to such locations upon graduation. This analysis is restricted to graduates who did not reside in a high-wage-growth CZ during high school or college. High-wage-growth CZs are defined analogously to Section 2.2, i.e., they correspond to the ten CZs (among the 50 largest CZs) with the highest predicted wage growth between 2012 and the individual's year of graduation.

[16]Our approach is conceptually related to identification strategies used in studies of intergenerational job referrals. For example, Kramarz and Skans (2014) and Staiger (2026) leverage variation in the timing of parents' firm transitions relative to when their children enter the labor market to identify the effect of parental employment at a firm on the child's likelihood of starting work there. The way we compare estimates for friends' moves before versus after an individual's own decision point is also related to mover designs that use the timing of moves to distinguish pre- and post-move exposure effects (e.g., Chetty and Hendren, 2018a; Brown et al., 2023). In those settings, however, the mover herself changes environments and the goal is to estimate place effects; here, we use the timing of friends' moves to estimate the effect of friends' locations on the individual's own migration decision.

may reflect unobserved factors that also shape whether *Alice* and *Bob* themselves leave Philadelphia, such as shared preferences or common shocks. To deal with this, our design relies on a symmetry-in-timing assumption: *Alice*'s friend moving before graduation should be no more (and no less) informative about these unobserved determinants than *Bob*'s friend moving after graduation. Unobservables therefore may exist and can even vary over time; what matters is that pre- and post-period friend moves are equally informative about these forces. Under this assumption, the difference in out-migration between *Alice* and *Bob* isolates the causal effect of having more friends outside Philadelphia at graduation.

Formalizing the idea of this stylized example, we extend equation 1 to include friends' moves:

$$Y_{i-j,t^*} = \sum_{t=early+1}^{T} b_{-j,t} \Delta n_{i-j,t} + X_{i-j,t^*} + \varepsilon_{i-j,t^*} \tag{3}$$

Here, a graduate's home CZ j is defined as the places in which they lived during adolescence and early adulthood, which we operationalize as the union of CZs in which they attended high school or college. We measure graduates' locations at t^*, defined as the last quarter of the graduation year,[17] and our event-time window begins with the fall (Q3) of one's sophomore year, referred to as *early*.

Central to the design is $\Delta n_{i-j,t} = n_{i-j,t} - n_{i-j,t-1}$, the change in the share of i's friends living outside j over time.[18] As mentioned in Section 2, we fix the friendship network to include only friends formed at least one year before i graduates from college, i.e. by the end of May of junior year. Holding the set of friends constant implies that $\Delta n_{i-j,t}$ captures friends' relocations rather than new friendships, and mitigates concerns about endogenous friendship formation, meaning the potential for individuals to systematically befriend people in response to where they will live upon graduation.[19]

[17] For example, if i is expected to graduate in May 2017, t^* corresponds to Q4 (i.e., October–December) of 2017. Panel (a) of Appendix Figure A3 indicates that by this point much of the post-graduation movement has occurred.

[18] We use a share-based measure to ensure that heterogeneity in Facebook usage—and thus in the total number of friends—does not mechanically affect the interpretation of the relevance of an individual's social network. In results not shown, we replicate the analysis using friend counts and find qualitatively very similar results.

[19] Because we fix the baseline network in May of the year before expected graduation, for this form of endogenous friendship formation to drive our main effects, graduates would need to have formed sufficiently specific post-graduation location plans roughly a year before graduation and then formed friendships in response to those plans. Survey evidence places the beginning of the average job search around six months before graduation, while career guidance recommends beginning active applications roughly four months before graduation (National Association of Colleges and Employers, 2025; LIV Consulting & Career Services,

We then study the effects of friends' moves during the twelve quarters prior to, as well as the twelve quarters following t^*. At baseline, we consider all t between 12 and 4 quarters before t^* as part of the pre-period (t^{pre}), and quarters between 4 and 12 quarters after t^* as part of the post-period (t^{post}). The remaining quarters constitute the "mid-period" ($\bar{t}$), during which graduates may already make post-graduation location decisions and can potentially anticipate friends' moves that occur shortly after graduation. We begin this period near the start of the expected graduation year, when students are increasingly likely to be actively searching and applying for post-graduation jobs: survey evidence places the beginning of the average job search around six months before graduation, while career guidance recommends beginning active applications roughly four months before graduation (National Association of Colleges and Employers, 2025; LIV Consulting & Career Services, 2023). The baseline window therefore excludes the quarters closest to graduation, where both anticipation and early post-graduation decisions are most likely to matter. We show that our results are robust to alternative definitions of the pre- and post-periods that shift these windows closer to graduation or use narrower windows around t^*.

Note that, while for expositional clarity we describe $\Delta n_{i-j,t}$ as arising from friends moving *away* from j, in practice we leverage variation coming from both moves away from and moves to j. In Section 4.3 we present evidence from an estimation that is based separately on moves away from and into j: we find little heterogeneity by move direction, justifying the more parsimonious definition employed here.

Equation 3 also includes a rich set of controls, X_{i-j,t^*}. In our main specifications, we include fixed effects for the full interaction of one's high school CZ, college CZ, graduating cohort, and quintiles of the share of friends outside j at $t = early$, with the quintiles defined within high-school-CZ-by-college-CZ-by-graduating-cohort cells. These controls implement the narrow comparisons in the stylized example: we compare graduates with similar location histories and similar initial network exposure who differ primarily in the timing of friends' moves. We show below that we reach similar conclusions when drawing even finer comparisons.

Applying the omitted-variables logic from equation 2, the coefficients from our dynamic design combine an event-time-specific causal effect and an event-time-specific bias: $b_{-j,t} = \beta_{-j,t} + \delta_{-j,t}$. Here, $\beta_{-j,t}$ captures the causal effect of friends' moves at t, while $\delta_{-j,t}$ reflects

2023), substantially later than our friendship cutoff. Nonetheless, we find similar results when we focus only on friends' moves that occur after the corresponding friendship had already formed.

bias from unobserved factors correlated with moves at that time: formally,

$$\delta_{-j,t} = \frac{\text{Cov}\left(\widetilde{\theta}_{i-j,t^*}, \widetilde{\Delta n}_{i-j,t}\right)}{\text{Var}\left(\widetilde{\Delta n}_{i-j,t}\right)},$$

where tildes denote residuals after partialling out the controls and all other event-time friend-move variables included in equation 3. The underlying $\Delta n_{i-j,t}$ may itself depend on unobservables, for instance because friends' and graduates' location choices may be driven by shared preferences or common shocks.[20] Crucially, friends' moves after graduation cannot affect the graduate's location choice at t^*, so $\beta_{-j,t} = 0$ in the post-period and the corresponding coefficients identify only the bias term, except to the extent that graduates' location choices themselves affect friends' subsequent moves, a reverse-causality possibility we discuss below. With this in hand, identification hinges on the following assumption about the timing of friends' moves.

Identifying Assumption. Conditional on observables and on experiencing friends' moves around t^*, whether those moves occur before or after t^* is orthogonal to graduates' potential outcomes.

Put differently, we do not require that where individuals' friends live or where they move to is quasi-random; rather, we only require that whether friends move before or after the graduate's own graduation is quasi-random. Under this assumption, friends' moves before versus after graduation are equally informative about unobserved determinants of the graduate's location choice at t^*, which implies that $\delta_{-j,t}$ is the same for event times in pre- and post-period. As a result, differencing the pre- and post-period $b_{-j,t}$ estimates removes the bias and recovers the causal network effect: for representative t's in t^{pre} and t^{post}, we have

$$\Delta b_{-j,t^{pre}} = b_{-j,t^{pre}} - b_{-j,t^{post}} = \beta_{-j,t^{pre}}. \tag{4}$$

Intuitively, this corresponds to comparing graduates like *Alice* and *Bob* who are similar ex ante but whose friends move slightly earlier versus later around graduation.

The formal assumption thus mirrors the symmetry-in-timing intuition described above. It allows for unobserved factors—including shared preferences and common shocks—and these factors may be persistent or time-varying. What would violate the assumption is a systematic

[20] More broadly, these unobservables may include persistent factors, such as latent location preferences or skills shared between individuals and their friends, as well as time-varying factors, such as local economic shocks or college- and field-specific migration pipelines.

difference in how informative friends' moves are about these unobservables in the pre-period versus the post-period. Put differently, we only require $\delta_{-j,t}$ to be the same for pre- and post-period, while $\theta_{i-j,t}$ is allowed to vary. We further discuss the plausibility of this assumption and present supporting evidence in Section 4.2.

As noted above, one caveat to using post-period friends' moves to quantify bias is the possibility of reverse causality: in principle, some friends' relocations in t^{post} may be driven by the graduates' location choices at t^*. To the extent this occurs, the $b_{-j,t}$ estimates for the post-period would not only reflect $\delta_{-j,t}$, but also a reverse-causality component capturing the graduate's influence on their friends. This concern would generate a downward bias in our estimated network effects. If friends follow graduates after t^*, then reverse causality raises the post-period coefficients. Since our estimator subtracts the average post-period coefficient from the average pre-period coefficient, this would lead us to over-correct and therefore attenuate the estimated network effect. Empirically, in Section 4.1 we show that the post-period coefficients are small relative to the pre-period coefficients, suggesting that this channel is not quantitatively important; Appendix C formalizes this argument and explains why, even if graduates affect their friends' future location choices, this need not generate large post-period coefficients and therefore implies only limited attenuation of the estimated network effect.

Directed Migration

The same conceptual framework also allows us to study directed migration, i.e., how the size of an individual's network in a given potential destination CZ k affects their probability of moving there upon graduation. Panel (b) of Appendix Figure A5 presents an illustrative example for the case of New York, similar in spirit to the example for out-migration discussed above. More formally, we estimate

$$Y_{ik,t^*} = \sum_{t=early+1}^{T} b_{k,t}\Delta n_{ik,t} + X_{ik,t^*} + \varepsilon_{ik,t^*} \tag{5}$$

where Y_{ik,t^*} is a binary indicator for whether individual i moves to CZ k upon graduation, and $\Delta n_{ik,t}$ corresponds to friends' moves to and from k at various points in time t. For clarity, we refer to $\Delta n_{ik,t}$ as capturing moves *to* k.[21] To capture network effects on active migration among graduates, for every potential destination k, we restrict to graduates who did not attend high school or college there. The control vector X_{ik,t^*} is defined analogously

[21] In Section 4.3 we estimate effects separately for friends moving to and away from k.

to the out-migration case, with fixed effects for the full interaction of high school CZ, college CZ, graduating cohort, and quintile of the share of friends in k at $t = early$, where the quintiles are defined within high-school-CZ-by-college-CZ-by-graduating-cohort cells.

Identification mirrors the out-migration case: we compare $b_{k,t}$ coefficients for pre- and post-period and assume that, conditional on having a friend move to k around t^*, the exact timing of that move is orthogonal to i's potential outcomes. Put differently, friends' moves to k before versus after graduation must be equally informative about any unobserved determinants that affect i's propensity to move to k upon graduation. As for out-migration, we probe this assumption in Section 4.2.

Before presenting our main results, note that while we began this section with the objective of wanting to estimate β_{-j} and β_k—or the causal effect of one's social network on location choice—we have now arrived at a framework that allows us to estimate versions of $\beta_{-j,t^{pre}}$ and $\beta_{k,t^{pre}}$ for various t. These parameters describe the effect of friends' moves at various points in time. The quantitative difference between them and our target parameters β_{-j} and β_k depends on the extent to which the network effects vary with tenure, i.e. the amount of time a friend has already spent outside j or in k. We return to this discussion in Section 4.1.

4 Results

We now present reduced-form evidence on the causal effect of social networks on residential choice using the identification strategy outlined above. We begin with out-migration as the outcome before turning to directed migration. We then assess the plausibility of the identifying assumption and present additional results on mechanisms, heterogeneity, and external validity.

4.1 Baseline Estimates

4.1.1 Out-migration

Figure 3 reports the main regression results for out-migration using the College Graduation Sample described in Section 2. Each dot is one $b_{-j,t}$ estimate from equation 3, capturing how friends leaving i's "home CZ" j at event-time t predict the graduate's probability of leaving j upon graduation. The horizontal axis is centered at t^*, the quarter in which we measure post-graduation location, i.e., Q4 of the year in which one graduates. The gray bar around t^* highlights the mid-period ($\bar{t}$) during which graduates likely decide where to live, and during

which they may anticipate friends' moves that occur shortly after their graduation. Red and blue dots surrounding $\bar{t}$ correspond to coefficient estimates for friends' moves in the pre- and post-period, respectively. The event window spans 25 quarters: nine in each of the pre- and post-periods, and seven in the mid-period.

The estimates depict a sharp drop around graduation: coefficients in the pre-period are large (between 0.32 and 0.45) and highly statistically significant, while coefficients in the post-period are much lower, hovering between 0 and 0.13. Thus, having a friend leave one's home CZ at any point prior to one's graduation is far more predictive of whether one leaves the home CZ upon graduation than having a friend leave at any point after one's graduation. As discussed in Section 3, each $b_{-j,t}$ *individually* does not represent a causal network effect, since it may combine a causal friend effect with a non-causal selection component. However, under our identifying assumption that post-period coefficients capture the same non-causal selection component but no causal effect, the difference between the pre- and post-period estimates isolates the causal friend effect. Since the post-period coefficients are small, this difference is close to the pre-period coefficients themselves; nevertheless, we use the pre–post difference as our baseline object because doing so does not require the pre-period coefficients to be unbiased, but only that any potential bias is comparable for friends moving before versus after graduation.

We summarize the friend effect in Table 3 by taking the average of the event-time coefficients in the pre-period and subtracting the corresponding average in the post-period. Column 1 shows that a one-percentage-point increase in the share of one's network outside j raises the probability of leaving j at graduation by 0.335 percentage points (SE = 0.007).[22] This estimate is about 60% as large as the corresponding cross-sectional slope of around 0.55, obtained by regressing an indicator for leaving the home CZ on the share of friends outside the home CZ at graduation.[23] The difference between the two estimates is consistent with positive selection in the raw relationship, but the timing-based design nevertheless implies that a substantial share of the cross-sectional association reflects causal network effects. Putting this magnitude into perspective, Panel (b) of Table 1 shows that 64% of the average graduate's friends live in the home CZ. Thus, if 10% of a graduate's friends living there left, this would imply a 2.1-percentage-point—or 17%—increase in the graduate's probability of leaving. This effect is comparable to the migration response to roughly a $7,300 decrease in annual local wages, when benchmarked against estimates from Sprung-Keyser et

[22]Appendix Table A2 reports the same estimate under alternative standard-error specifications. The estimate remains highly statistically significant when standard errors are clustered at the high-school-CZ-by-college-CZ-by-cohort level, at the college-by-cohort level, or two-way clustered along both dimensions.

[23]Appendix Figure A4 presents this cross-sectional relationship graphically.

al. (2022).[24]

Appendix Table A3 shows that the out-migration results are similar when splitting graduates by whether their high school and college CZs coincide, and when separately examining moves away from the high school and college CZ for graduates for whom these locations differ. The estimates are positive in all cases, indicating that friends' locations shape moves away from both margins of the home CZ definition.[25]

Figure 3 also shows that coefficients are relatively flat within the pre- and post-periods and fall mainly in the narrow window around t^*, suggesting that our results are not driven by the precise definition of the pre- versus post-period. Appendix Table A1 confirms this directly: the estimates remain positive, highly statistically significant, and economically similar when we shift the pre- and post-period windows closer to graduation or use narrower windows around t^*. The smallest estimate across these alternatives is 0.26, more than three-quarters of the baseline estimate of 0.335.[26] The same table also shows that defining the pre-period using only quarters five and four before t^*—so that all moves used to construct the pre-period occur after the corresponding friendship was formed—yields an estimate that is virtually identical to the baseline result. This finding further mitigates the residual concern about endogenous friendship formation potentially affecting our results, discussed in more detail in Section 3.

The stability of the estimates across alternative pre- and post-period definitions indicates that "tenure"—how long a friend has already been outside j—is not a major source of heterogeneity. We therefore view the effects identified from friends' moves as a reasonable approximation to β_{-j}, the more general parameter capturing the effect of having a larger network outside j on out-migration, abstracting from tenure effects. In what follows, we commonly refer to the difference between the average pre- and post-period coefficients as an estimate of this general network effect.

[24]Sprung-Keyser et al. (2022) find that a roughly \$800 wage increase in one's childhood CZ increases the probability of staying there through age 26 by 0.23 p.p. Thus, 2.1 p.p. / 0.23 p.p. $\times$\$800 $\approx$ \$7,300.

[25]Column 1 restricts to graduates whose high school and college CZs coincide and estimates effects on leaving this common home CZ. Column 2 restricts to graduates whose high school and college CZs differ and estimates effects on leaving the union of the two CZs, corresponding to our baseline home CZ definition. Columns 3 and 4 keep this latter sample but separately use indicators for leaving the high school CZ and leaving the college CZ as outcomes. When comparing magnitudes across columns, it is important to account for differences in the underlying migration rates: in particular, baseline rates of leaving the high school and college CZs are mechanically much higher when these locations differ than the rate of leaving the union of the two CZs.

[26]We do not view these narrower windows as conceptually preferable to the baseline. As the post-period moves closer to t^*, the contrast may become attenuated because graduates may anticipate friends' moves that occur shortly after graduation. This is the reason our baseline specification excludes the quarters closest to t^*.

Lastly, note that the small post-period coefficients in Figure 3 also indicate that concerns over reverse causality, as discussed in Section 3, are unlikely to play a quantitatively important role. If graduates' own location choices caused friends to follow them after t^*, this would primarily raise the post-period estimates. Since our estimator subtracts the average post-period coefficient from the average pre-period coefficient, this channel would lead us to overcorrect and therefore attenuate the estimated network effect. The fact that the post-period estimates are small relative to the pre-period estimates therefore implies that any resulting downward bias in the pre–post difference is likely limited.[27]

4.1.2 Directed Migration

Turning to directed migration, we find that networks shape not only whether graduates leave their home CZ, but also where they go. Panel (a) of Figure 4 illustrates this for New York: we estimate equation 5 relating a graduate's probability of moving to New York upon graduation to friends' moves to New York at different points in time, restricting the sample to graduates who did not live there during high school or college. While the estimates are noisier than those in Figure 3, the event-time profile is similar: coefficients are elevated in the pre-period, fall sharply during the mid-period, and are much lower in the post-period. Under our identifying assumption, the difference between the average pre- and post-period coefficients yields that having a one-percentage-point larger network in New York raises the probability of moving there at graduation by 0.36 percentage points, or 35%.

Panel (b) generalizes this exercise, plotting the distribution of destination-specific network effects, i.e., the difference between the average pre- and post-period estimates. To mitigate concerns over noise in the estimates, here—as well as in the remainder of the paper—we only estimate effects for the 50 largest CZs[28] and apply an empirical Bayes shrinkage procedure following Chetty and Hendren (2018b) described in more detail in Appendix D.[29] We find positive effects for every CZ considered—both before and after empirical Bayes shrinkage—implying that a larger share of one's network in any of these destinations increases the likelihood of moving there at graduation.[30] The average estimate across destinations, shown

[27] Small post-period coefficients do not imply that graduates have no effect on their friends' later location choices. Even if reverse causality contributes to the post-period coefficients, these coefficients are not a direct measure of all graduate-to-friend influence. Appendix C formalizes this distinction.

[28] Appendix Figure A2 presents the included CZs and Table 1 shows that 58% of individuals in our College Graduation Sample live in one of these CZs at graduation.

[29] Appendix Figure A6 presents the distribution of raw, unshrunk effects.

[30] Appendix Table A5 reports inference for the same destination-specific estimates under alternative standard-error specifications. The results are very similar when standard errors are clustered at the high-school-CZ-by-college-CZ-by-cohort level, at the college-by-cohort level, or two-way along both dimensions: 43–44 of the 50 estimates are significant at the 5% level and 38 at the 1% level, compared with 45 and 43

in column 1 of Table 4, is 0.104 percentage points: a one-percentage-point increase in the share of one's network in a destination raises the probability of moving there by about 0.104 percentage points on average. Expressed relative to each destination's baseline migration rate, the average effect of this same one-percentage-point increase is 89.1%, with destination-specific effects ranging from 27.9% to 297.2%. These results are robust to alternative definitions of the pre- and post-periods.[31] Notably, effects are positive even for a lower-wage destination like Detroit, highlighting that social ties can steer location choices even toward places with limited economic opportunities. In Section 4.4, we explore heterogeneity across CZs more systematically.

4.2 Evidence Supporting the Identifying Assumption

Interpreting the pre–post differences as causal network effects rests on the identifying assumption that whether friends move before or after the graduate's own graduation is quasi-random. As discussed in Section 3, this assumption allows for unobserved factors, including shared preferences, college-specific pipelines, and shocks affecting both graduates and their friends. What would violate the assumption is not the presence of such forces, but a systematic difference in how strongly they are reflected in the moves of friends who relocate before versus after t^*. The fact that the coefficients in Figure 3 and Panel (a) of Figure 4 are elevated throughout the pre-period and fall sharply around the graduate's own decision point further narrows the scope for this: for unobservables to drive the effects, friends moving at any time before graduation—including those who move several years earlier—would need to be systematically more similar to the graduate in their unobserved location preferences, or more exposed to shocks that also affect the graduate, than friends who move after graduation, even when the latter move relatively soon after. Relatedly, Appendix Tables A1 and A4 as well as Panel (a) of Appendix Figure A7 show that the estimates remain similar when we shift the pre- and post-period windows closer to t^*. These specifications compare friends' moves that occur closer together in time, making it less likely that the pre–post difference reflects changes over event time in why friends move or in how their moves are related to graduates' own location choices. While these results already limit the scope for systematic

under the baseline standard errors.

[31] Appendix Table A4 and Panel (a) of Appendix Figure A7 repeat the destination-specific analysis using alternative definitions of the pre- and post-period. The average estimates remain positive and relatively similar across specifications, and the destination-specific estimates remain strongly aligned with the baseline estimates. As with the out-migration results, estimates using windows closer to t^* may be attenuated by anticipation of friends' post-graduation moves. The same figure and table also show that defining the pre-period using only quarters five and four before t^*—so that all moves used to construct the pre-period occur after the corresponding friendship was formed—yields an average estimate similar to the baseline effect, and 48 out of the 50 destination-specific estimates remain positive.

differences in how strongly unobservables are reflected in earlier versus later friends' moves, we now probe this possibility more directly in the data.

We present four pieces of supporting evidence. First, friends moving in the pre- and post-period look similar on observable characteristics. Second, the estimates are robust to richer controls that directly absorb increasingly narrow versions of college-, major-, and calendar-time-specific migration patterns; the estimates also remain comparable when we exclude friend types most likely to capture narrow type-specific pipelines, whereby individuals with similar skills or preferences tend to move to the same destinations around the same time. Finally, two placebo exercises using "sister CZs" and future friends provide little evidence that the timing of friends' moves proxies for omitted shocks in settings where causal network effects are absent.

Differences in Observable Characteristics

We first examine whether friends who move before graduation are systematically more similar to the graduate on observable characteristics than friends who move afterward. By design, this concern is limited: friendship networks are fixed at pre-graduation levels, so that the set of friends under study is the same throughout the event window. Panel (a) of Figure 5 further shows that, within this fixed set of friends, those who move in the pre-period are no more similar to the graduate on observables (e.g., socioeconomic status) than friends who move in the post-period. For each observable, we compute the average difference between graduates and their friends (in standard deviations) by quarter of move and regress these differences on an indicator for whether friends moved in the pre- rather than the post-period.[32] All estimates are statistically insignificant and very small in magnitude, never exceeding 0.05 standard deviations. If anything, the point estimates suggest that friends moving in the post-period are marginally more similar to graduates on some characteristics, including the college attended, rather than the reverse. Overall, friends moving earlier and later look similarly related to graduates on observed characteristics.

Of course, this analysis cannot rule out differences along unobserved dimensions: friends who move earlier could, for unobserved reasons, still be more similar to the graduate in latent preferences or more exposed to shocks that also matter for the graduate's location choice. Similarly, even if friends moving in the pre- and post-period look similar on average, one might still worry that the identifying variation is concentrated among particular types of ties whose moves are differentially informative about graduates' own location choices. We therefore next examine whether the estimates are robust across different subsets of

[32]For the purpose of this exercise, we exclude moves happening in the mid-period.

friends.

Robustness to Potential Migration Pipelines

A related concern is that friends' moves may reflect college-, major-, or destination-specific pipelines. Much of this concern is already addressed by our baseline specification, which compares graduates within fully interacted cells defined by high school CZ, college CZ, graduating cohort, and initial network exposure. These controls already absorb broad differences across cohorts and high school/college places, as well as differences in baseline network exposure—the share of friends outside the home CZ for out-migration, or in the destination CZ for directed migration. In principle, the pre–post design should further account for pipeline-related forces in the same way that it accounts for other shared unobservables: if such pipelines affect both graduates and their friends, then friends moving after graduation should proxy for them just as friends moving before graduation do. The remaining identifying concern is therefore narrower: friends who move before graduation may be differentially exposed to some pipeline-relevant component of these forces relative to friends who move after graduation, even within the baseline comparison cells.

We conduct two exercises to probe this possibility. To begin with, we re-estimate the baseline specification with increasingly rich controls designed to absorb narrower versions of such pipeline effects. Specifically, columns 2–6 of Table 3 report five variants for out-migration. First, we interact the baseline fixed effects with gender, so that comparisons are made separately by gender within the baseline cells. Second, we interact the baseline fixed effects with major-group indicators, allowing for broad field-specific migration patterns within the same baseline cells.[33] Third, we add college-by-cohort fixed effects, absorbing institution-specific migration patterns for each graduation cohort. Fourth, we add college-by-major-group fixed effects, allowing different fields within the same college to have different location pipelines. Finally, we control for friends' moves in absolute calendar time, so that identification comes from the timing of friends' moves relative to a graduate's own graduation date rather than from aggregate time-varying migration patterns.[34]

The estimates are very similar across specifications. Panel (b) of Figure 5 presents analogous robustness checks for directed migration. For each richer-control specification, we plot the

[33]We assign majors to five groups based on the occupational distributions of recent graduates. Appendix B describes the construction in detail.

[34]This specification also helps address concerns that some post-period friends' moves occur during the COVID-19 period. Because the control absorbs the predictive content of friends' moves in each calendar period, our main estimates are not identified from broad differences in mobility patterns between pre-period and post-period years. This mitigates potential concerns that the post-period coefficients are mechanically different because they are, in part, measured in unusual calendar years.

resulting destination-specific estimates against the baseline estimates. The estimates lie close to the 45-degree line, indicating that destinations with larger baseline network effects continue to have larger effects after adding these controls.[35]

In a second exercise to address pipeline effects, we ask whether the estimates are driven by the friends who are most likely to share college- or field-specific migration pathways with the graduate. Specifically, we re-estimate the out-migration and directed-migration specifications after excluding, in turn, friends who attended the same college and studied a similar major, friends who studied a similar major, and friends who attended the same college.[36] We would not necessarily expect the estimates to be identical across these specifications: the excluded friends may be closer or more relevant ties, and the remaining network measures are based on a more restrictive subset of friendships. The relevant question is instead whether the estimates disappear once these potentially most concerning ties are removed. They do not. Appendix Table A6 shows that the out-migration estimates remain positive, economically large, and highly statistically significant across all three exclusions. Panel (b) of Appendix Figure A7 shows the analogous results for directed migration, and Appendix Table A7 summarizes the corresponding destination-specific estimates: across all three exclusions, the estimates remain strongly positively related to the baseline estimates and lie relatively close to the 45-degree line. Moreover, for both outcomes, the estimates are quite similar to the baseline when excluding the ties that are arguably most likely to reflect shared college and field-specific pipelines—friends from the same college and those studying a similar major. Thus, the baseline results are not driven by a narrow set of friends who share the most obvious college- or field-specific pipelines with the graduate.

Together, the richer-control and friend-exclusion exercises suggest that the results are not driven by fine-grained college-, major-, or destination-specific pipelines—nor by aggregate calendar-time-specific migration patterns—beyond those already addressed by the baseline specification and the pre–post design.

Placebos

The preceding exercises show that the estimates are robust to adding increasingly rich controls designed to absorb such pipeline-related differences, and to excluding the friends most

[35] Columns 2–6 of Table 4 report corresponding summary statistics. The average destination-specific estimates remain between 0.095 and 0.105 p.p. across the richer-control specifications, compared to 0.104 p.p. in the baseline.

[36] We classify two majors as similar when they belong to the same major group used in the control specifications above. Appendix B describes the grouping and the construction of these alternative friend definitions in detail.

likely to share college- or field-specific migration pipelines with the graduate. We next ask whether the design produces effects in settings where shared unobservables may generate correlated location choices but causal network effects should be absent. If the pre–post breaks underlying our main results in Figures 3 and 4 were driven by shared unobservables that are captured more strongly by friends moving earlier, then similar patterns should emerge in placebo settings where such confounding remains possible, but causal network effects are ruled out by construction.

Our first placebo is based on "sister CZs", i.e., CZs with industry compositions very similar to that of the focal CZ.[37] This approach draws on the intuition behind shift-share designs à la Bartik (1991): places with similar industry structure tend to face similar economic shocks.[38] If our results so far reflect common shocks that differentially load onto friends moving earlier, then friends' moves to a sister CZ should generate apparent effects on whether graduates choose to live in the focal CZ, insofar as those moves are driven by shocks similar to those affecting the focal CZ. By contrast, if the design isolates causal network effects, then friends' moves to sister CZs should not make graduates more likely to choose the focal CZ. Because this test concerns destination choice, we implement it only in the directed-migration setting.

Panel (c) of Figure 5 plots the resulting estimates for the effects of friends' moves to sister CZs against those for moves to the focal CZs.[39] Reassuringly, we find no meaningful effects for sister CZs. For example, having a one-percentage-point larger network in Detroit increases the probability of moving to Detroit by 80.6%, whereas the same one-percentage-point increase in one's network in Detroit's sister CZ—Cleveland—has no effect on moving to Detroit. More generally, sister CZ coefficients are always essentially, if not literally, zero.

Our second placebo exercise considers moves among individuals whom graduates eventually

[37] To find sister CZs, we pair each focal CZ with its nearest neighbor among the set of 50 largest CZs based on 2012 industry employment shares. We provide additional details regarding the construction of sister CZs in Appendix E.

[38] Consistent with this, across focal–sister pairs in our setting, the correlation in percentage employment (wage) growth between 2012 and 2024 is 0.61 (0.58).

[39] The estimation equation is:

$$Y_{ik,t^*} = \sum_{t=early+1}^{T} b_{k,t}\Delta n_{ik,t} + \sum_{t=early+1}^{T} b^{k'}_{k,t}\Delta n_{ik',t} + X_{ik,t^*} + \varepsilon_{ik,t^*}$$

Here, $\Delta n_{ik',t}$ denotes friends' moves to sister CZ k'; all other terms are defined as before. We continue to include friends' moves to the focal CZ in light of Danieli et al. (2026). The estimates shown in Panel (c) of Figure 5 correspond to the difference between the average pre- and post-period estimates for friends moving to the focal and sister CZ, respectively.

befriend, but who are not yet friends at the time of graduation. In our analyses so far we have focused on "pre-graduation friends", i.e., friendships formed at least one year before graduation; here, we instead study "post-graduation friends", i.e., friendships formed at least four years after graduation between individuals who did not overlap in the same high school or college, so that they befriended each other only after the event window.[40] These future friends cannot causally affect location choices at t^*, but are still likely to have location preferences similar to the graduate and to respond to similar location-specific shocks.[41] Thus, if shared unobservables interacting with the timing of friends' moves were driving the main results, moves among these future friends should generate similar event-time patterns.

Panel (d) of Figure 5 reports the corresponding results, comparing the effects of moves among pre- and post-graduation friends.[42] In both the out-migration and directed-migration settings, coefficients on post-graduation friends' moves are very close to zero. Together, these placebo exercises therefore provide no evidence that shared preferences or common shocks—even if present—load differentially onto friends moving before versus after graduation, supporting the interpretation that the results in Figures 3 and 4 reflect causal network effects.

4.3 Mechanisms

Having demonstrated that social networks shape residential choices, we next explore the mechanisms underlying these effects, focusing in particular on the role of information. A large literature highlights that social networks shape access to job opportunities, housing, and knowledge about local conditions more broadly (e.g., Rees, 1966; Granovetter, 1973; Bayer et al., 2008; Desmond, 2012). In turn, these informational advantages may induce individuals to live in places where they have a larger network. Relatedly, friends could shape

[40] In practice, in some cases, "post-graduation" friends and graduates may have known each other since well before they connect on Facebook. Importantly, any remaining misclassification would likely bias the placebo coefficients away from zero, making it more difficult to obtain null effects.

[41] Consistent with this premise, among those with at least five post-graduation friends, the correlation between the proportion of a graduate's pre- and post-graduation friends across the 50 largest CZs is 0.68, indicating that the two sets of friends exhibit similar geographic sorting across destinations.

[42] For out-migration, the estimation equation is given by:

$$Y_{i-j,t^*} = \sum_{t=early+1}^{T} b_{-j,t}\Delta n_{i-j,t} + \sum_{t=early+1}^{T} b_{-j,t}^{PostGrad}\Delta n_{i-j,t}^{PostGrad} + X_{i-j,t^*} + \varepsilon_{i-j,t^*}$$

$\Delta n_{i-j,t}^{PostGrad}$ corresponds to moves among post-graduation friends, and all other terms are defined as before. For directed migration, the estimation equation is defined analogously. As in the case of sister CZs, we continue to include friends' moves in the regression in light of Danieli et al. (2026). The estimates shown in Panel (d) of Figure 5 correspond to the difference between the average pre- and post-period estimates for pre-graduation and post-graduation friends, respectively.

location choices by making particular places more salient, leading graduates to consider or search for jobs in locations that they otherwise would not have considered. While our data do not allow us to directly test such informational or consideration-set channels, we present evidence suggesting that information alone is unlikely to be the primary driver of the network effects we document.

We first consider job-related information. One reason to doubt that such information is the main mechanism comes from the robustness exercises in Section 4.2. If the effects were primarily driven by job-search networks, one might expect them to be especially sensitive to friends who share similar fields of study and are therefore more likely to provide relevant information about job opportunities. Instead, the estimates remain similar when we exclude friends studying a similar major, and even when we exclude friends who attended the same college and studied a similar major. These results do not rule out any role for job-related information, but they suggest that the baseline effects are not driven by the types of ties most likely to transmit occupation-specific job information.

In a second exercise, we study whether network effects depend on friends' contemporaneous presence in a location, as opposed to past exposure. If friends primarily matter because they convey information about a place—and if they can still provide such information even after having left—then friends' departures should matter less for location choices. By contrast, if network effects depend on ongoing interactions or other benefits that require friends' contemporaneous presence, friends' departures should exert effects on where graduates choose to live. To assess this distinction, we separately estimate the effects of friends' moves that increase versus decrease their contemporaneous presence in a location: for out-migration, we distinguish friends leaving the graduate's home CZ from friends moving into it; for directed migration, we distinguish friends moving to a potential destination CZ from friends moving away from it.[43]

Panels (a) and (b) of Figure 6 present the corresponding results.[44] Beginning with out-

[43] One caveat is that a friend's decision to move away might itself convey information about a place (e.g., poor job prospects). While we cannot rule this out directly, such information would likely be transmitted before the friend moves. This would tend to generate gradual changes in coefficients prior to t^*, whereas the event-time profiles in Panel (a) of Figure 6 exhibit relatively flat pre-period estimates followed by sharp changes around $\bar{t}$.

[44] For out-migration, the estimation equation is given by:

$$Y_{i-j,t^*} = \sum_{t=early+1}^{T} b_{-j,t}^{LeaveHomeCZ} n_{i-j,t}^{LeaveHomeCZ} + \sum_{t=early+1}^{T} b_{-j,t}^{MoveToHomeCZ} n_{i-j,t}^{MoveToHomeCZ} + X_{i-j,t^*} + \varepsilon_{i-j,t^*}$$

$n_{i-j,t}^{LeaveHomeCZ}$ and $n_{i-j,t}^{MoveToHomeCZ}$ capture the proportion of a graduate's friends leaving—or moving

migration in Panel (a), the patterns for friends leaving the graduate's home CZ and friends moving there are close to mirror images: while the series for friends leaving looks very similar to that in Figure 3, estimates for friends moving to the home CZ are negative in the pre-period, increase steeply around $\bar{t}$, and are close to zero in the post-period. The difference between the average pre- and post-period estimates is similar in absolute magnitude to that for friends leaving the home CZ, 0.31 versus 0.26. Panel (b) shows analogous patterns for directed migration: effects for friends moving away from a destination are again negative and often of similar absolute magnitude as those for friends moving to that destination. Thus, network effects are tightly linked to friends' contemporaneous presence, which further narrows the class of information-based explanations that could be plausible mechanisms.

In principle, the emphasis on contemporaneous presence could reflect romantic relationships as the main driver of the network effects. The evidence does not point in this direction. For out-migration, specifications analogous to column 1 of Table 3, but distinguishing same- from different-gender friends, yield coefficients of 0.22 and 0.17, respectively.[45] Appendix Figure A8 shows a similar pattern for directed migration: destination-specific estimates for same- and different-gender friends lie relatively close to the 45-degree line. These results suggest that romantic relationships alone are unlikely to explain the main effects.

The importance of friends' contemporaneous presence also helps interpret the distinction between our design and the peer-composition approaches discussed in Section 3. Peer-composition designs are well suited to study how exposure to peers with particular backgrounds or place-specific experiences affects later choices. Our design instead isolates how location choices respond to the current locations of individuals' actual friends. The results above suggest that this distinction matters in our setting: the network effects appear to be dominated by the value of having friends contemporaneously present in the relevant location.

Finally, to shed light on what individuals value about their friends' contemporaneous pres-

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to—home CZ j between $t-1$ and t. Other terms are unchanged.

For directed migration, the estimation equation is given by:

$$Y_{ik,t^*} = \sum_{t=early+1}^{T} b_{k,t}^{To} n_{ik,t}^{To} + \sum_{t=early+1}^{T} b_{k,t}^{Away} n_{ik,t}^{Away} + X_{ik,t^*} + \varepsilon_{ik,t^*}$$

where $n_{ik,t}^{To}$ ($n_{ik,t}^{Away}$) captures the proportion of a graduate's friends moving to (away from) k between $t-1$ and t. Other terms are unchanged.

[45] In these specifications, moves of same- and different-gender friends enter jointly, so each coefficient is estimated conditional on the other. The magnitudes are not directly comparable to the all-friends estimate, since splitting friends by gender substantially subsets the set of relevant friends and therefore changes the scale of the underlying variation.

ence, we complement these analyses with evidence from an off-platform survey, described in more detail in Appendix F. The responses are consistent with social networks being an important self-reported driver of location choices: over 80% agree that friends and family are one of the main reasons they live in their current city. They also point to the "social amenity" value of these ties. Over 70% report that they see the people they consider most important for their location choice always or most of the time for leisure, and 85% indicate that they value the companionship of these ties far more than any practical assistance they provide. Because the survey is drawn from a selected sample of respondents who live near where they grew up and relies on self-reported responses, we view these results as suggestive. At the same time, they indicate that the social amenity value of having friends and family nearby may be an important driver behind the observed network effects.

4.4 Average Network Effects and Heterogeneity

To summarize the effects documented so far in a single object—and to assess how they vary—we next estimate a discrete choice model using a Poisson pseudo-maximum likelihood (PPML) specification. Rather than estimating separate regressions for each commuting zone, the PPML approach models graduates' location choices over the full set of potential destinations within a single choice framework. This parsimonious characterization will also serve as a direct input into the spatial equilibrium model we develop in Section 5.

We implement this PPML specification using a stacked version of the directed-migration design described in equation 5. As is common in discrete choice settings, we require that all individuals face the same choice set and choose exactly one option. For computational feasibility, we therefore restrict the sample to graduates who live in one of the 50 largest CZs upon graduation.[46] For each of these graduates, the data contain one observation for each CZ in the choice set. We adapt the same fixed-effect structure as in the baseline OLS specifications to this stacked choice setting by interacting those fixed effects with potential-destination fixed effects and adding individual fixed effects. As in the preceding analyses, the main regressors of interest are changes in the share of a graduate's friends residing in a given destination at different event times.

Panel (a) of Figure 7 presents the resulting estimates. Again, the coefficients exhibit a relatively stable pre-period, a sharp decline around graduation, and a substantially lower

[46]Unlike in the preceding directed migration analyses, here, we do not exclude individuals who went to high school or college in a given potential destination CZ, as our objective is to summarize network effects broadly, and to capture effects on both out-migration and directed migration since our results so far have shown that network effects exist for both of these margins.

post-period plateau. Summarizing this pattern by the difference between average pre- and post-period coefficients yields an estimate of 4.0 (SE = 0.10), meaning that a 10-percentage-point increase in the share of one's friends in a given CZ raises the relative likelihood of choosing that CZ upon graduation by 0.4 log points, on average.[47]

We next examine how network effects vary across different groups of graduates. To do so, we again draw on the PPML framework which allows us to effectively take underlying differences in migration rates into account. Panel (b) of Figure 7 shows heterogeneity by gender, by selectivity of one's college (proxied by the average SAT score of the student body), and by socioeconomic background (proxied by parental SES).[48] Across these dimensions, we find relatively little heterogeneity. Network effects are highly significant and similar in magnitude across groups, and most differences in point estimates are statistically indistinguishable from zero. Thus, meaningful network effects appear to exist across different groups of graduates and are not concentrated among specific subpopulations.

Finally, we ask whether network effects differ across destinations with stronger versus weaker economic conditions. Specifically, we estimate PPML specifications that interact friends' moves with indicators for whether the destination has above-median pay, or above-median predicted pay growth. Appendix Figure A9 reports the resulting estimates. In both cases, the main network effect remains large and highly significant, while the interaction term is quite small in comparison. Thus, we find little evidence that social ties matter only for economically stronger destinations. Rather, network effects appear similar for destinations with higher and lower pay, consistent with the destination-specific estimates showing meaningful effects even for lower-wage places such as Detroit.

4.5 External Validity

To assess whether the network effects we identify extend beyond college graduates, we next adapt our estimation strategy to a broader population. While this modification allows us to estimate network effects for a more representative set of individuals, it comes at the cost of a stronger identifying assumption. We therefore view these results as suggestive.

[47] The causal interpretation of this contrast requires the identifying assumption in Section 3 to hold on the scale of the PPML index. In particular, the effect of the share of a graduate's friends living in a destination must be additive on that index, and unobserved determinants of destination choice must be similarly related to pre- and post-period friends' moves on the same scale.

[48] We proxy for college selectivity by the average SAT score of the student body as of 2013 using data from Chetty et al. (2020). We proxy for socioeconomic background using parents' predicted SES, see Section 2 for details. The corresponding analysis is conducted on the 70% of graduates in our sample for whom we are able to observe parental SES.

At a high level, we again compare observably similar individuals who primarily differ in when their friends relocate. Intuitively, we compare two individuals who lived in the same CZ in Q1 of 2012 and also attended high school—and if they went to college, also college—in the same CZ. We then ask how their locations in late 2017 differ given that, again, the first person had a friend relocate before that date, while the second person's friend relocated only afterwards. We implement this thought experiment by returning to our *General Sample* and amending equations 3 and 5, so that *early* now corresponds to Q1 of 2012 and t^* is the last quarter of 2017.[49] X_{i-j,t^*} and X_{ik,t^*} are fully interacted fixed effects for one's high school CZ, college CZ, birth cohort, the CZ one lives in as of Q1 of 2012, and quintiles of the share of friends outside j (for out-migration), or in k (for directed migration) at the same point in time.[50] As before, our object of interest is the difference between coefficient estimates for friends' moves prior to versus after t^*.

For the difference between pre- and post-period estimates to recover causal network effects, we rely on the same identifying logic as in the college graduation setting, but it now implies a somewhat stronger condition. Identification again requires that friends' moves before and after t^* are equally informative about any unobserved factors that may influence both individuals' and friends' location choices. For college graduates, this was a relatively weak assumption. The design centered on a clearly defined transition point, meaning that graduates were typically still in college during the pre-period and therefore unlikely to have already made their post-college location choice. Friends' moves in the pre-period thus generally preceded graduates' post-college relocations, whereas friends' post-period moves occurred afterward, giving the event study a relatively clear temporal ordering. As discussed in Section 4.2, this ordering, together with the patterns in Figure 3 and Panel (a) of Figure 4—elevated coefficients throughout the pre-period, a steep decline around graduation, and consistently lower estimates in the post-period—leaves relatively limited scope for unobserved factors to drive our estimates. For unobserved factors to explain this pattern, friends moving at any point before graduation, including those who move several years earlier, would need to be systematically more similar to the graduate in unobserved location preferences, or more exposed to shocks that also affect the graduate than friends who move after graduation, even when the latter move relatively soon after.

In contrast, in the more general setting, individuals are free to relocate at any point between *early* and t^*. As a result, an individual's location measured at t^* may reflect moves made well

[49] We choose this point in time because Q4 of 2017 corresponds to the first point at which we measure location outcomes in the college graduation research design.

[50] For individuals who do not list a college, we create one residual college CZ in order for these individuals not to be omitted from the regressions.

before that date, i.e., in the pre-period, potentially in response to the same shocks that also affected their friends moving at that time, so the relative timing between friends' moves and individuals' location choices is less cleanly ordered for the pre-period. The assumption that friends' moves before and after t^* are equally informative about unobserved determinants of location choices therefore now implies that such unobservables must be relatively stable over time. This is a stronger requirement.

With this stronger condition in mind, Figure A10 presents the estimation results for this more general setting.[51] The results closely mirror the patterns observed for college graduates. In the out-migration case in Panel (a), estimates are positive, highly significant, and relatively stable prior to t^*, then decline sharply for friends' moves around t^* and remain much lower thereafter. The resulting pre–post difference (0.63) is similar in magnitude to that observed for college graduates when each estimate is expressed relative to the respective baseline out-migration rate: the implied network effects are 3.0% and 2.7% of baseline out-migration in the general and college graduation samples, respectively. Panel (b) further shows that we also observe similar network effects for directed migration. When we plot the effects from this more general setting against those obtained for college graduates, the two line up relatively closely, with a correlation of 0.56 and a slope of 0.48, suggesting, if anything, stronger network effects for the more general sample. We interpret this comparison cautiously, however, since the exercise using the broader sample is less insulated from unobserved forces that jointly affect individuals and their friends and could therefore bias the estimates upward relative to the design focused on college graduates. Lastly, Panel (c) highlights that these general-sample estimates are not concentrated in a particular subgroup. In a PPML specification analogous to the one in Section 4.4, estimated on the broader sample, we find positive and relatively consistent network effects across individuals with and without a listed college, men and women, different college selectivity groups, parental SES quartiles, and birth cohorts. Under the stronger identifying assumption described above, these results suggest that substantial network effects also exist for this more general population. While we continue to view the college graduation design as our cleanest source of causal evidence, the broader-sample results suggest that similar forces operate outside that setting as well. We next ask whether these individual-level effects are large enough to help explain the broader patterns of residential choice that motivate this paper.

[51] For computational reasons, our estimation uses a 50% random sample.

5 Model

To quantify the aggregate role of social networks, we embed the quasi-experimental evidence above in a quantitative spatial model of residential choice. The purpose of this exercise is not simply to improve the fit of a Rosen-Roback model by adding another flexible, unobserved force, but to ask whether an observed, microfounded source of local attachment can reduce the burden often placed on unobserved amenities or very high moving costs. We proceed in two steps. First, holding the observed distribution of social networks fixed, we examine how much the estimated network effects from Section 4 can account for static patterns, i.e., why CZs differ in population size, and why even CZs with limited economic opportunities often have large populations. Second, we assess whether the same framework helps us understand heterogeneous responses to local shocks, namely why more- and less-educated individuals respond differently to changing economic conditions. Methodologically, in the first step, we extend a standard Rosen–Roback spatial choice framework to incorporate social ties in a static manner; in the second step, we add standard production and housing-market blocks and simulate local productivity shocks to study heterogeneous migration responses. For the purpose of this section, we draw on the Model Sample described in Section 2 and focus again on location choices among the 50 largest CZs.

The model structure builds on Zabek (2024), who incorporates birthplace-based local attachment into a spatial equilibrium model. In our setting, we instead model local *social* attachment: we measure attachment using the observed locations of each individual's friends, and discipline its strength using the quasi-experimental evidence from Section 4.

5.1 Model Setup

We posit that individual i's utility associated with living in CZ j is given by:

$$U_{ij} = \alpha_w \ln(w_j) - \alpha_h \ln(r_j) + A_j + \tilde{\beta} n_{ij} + \epsilon_{ij} \tag{6}$$

where $\ln(w_j)$ and $\ln(r_j)$ denote the natural logarithm of local wages and rents, A_j captures local amenities,[52] and ϵ_{ij} represents idiosyncratic preferences, assumed to follow a Type I extreme value distribution, yielding a standard logit structure (McFadden, 1974). The main innovation is the inclusion of $\tilde{\beta} n_{ij}$.[53] As in Sections 3 and 4, n_{ij} denotes the share of i's friends residing in CZ j. Note that n_{ij} is an endogenous object because people choose their friends

[52] We assume that everyone living in j has equal access to these amenities and values them equally.

[53] This simple additive specification is consistent with the reduced-form evidence in Section 4.4, where we find little evidence that network effects differ systematically between higher- and lower-pay destinations.

and because friends' location choices are themselves endogenous. Endogenizing the network would require modeling both friendship formation and the joint location choices of connected individuals. We leave this extension to future research. For the exercises presented here, we instead hold the observed network distribution fixed and ask how much the estimated network effects can account for empirically observed patterns of residential choice, taking people's networks as given.

The key parameter in equation 6 is $\tilde{\beta}$, which captures the utility value of living near one's social ties. Importantly, $\tilde{\beta}$ is a structural parameter and therefore distinct from the reduced-form parameters estimated in Section 4.[54] However, under the logit assumption imposed on ϵ_{ij}, the coefficient on n_{ij} from a multinomial logit—or PPML—specification on observational data coincides with $\tilde{\beta}$, provided that this coefficient is causally identified. Across various parts of Section 4, we presented PPML estimates of our quasi-experimental design for college graduates and provided evidence supporting their causal interpretation; we further showed that, under somewhat stronger assumptions, the effects identified for college graduates likely generalize to a broader population. We therefore now leverage this well-identified micro-estimate to calibrate $\tilde{\beta} = 4.0$, our main PPML estimate from Section 4.4.[55]

To find the remaining model parameters, we use an iterative procedure that targets two objectives. First, we require the model to generate an average migration elasticity with respect to wages of one, which falls in the middle of the range of empirical estimates (e.g., Beaudry et al., 2014; Wozniak, 2010; Hornbeck and Moretti, 2024). Second, we require that the model-implied population sizes match those observed for our sample. Given these targets, we then solve iteratively for α_w and the vector of A_j terms using the observed distribution of social networks together with data on skill-adjusted wages and rents.[56] We

[54] We use tildes to denote the conceptual difference between $\tilde{\beta}$ and β_{-j} and β_k.

[55] As an alternative, one could calibrate $\tilde{\beta}$ using a simulated method of moments (SMM) approach. We instead use the PPML estimate from Section 4.4 because it disciplines the network parameter using quasi-experimental variation. By contrast, an SMM approach would choose $\tilde{\beta}$ based on the model's ability to match observed location choices, where the locations of one's social network are correlated with many other forces that also affect residential choice. This is precisely the concern that motivates our quasi-experimental design. In addition, the scale of the Facebook data would make a full SMM exercise computationally costly. The PPML specification includes destination fixed effects, which absorb average destination-level forces corresponding to wages, rents, and amenities in the model, as well as additional fixed effects, such as cohort and home CZ controls, that do not have a direct counterpart in the model. To the extent these additional controls reflect systematic differences in destination attractiveness for particular groups, our model with a single destination-level residual A_j can capture only their average contribution to destination population shares, not the underlying heterogeneity itself. We therefore use the PPML estimate to discipline the strength of the network term, while recognizing that the mapping between the reduced-form specification and the model is not exact.

[56] Both measures are constructed using 2012 ACS data obtained through the IPUMS National Historical Geographic Information System (Manson et al., 2023). We construct skill-adjusted wages by first computing national shares across five education groups: less than high school, high school, some college or associate's

calibrate $\alpha_h = \alpha_w/3$, implying that individuals spend one-third of their income on housing, consistent with the housing share in the Consumer Expenditure Survey.[57] The resulting main parameters are shown in Appendix Table A8.

The baseline calibration implies $\widetilde{\beta}/\alpha_w = 2.12$, so that a one-percentage-point increase in the share of an individual's friends in a CZ has the same effect on utility as approximately a 2.1% increase in wages, or roughly an $800 increase in annual wages evaluated at the average wage. This is somewhat smaller than, but of the same order of magnitude as, the wage-equivalent comparison based on our OLS estimates in Section 4.1, suggesting that our baseline parameterization is conservative about the relevance of social networks. As an untargeted check on the calibration, we compare the observed share of each individual's friends in each CZ to the corresponding share implied by the model's predicted location choices. Although this moment is not targeted, we observe a correlation of 0.94 between the two series, providing reassurance that the calibrated network effect is broadly consistent with the observed relationship between where people live and where their friends are located. Appendix Figure A11 further shows that the results discussed below are robust to varying the wage-equivalent value of social networks around our baseline calibration.

Importantly, the procedure used to calibrate α_w, α_h, and the A_j terms implies that the latter are, in practice, residual location-specific components required to reconcile the model with the stated objectives, rather than direct measures of "true" amenities. This feature is not unique to our setting, but holds more generally for spatial choice models that solve for amenities in this way; this insight will become relevant when we assess the model's ability to explain observed location choices.

To benchmark the performance of our "network model", we compare it to a parsimonious "basic" model which does not include social ties, so that an individual's utility function only depends on local wages, rents, amenities, and idiosyncratic preferences.[58] We intentionally compare the network model to such a basic model to highlight how much the addition of a single factor, social networks—for which we have provided well-identified reduced-form

degree, bachelor's degree, and graduate or professional degree. We then combine these national shares with county-level median earnings by education group, yielding a county-level wage measure that holds fixed the national distribution of education groups. For rents, we use median gross rent, annualized by multiplying monthly rent by 12. We aggregate the county-level wage and rent measures to CZs using population weights.

[57] Housing accounts for 33.4% of total annual expenditures in the 2024 Consumer Expenditure Survey (U.S. Bureau of Labor Statistics, 2025).

[58] In the basic model, individual i's utility function is given by:

$$U_{ij}^{Basic} = \alpha_w^{Basic} \ln(w_j) - \alpha_h^{Basic} \ln(r_j) + A_j^{Basic} + \epsilon_{ij}^{Basic} \tag{7}$$

The superscript *Basic* indicates that the parameters can differ relative to those shown in equation 6.

evidence—can help to understand location choices without the need to incorporate other, potentially less well-identified forces. We also calibrate the basic model to match local population sizes and have an average wage elasticity of one using an analogous iterative approach. The corresponding parameter values can again be found in Table A8.

5.2 Model Results

Explaining Differences in Population Size Across CZs

Incorporating social networks greatly helps explain differences in population size across CZs. To illustrate this, in Panel (a) of Figure 8 we compare the models' respective vectors of local amenities—A_j in the network model and A_j^{Basic} in the basic model—which, as discussed above, correspond to residuals. We scale these residuals so that they can be interpreted as log-wage-equivalent willingness to pay.[59] In the basic model, the distribution of A_j^{Basic} ranges from around -0.4 to about $+0.8$. At the extremes, this corresponds to an amenity difference equivalent to nearly 1.2 log points of annual wages, reflecting a very large residual. In contrast, the distribution of A_j is much tighter, with a standard deviation less than 40% of that of A_j^{Basic}. To benchmark the size of this reduction, we also ask how large a conventional moving-cost term would have to be to generate the same reduction in residual amenity dispersion. We augment the basic model with a uniform cost of choosing a CZ other than one's initial CZ and recalibrate the model to match the same population and wage-elasticity targets. We find that, in order to reduce the standard deviation of residual amenities by as much as observed social networks do, the model requires a very sizable moving cost of 1.48 log-wage points.[60]

The large reduction in the dispersion of the residuals in the network model relative to the basic model comes from two forces. First, Section 4 showed that network effects are large: even modest differences in where one's friends live meaningfully affect residential choices. Second, Section 2 highlighted that networks tend to be highly spatially concentrated, with the average individual in our Model Sample having around 63% of their friends in the CZ where they currently reside. Together, these forces make the network model much less reliant on very large or very small residual amenity values to explain differences in population sizes

[59] For each model, we convert the residual amenity term into wage-equivalent units by dividing by that model's estimated wage coefficient, α_w, and demeaning the resulting values separately within the network and basic models.

[60] Specifically, we augment the basic model with a uniform utility cost μ of choosing a CZ other than the individual's initial CZ, and choose μ so that the standard deviation of the scaled residual amenity vector, $A_j^{\text{move-cost}}/\alpha_w^{\text{move-cost}}$, falls by the same amount as when adding observed social networks. The required value is $\mu/\alpha_w^{\text{move-cost}} = 1.48$.

given wages and rents.

Not only do residuals in the network model matter much less, they also align more closely with their interpretation as amenities. In Panel (b) of Figure 8, we correlate the A_j and A_j^{Basic} terms with place characteristics commonly considered local amenities: air quality, weather and retail availability (e.g., Chay and Greenstone, 2005; Rappaport, 2007; Glaeser et al., 2001). While the A_j^{Basic} terms only exhibit correlations between -0.11 and 0.24, we find much larger correlations for the network model, between 0.21 and 0.51. This result reflects the fact that the degree to which people's networks are locally concentrated is negatively correlated with local economic conditions.[61] For instance, residents of Detroit have, on average, 72% of their friends locally, whereas residents of Austin and San Diego have only 49% and 53% of their friends in the same CZ. Incorporating this observation reduces the demeaned residual amenity value assigned to Detroit, and increases it for Austin and San Diego, which in turn raises the correlations in Panel (b) of Figure 8. Thus, accounting for people's social networks and their variation leaves residuals (A_j) that align more closely with their interpretation as local amenities.

Conceptually, the results from Panel (b) of Figure 8 imply that social networks are not merely a different label for moving costs or preferences to stay—factors considered in prior work (e.g., Kennan and Walker, 2011; Diamond, 2016; Piyapromdee, 2021). While adding a sufficiently large moving-cost term to the basic model can—as shown above—produce reductions in the dispersion of the residual amenity terms similar to those found for the network model in Panel (a), such a term would not yield a similarly strong increase in the correlations with observable amenities shown in Panel (b) without additional assumptions about how moving costs vary across places. Thus, while social networks may to some extent generate patterns similar to moving costs or preferences to stay, they are distinct because they vary across places in ways that allow the model to account for observed differences in CZ population sizes in a much more direct manner. Moreover, as we highlight below, variation in social networks also helps explain why migration responses to local shocks differ across individuals.

Why More-Educated People Respond More Strongly to Local Shocks

A large literature has highlighted that migration responses to changing local economic conditions vary substantially across socioeconomic groups (e.g., Bound and Holzer, 2000; Gregg et al., 2004; Wozniak, 2010; Sprung-Keyser et al., 2022). In particular, more-educated in-

[61] The correlation between the average share of residents' friends who live in the same CZ and CZ-level rents is -0.61 when controlling for 2012 population size; for wages the same correlation is -0.20.

dividuals are more likely to move across local labor markets and respond more strongly to local economic shocks. We now show that differences in the geographic dispersion of social networks provide a natural explanation for part of this pattern.

The intuition is straightforward. If social ties enter residential choice, then the response to a local shock depends not only on the economic consequences of the shock, but also on where one's friends live. Individuals whose networks are concentrated in their current CZ lose more social ties by leaving, while individuals with more friends elsewhere have both less to give up locally and more social ties in potential destinations. Appendix Figure A12 shows that these differences in network dispersion are strongly related to education. Following prior work using the same Facebook data (e.g., Bailey et al., 2024), we measure educational background using whether an individual lists a college and, among those who do, the average SAT score of the student body at that college. For brevity, we refer to the resulting patterns as differences by education or between more- and less-educated individuals. Using this measure, more-educated individuals have substantially larger shares of friends outside the CZ they live in, whereas less-educated individuals have more geographically concentrated networks: individuals without listed college information have only 33% of their friends outside the CZ they live in, compared to nearly 55% among those listing the most selective colleges.

To study whether these differences can translate into different migration responses, we extend the static model above by adding standard local production and housing-market blocks, described in Appendix G. On the production side, local productivity determines wages through a standard local labor-demand system, so an adverse productivity shock lowers local wages and induces population responses. On the housing side, rents adjust with local population through an upward-sloping housing supply curve. Together, these additional blocks allow local shocks to affect wages, rents, and population shares in equilibrium, while leaving the residential-choice component of the model unchanged.

We simulate adverse local productivity shocks by shocking each CZ one at a time. For each CZ, we choose the size of the productivity decline so that, absent any migration response, wages in the shocked CZ would fall by 15%. This normalization puts shocks in comparable wage units across CZs and corresponds to a sizable local downturn; the qualitative comparisons across individuals are similar when we use smaller or larger shocks. For simplicity, and to avoid introducing additional assumptions about network formation, we continue to hold the observed network distribution fixed throughout the simulation. We therefore interpret these simulations as fixed-network equilibrium responses, taking the observed geography of social ties as given. For any given shocked CZ, individuals face the same wages, rents,

amenities, and productivity shock, and we impose common preference parameters across individuals. Thus, within a shocked CZ, differences in predicted out-migration responses arise only from differences in individuals' observed networks. For each individual initially living in the shocked CZ, we compute the model-implied percentage decline in the probability of living in that CZ relative to the baseline probability, which we interpret as the predicted out-migration response to the shock.

Figure 9 reports the results. To focus on differences among individuals facing the same shocked CZ, the figure residualizes both axes with respect to shocked-CZ fixed effects and recenters them at their sample means. Panel (a) shows a strong positive relationship between network dispersion and predicted out-migration responses. Individuals with few friends outside their current CZ are predicted to respond only weakly to the shock, while those with many friends elsewhere are much more likely to leave. This relationship illustrates the central mechanism: when networks are more dispersed, leaving the current CZ entails a smaller loss of local social ties and provides access to more ties elsewhere.

Panel (b) then shows that this mechanism maps closely into differences by education. Among individuals who list a college, predicted out-migration responses rise strongly with the average SAT score of that college. The predicted response for individuals at the most selective colleges is nearly 70% higher than the average response among individuals without listed college information. Since the simulation imposes the same structural parameters across groups, this gradient reflects differences in the geography of social networks rather than education-specific differences in preferences or wage sensitivity.

Together, these results show that social networks can generate substantially larger responses to local shocks among more-educated individuals without imposing education-specific preferences, moving costs, or wage sensitivities. We view this mechanism as complementary to other explanations: the geography of social networks provides one concrete source of the heterogeneous attachment to places that prior work has often captured through moving costs or preferences to stay. This perspective also changes how one should interpret low migration rates. Low mobility need not simply reflect a barrier to moving; it may also reflect the value individuals place on remaining near their social ties.

6 Conclusion

Despite large regional differences in economic opportunity, migration is limited, and even those who move often do not go to places offering greater opportunities. These patterns are especially pronounced among less-educated individuals. This paper shows that social

networks are a first-order force in explaining these observations.

Using individual-level data from Facebook on the social networks and locations of millions of users in the United States, we document that social networks are highly spatially concentrated. We then present quasi-experimental evidence showing that friends' locations shape residential choice: they affect both whether individuals remain in their home region and where they move.

By incorporating our estimates of network effects and the observed geography of individual networks into a quantitative spatial model, we show that the estimated per-friend effects translate into a large aggregate force on residential choice. This force helps explain why many individuals remain in regions with limited economic opportunities, even when other places offer better economic prospects. Because less-educated individuals tend to have more geographically concentrated networks, the same mechanism also helps explain why they respond less strongly to adverse local economic shocks.

Overall, our findings highlight the importance of social networks in shaping access to economic opportunity. By influencing migration decisions, social ties affect which places—and therefore which opportunities—people can access. Locally concentrated ties can make leaving economically distressed areas less attractive, while ties elsewhere can provide pathways to new destinations. Social networks can thus both facilitate and inhibit mobility toward better prospects. Their role in residential choice also changes how low migration rates should be interpreted: they may arise not only from barriers to moving, but also from the benefits of living close to friends and family.

A natural direction for future research is to study how policymakers can account for—and potentially leverage—these ties when responding to adverse local economic shocks and seeking to broaden access to economic opportunity. On the one hand, our findings suggest that the case for place-based interventions may be stronger when moving entails giving up valuable social ties. At the same time, because social ties are not inherently place-based, policies that help individuals relocate together with their ties may make moving more attractive and therefore generate larger mobility responses than policies targeting individuals in isolation.

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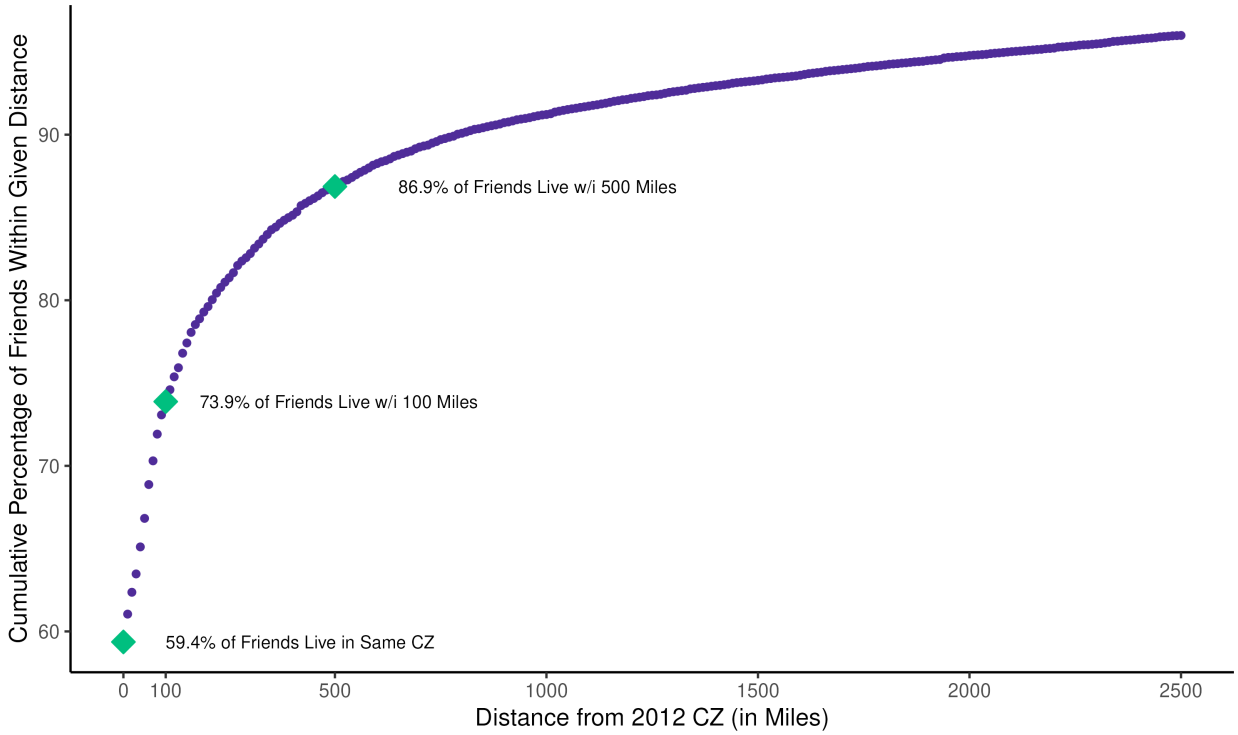
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Figures

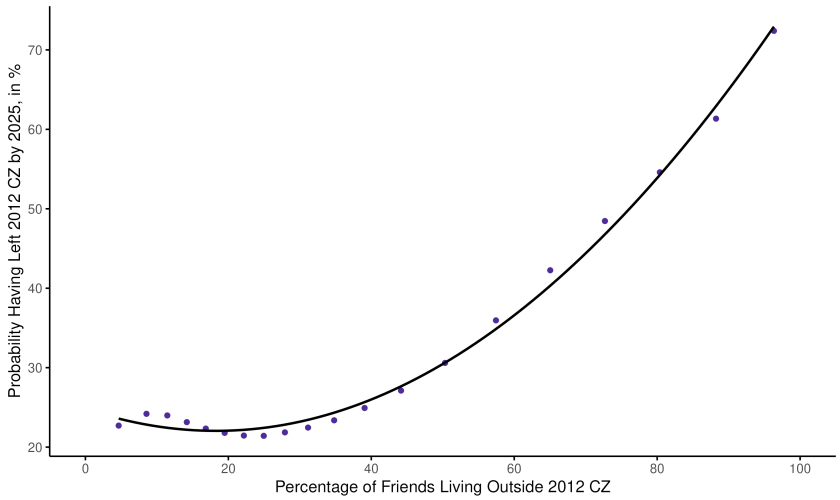
Figure 1: Spatial Concentration of Social Networks



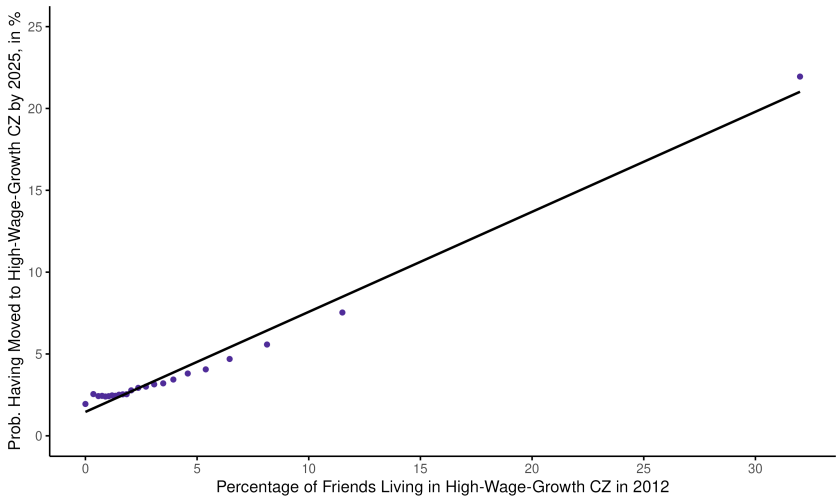
Notes: This figure shows the cumulative distribution of friends by distance (in miles) between an individual's CZ of residence in Q1 of 2012 and the CZs of their friends. For each individual, we compute the cumulative percentage of friends located within the indicated distance at 10-mile intervals and then average this CDF across individuals. The point at zero reports the percentage of friends living in the individual's own CZ. The analysis draws on the General Sample described in Section 2 and restricts to friendships formed prior to 2012; friends' locations are measured in Q1 of 2012. Distances are calculated using CZ centroids.

Figure 2: Migration Behavior and Location of Social Networks

(a) Probability of Leaving 2012 CZ

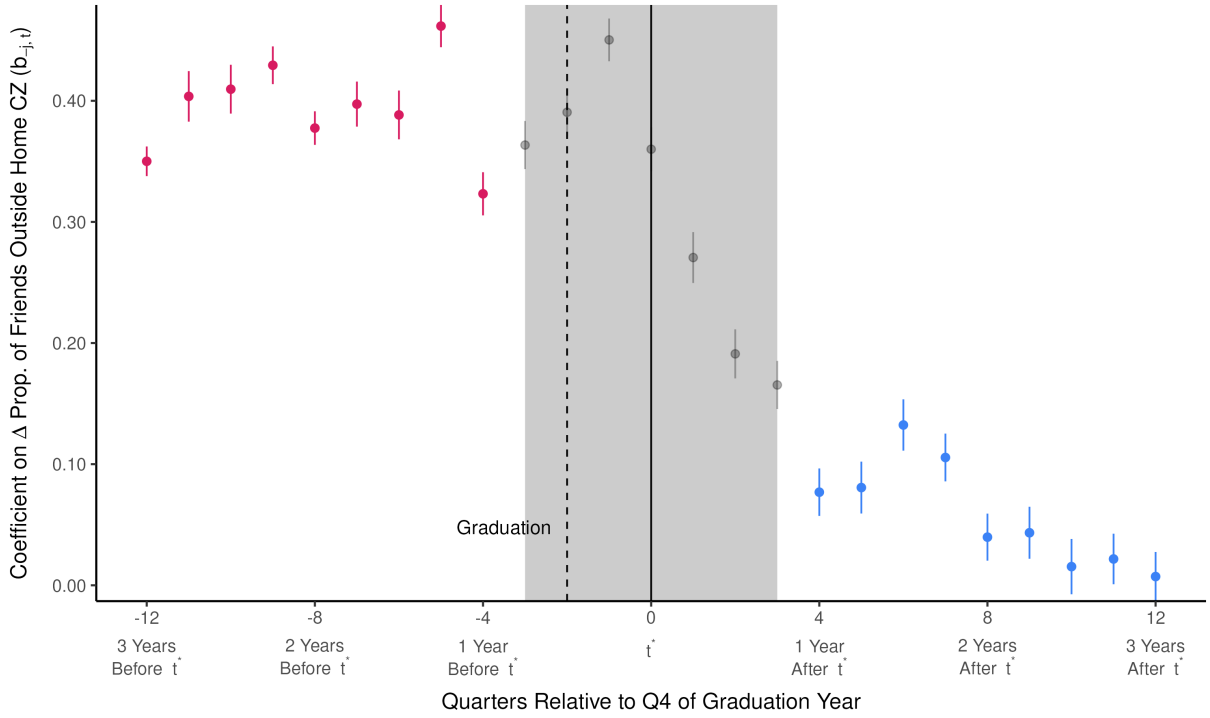


(b) Probability of Moving to CZ with High Wage Growth



Notes: This figure presents descriptive evidence on the relationship between social networks and migration in the General Sample described in Section 2. In both panels, friends' locations are measured in Q1 of 2012, and the analyses restrict to friendships formed prior to 2012. Panel (a) shows a binned scatter plot where the vertical axis corresponds to the probability (in %) that an individual lives in a different CZ in Q4 of 2025 than in Q1 of 2012. The horizontal axis displays the percentage of friends living outside the individual's 2012 CZ. The black line represents the best fit from a quadratic regression on the binned series. Panel (b) shows a binned scatter plot where the vertical axis corresponds to the probability (in %) that an individual resides in a high-wage-growth CZ in Q4 of 2025, and the horizontal axis displays the percentage of their friends living in such CZs in Q1 of 2012. High-wage-growth CZs are defined as the ten CZs among the 50 largest CZs (shown in Appendix Figure A2) with the highest predicted wage growth between 2012 and 2024, based on a shift-share approach using local industry composition; see footnote 13 for details. The analysis is restricted to individuals who did not live in a high-wage-growth CZ in Q1 of 2012. The black line represents the best fit from a linear regression on the binned series.

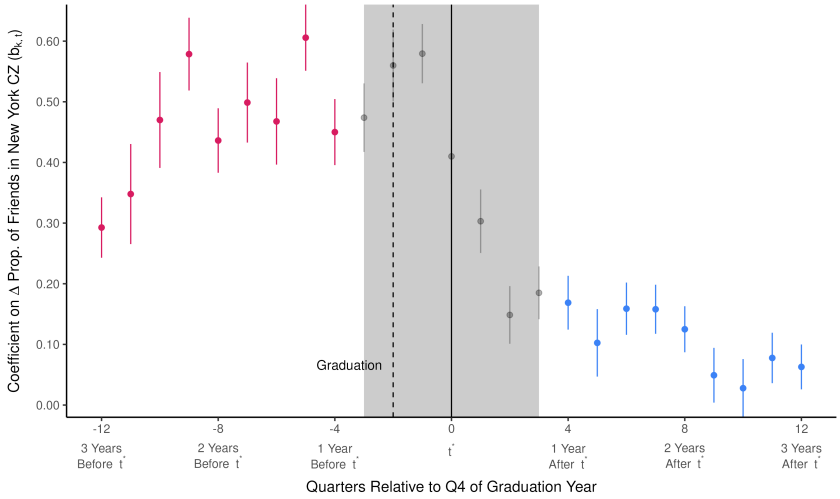
Figure 3: Network Effects on Out-Migration



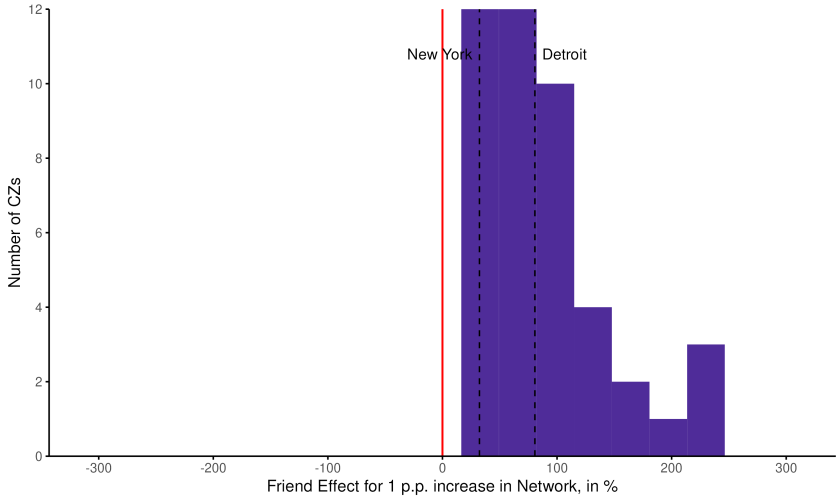
Notes: This figure presents coefficient estimates from equation 3 based on the College Graduation Sample described in Section 2. The horizontal axis is measured in quarters relative to t^*, defined as Q4 of the year in which graduates are expected to graduate, when their locations are measured. Each plotted estimate is a $b_{-j,t}$ coefficient corresponding to changes in the share of a graduate's friends living outside their home CZ at event-time t. A graduate's home CZ is defined as the union of the CZs in which they attended high school or college. The outcome is an indicator for whether, at t^*, the graduate lives outside their home CZ. The regression includes fixed effects for the full interaction of high school CZ, college CZ, graduation cohort, and quintile of the initial share of friends living outside the home CZ, with the quintiles defined within high-school-CZ-by-college-CZ-by-graduation-cohort cells. Red coefficient estimates correspond to friends' moves occurring between twelve and four quarters prior to t^*. Blue coefficient estimates correspond to friends' moves occurring between four and twelve quarters after t^*. Gray coefficient estimates correspond to friends' moves occurring during $\bar{t}$, i.e., the window between three quarters prior to t^* and three quarters after t^*. During this window, graduates are likely making location decisions or may anticipate friends' moves occurring shortly after graduation. Vertical lines denote 95% confidence intervals. The dashed gray line indicates the time of graduation, i.e., the second quarter of senior year.

Figure 4: Network Effects on Directed Migration

(a) Moving to New York



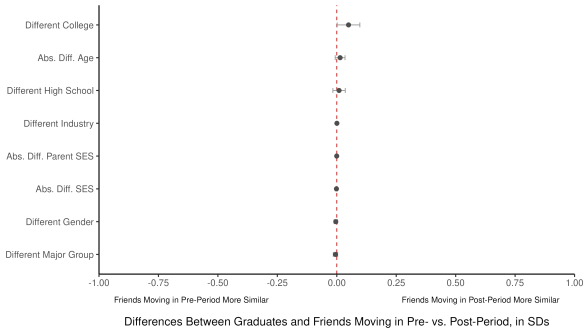
(b) Distribution of CZ-Specific Estimates



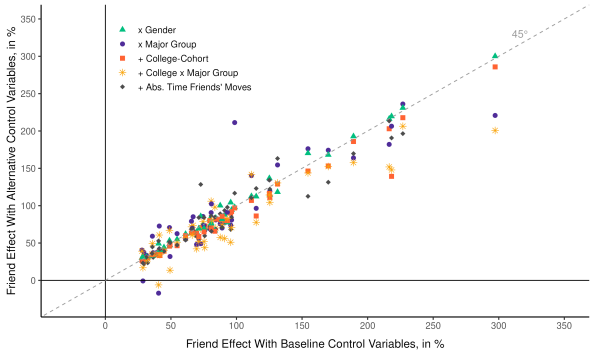
Notes: This figure presents coefficient estimates based on equation 5, drawing on the College Graduation Sample described in Section 2. Panel (a) reports event-time coefficients for New York, restricting the analysis to graduates who did not attend high school or college in the New York CZ. Each plotted estimate is a $b_{k,t}$ coefficient corresponding to changes in the share of a graduate's friends living in New York at event-time t. The outcome is an indicator for whether a graduate lives in New York at t^*. The figure uses the same visual convention as Figure 3. Panel (b) shows a histogram of destination-specific network effects for the 50 largest CZs, where each effect is given by the difference between the average pre- and post-period estimates from equation 5. The pre-period comprises quarters twelve to four before t^*, and the post-period comprises quarters four to twelve after t^*. Each effect is scaled to represent the percentage change in the baseline migration rate to the corresponding destination associated with a one-percentage-point increase in the share of friends living there. For each destination-specific estimate, the analysis is restricted to graduates who did not attend high school or college in that CZ. The regressions include fixed effects for the full interaction of high school CZ, college CZ, graduation cohort, and quintile of the initial share of friends living in the destination CZ, with the quintiles defined within high-school-CZ-by-college-CZ-by-graduation-cohort cells. Estimates are shrunk using the empirical Bayes procedure described in Appendix D. Appendix Figure A6 shows the distribution of unshrunk estimates. The solid red line marks zero, and the dashed vertical lines mark the estimates for New York and Detroit, respectively.

Figure 5: Evidence Supporting the Identifying Assumption

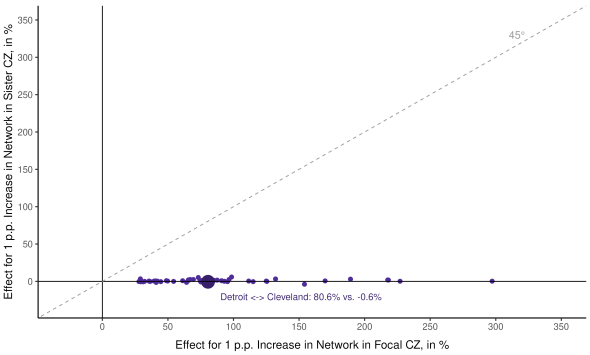
(a) Differences in Friend Characteristics



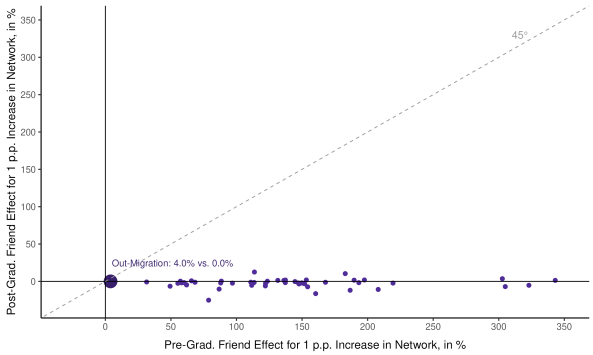
(b) Robustness to Pipeline-Related Controls



(c) Sister CZ Placebo



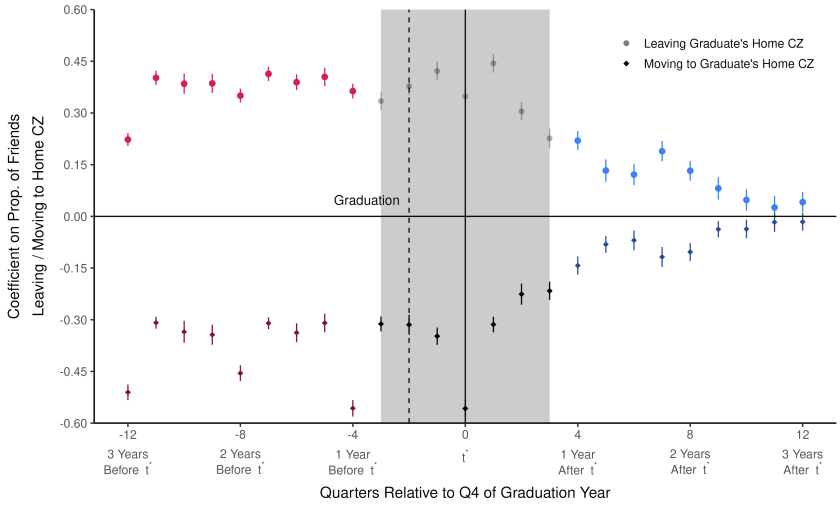
(d) Post-Graduation-Friend Placebo



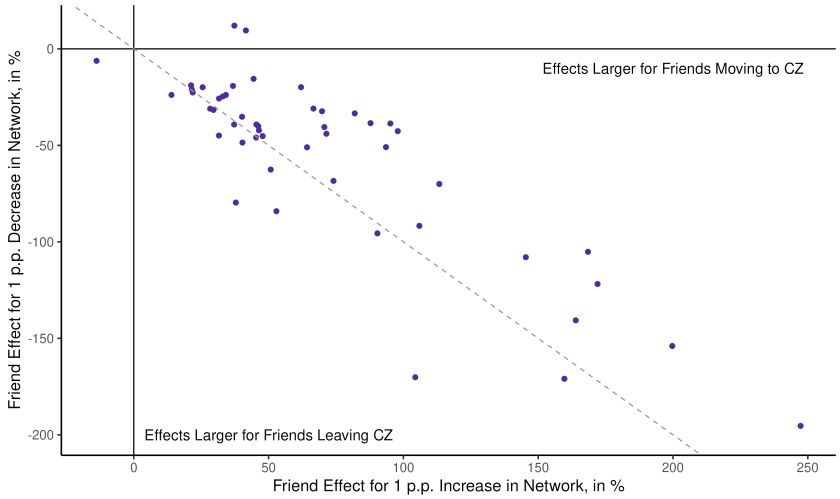
Notes: This figure presents evidence supporting the identifying assumption. Panel (a) examines whether friends who move in the pre-period are observably more similar to the graduate than friends who move in the post-period. For each observable, we construct a standardized measure of graduate–friend dissimilarity, compute its average by quarter of move, and regress these quarter-specific averages on an indicator for whether the quarter belongs to the pre- rather than the post-period. Pre-period quarters are those between twelve and four quarters before t^*; post-period quarters are those between four and twelve quarters after t^*. t^* is defined as Q4 of the year in which graduates are expected to graduate. Quarters corresponding to the mid-period are excluded from the analysis. The points report the resulting coefficient estimates. Negative values indicate that friends moving in the pre-period are more similar to the graduate; positive values indicate that friends moving in the post-period are more similar. Vertical lines denote 95% confidence intervals. Panel (b) reports robustness of the destination-specific network effects to alternative control variables. The horizontal axis shows the baseline destination-specific estimates, and the vertical axis shows the corresponding estimates obtained when modifying the control vector. The five alternative specifications correspond to additionally interacting the baseline fixed effects with gender or major-group indicators, adding college-by-cohort or college-by-major-group fixed effects, and controlling for friends' moves in absolute calendar time. Major groups classify majors into five groups based on the occupational distributions of recent graduates; see Appendix B for details. Panels (c) and (d) report results from two placebo exercises. Panel (c) plots, for each focal CZ, the effects of friends' moves to the focal CZ (horizontal axis) against those of friends' moves to sister CZs (vertical axis) on a graduate's likelihood of living in the focal CZ at t^*. Sister CZs are CZs with industry compositions very similar to those of the focal CZs; see Appendix E for more details. The larger, darker circle indicates the estimates for Detroit and its sister CZ, Cleveland. Panel (d) plots effects for pre-graduation friends (horizontal axis) against those for post-graduation friends (vertical axis). Pre-graduation friends follow the definition of friends in Figures 3 and 4; post-graduation friends are people whom graduates befriend at least four years after graduation and who did not overlap with the graduates in high school or college; see Appendix B for details. The larger, darker triangle indicates the estimate for out-migration. All estimates in Panels (b)–(d) correspond to the difference between the average pre- and post-period coefficients and are scaled to represent the percentage change in the relevant baseline migration rate associated with a one-percentage-point increase in the network measure. Destination-specific estimates use the same sample restriction and empirical Bayes shrinkage procedure as in Figure 4.

Figure 6: Mechanisms

(a) Friends Leaving vs. Moving to Home CZ



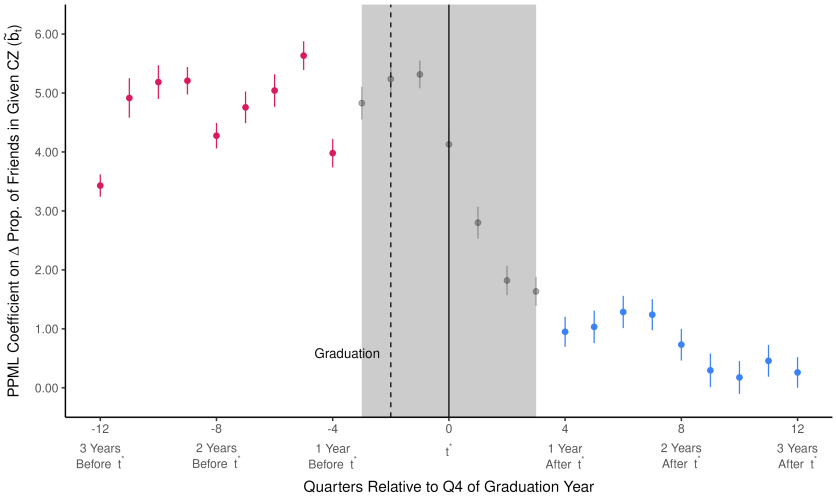
(b) Friends Moving to vs. Leaving Specific CZ



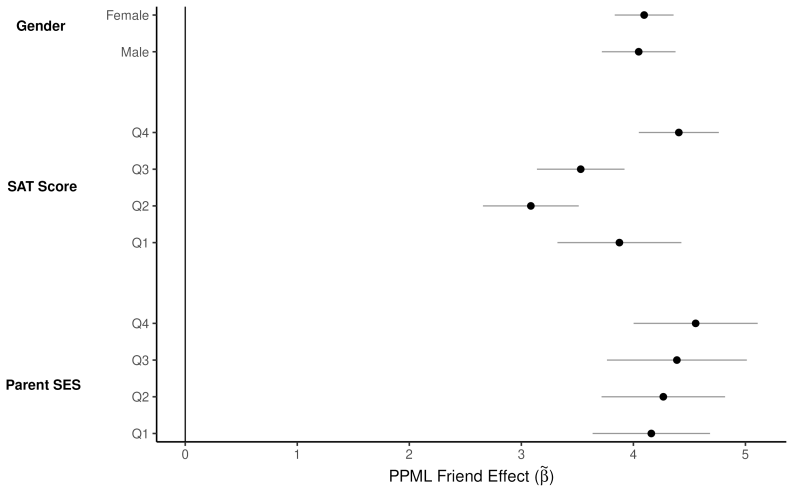
Notes: This figure presents evidence on the mechanisms underlying network effects. Panel (a) presents event-time coefficient estimates for out-migration when separately including friends who leave the graduate's home CZ and friends who move into it in the same regression. The definition of event-time and the visual coding of pre-, mid-, and post-period coefficients follow Figure 3. Effects for friends leaving the graduate's home CZ are shown with lighter shaded circles, whereas effects for friends moving into the home CZ are shown with darker shaded diamonds. Panel (b) reports destination-specific estimates, plotting effects for friends moving to a given CZ on the horizontal axis against effects for friends moving away from that CZ on the vertical axis. The two effects are estimated jointly for each destination. Each estimate is given by the difference between the average pre- and post-period coefficients and is scaled to represent the percentage change in the baseline migration rate to the corresponding destination associated with a one-percentage-point increase in the relevant network measure. Estimates use the same sample restriction and empirical Bayes shrinkage procedure as in Figure 4. The dashed line indicates effects of equal magnitude and opposite sign.

Figure 7: Average Network Effects and Heterogeneity

(a) PPML Estimates



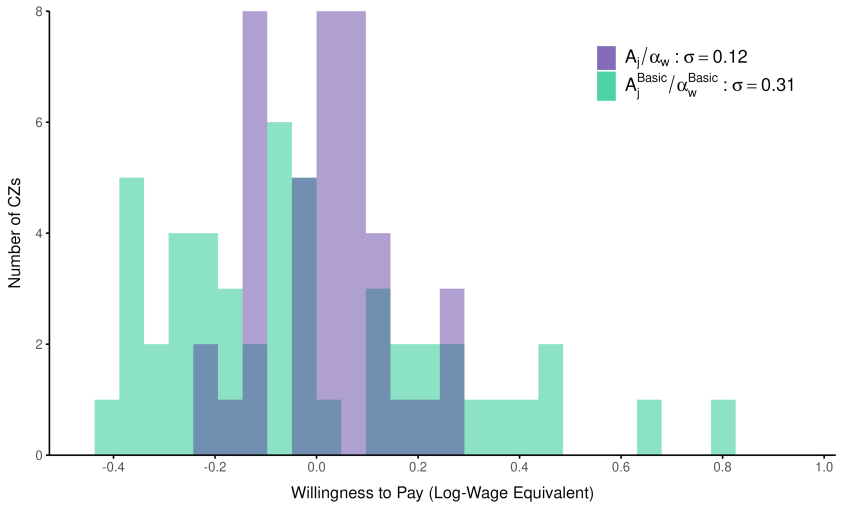
(b) Heterogeneity



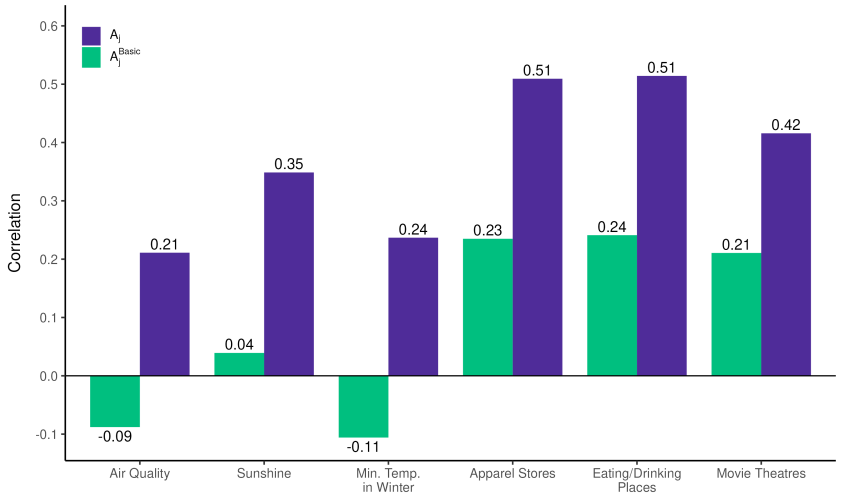
Notes: This figure presents PPML estimates of average network effects and their heterogeneity. Panel (a) reports event-time coefficient estimates from a stacked PPML specification that models graduates' location choices over the 50 largest CZs. The analysis is restricted to graduates who live in one of these CZs at t^*. The figure uses the same visual convention as Figure 3. Panel (b) reports subgroup-specific PPML estimates obtained from separate regressions for each subgroup, shown as the difference between the average pre- and post-period coefficients, with horizontal lines denoting 95% confidence intervals. The SAT score measure corresponds to the 2013 average SAT score of the student body at the college attended, using data from Chetty et al. (2020). Q1–Q4 denote quartiles of SAT scores or predicted parental SES; parental-SES quartiles are formed separately by graduation cohort. Estimates for parental SES are based on the subset of graduates for whom parental information can be linked; see Section 2 for details.

Figure 8: Local Amenities: A_j, A_j^{Basic}

(a) Willingness to Pay



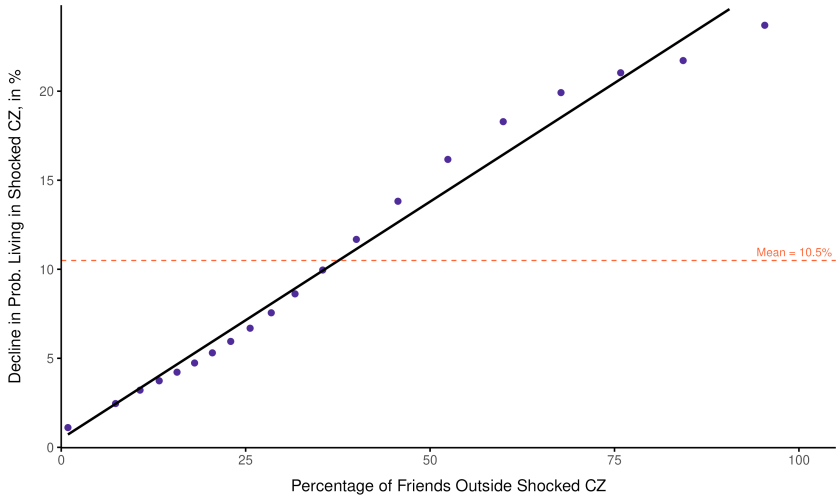
(b) Correlations: A_j, A_j^{Basic} vs. Place Characteristics



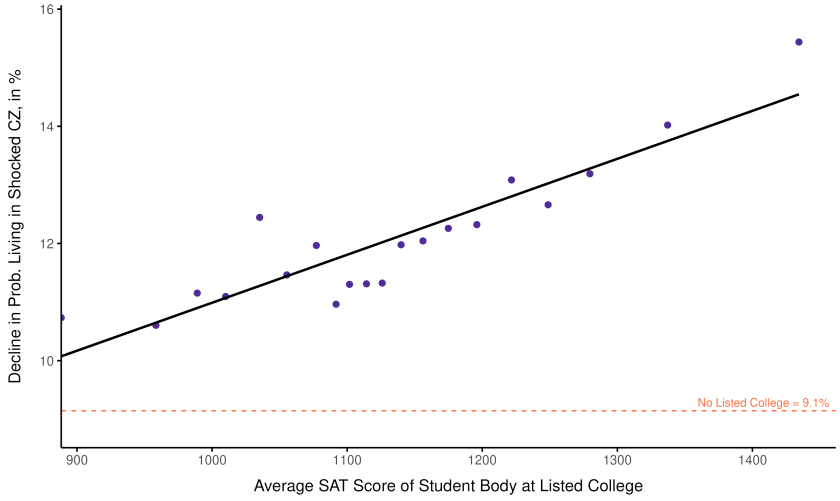
Notes: This figure compares the residual location-specific components required to match observed CZ population sizes in the network and basic models described in Section 5. We denote these components by A_j in the network model and A_j^{Basic} in the basic model. Both models are calibrated to generate an average migration elasticity with respect to wages of one. Panel (a) shows the distributions of A_j and A_j^{Basic} across the 50 CZs. For each model, the residuals are expressed as log-wage-equivalent willingness to pay by dividing by the corresponding model-specific wage coefficient, α_w, and demeaning the resulting values separately within the network and basic models. The legend reports the standard deviation of each distribution. Panel (b) reports population-weighted correlations between the residuals from each model and six place characteristics commonly considered local amenities. Air quality is defined as the negative of 24-hour $PM_{2.5}$ concentration, so that higher values correspond to better air quality, using data from the U.S. Environmental Protection Agency. Sunshine is average daily sunlight in kJ/m^2, and minimum winter temperature is average daily minimum air temperature in degrees Fahrenheit during meteorological winter (December through February), both measured over 1979–2011 using NASA NLDAS data obtained through CDC WONDER. Apparel stores, eating and drinking places, and movie theaters are measured as establishments per capita using 2023 County Business Patterns data from the U.S. Census Bureau. The corresponding NAICS codes are 448 for apparel stores, 7224 and 7225 for eating and drinking places, and 512131 for movie theaters. County-level measures are aggregated to CZs using population weights where applicable.

Figure 9: Heterogeneity in Migration Responses to Local Productivity Shocks

(a) Network Dispersion



(b) College Selectivity



Notes: This figure presents model-implied responses to adverse local productivity shocks using the Model Sample described in Section 2 and the network model described in Section 5. We shock each of the 50 CZs one at a time, choosing the productivity decline so that wages in the shocked CZ would fall by 15% absent any migration response. Wages, rents, and population shares are then allowed to adjust in equilibrium, while the observed distribution of social networks is held fixed. For each individual living in the shocked CZ in Q1 of 2012, the vertical axis reports the model-implied percentage decline in the probability of living in that CZ relative to the baseline probability, which we interpret as the predicted out-migration response. Panel (a) plots this response against the percentage of the individual's friends living outside the shocked CZ. Panel (b) plots the same response against the 2013 average SAT score of the student body at the college listed, using data from Chetty et al. (2020). All variables plotted in both panels are residualized with respect to shocked-CZ fixed effects and recentered at their corresponding sample means. The binned series in Panel (b) is restricted to individuals with a college listed for which average SAT scores are observed. Dots report means within 20 equally sized bins, and black lines show linear best fits through the binned series. The dashed horizontal line in Panel (a) marks the mean response in the Model Sample; the dashed line in Panel (b) marks the mean response among individuals without listed college information. Network measures are based on friendships formed prior to 2012, with friends' locations measured in Q1 of 2012.

Tables

Table 1: Summary Statistics

Panel (a): General Sample

	Mean	SD	P5	P10	P25	P50	P75	P90	P95
Left 2012 CZ by 2025	0.32	0.47	0	0	0	0	1	1	1
N 2012 Friends	138	101	53	57	73	107	169	259	329
Share Friends Outs. 2012 CZ	0.40	0.27	0.07	0.10	0.18	0.33	0.61	0.85	0.92
N Largest 50 CZs w/ 1+ Friend	9	6	2	3	5	8	12	18	21
Lists College	0.59	0.49	0	0	0	1	1	1	1
Female	0.63	0.48	0	0	0	1	1	1	1

Panel (b): College Graduation Sample

	Mean	SD	P5	P10	P25	P50	P75	P90	P95
Left Home CZ Upon Grad.	0.14	0.34	0	0	0	0	0	1	1
Moved Upon Grad.	0.23	0.42	0	0	0	0	0	1	1
N Pre-Grad Friends	231	207	16	30	79	176	326	512	623
N CZs w/ 1+ Friend	34	23	5	8	17	31	49	67	78
N Largest 50 CZs w/ 1+ Friend	15	9	2	4	8	14	21	28	32
Share Friends Outs. Home CZ	0.36	0.15	0.14	0.19	0.26	0.35	0.45	0.55	0.62
Share Friends in Largest 50 CZs	0.58	0.31	0.09	0.12	0.25	0.71	0.86	0.92	0.95
Has Parent SES	0.70	0.46	0	0	0	1	1	1	1
Lists Major	0.37	0.48	0	0	0	0	1	1	1
Female	0.62	0.49	0	0	0	1	1	1	1
Lives in Largest 50 CZ Upon Grad.	0.58	0.49	0	0	0	1	1	1	1

Panel (c): Model Sample

	Mean	SD	P5	P10	P25	P50	P75	P90	P95
Left 2012 CZ by 2025	0.30	0.46	0	0	0	0	1	1	1
N 2012 Friends	136	102	52	57	72	106	165	255	322
Share Friends Outs. 2012 CZ	0.37	0.27	0.06	0.08	0.15	0.29	0.57	0.81	0.90
N Largest 50 CZs w/ 1+ Friend	10	6	2	3	5	9	13	19	22
Lists College	0.61	0.49	0	0	0	1	1	1	1
Female	0.62	0.48	0	0	0	1	1	1	1

Notes: This table presents summary statistics for the three main samples used in the paper, described in more detail in Section 2 and Appendix B. Panels (a)–(c) report statistics for the General Sample, the College Graduation Sample, and the Model Sample, respectively. Friend-related measures in Panel (a) and (c) are based on friendships formed prior to 2012 and friends' locations are measured in Q1 of 2012. Friend-related measures in Panel (b) are based on pre-graduation friendships, i.e., those formed at least one year before graduation (described further in Appendix B) and friends' locations are measured in Q4 of the expected graduation year. All friend-related measures use U.S.-based friends only. "Left 2012 CZ by 2025" is an indicator for whether an individual lives in a different CZ in Q4 of 2025 than in Q1 of 2012. "Largest 50 CZs" refers to the 50 largest CZs by Facebook user count shown in Appendix Figure A2. "Left Home CZ Upon Grad." is an indicator for whether a graduate lives, in Q4 of the expected graduation year, outside the union of the CZs in which they attended high school or college. "Moved Upon Grad" is an indicator for whether a graduate lives, at that same point in time, in a CZ other than the one in which they lived in Q2 of their expected graduation year. "Has Parent SES" indicates that we are able to link the graduate to parental information used to construct parental SES; see Section 2 for details.

Table 2: Migration Behavior, Location of Social Networks, and Covariates

Panel (a): Probability of Leaving 2012 CZ

	Has Left 2012 CZ by 2025					
	(1)	(2)	(3)	(4)	(5)	(6)
Prop. Friends Outs. 2012 CZ	0.510***				0.485***	0.450***
	(0.000)				(0.001)	(0.006)
Age		-0.009***			-0.016***	-0.016***
		(0.000)			(0.000)	(0.000)
Lists College			0.101***		0.037***	0.037***
			(0.000)		(0.000)	(0.003)
Grew Up in 2012 CZ				-0.301***	-0.133***	-0.143***
				(0.000)	(0.000)	(0.005)
2012 CZ FEs						✓
N	16,699,759	16,725,808	16,725,808	16,725,808	16,699,759	16,699,759
Mean	0.323	0.324	0.324	0.324	0.323	0.323
R^2	0.088	0.023	0.011	0.083	0.165	0.172

Panel (b): Probability of Moving to CZ with High Wage Growth

	Has Moved to High-Wage-Growth CZ by 2025					
	(1)	(2)	(3)	(4)	(5)	(6)
Prop. Friends in HWG CZ	0.624***				0.660***	0.601***
	(0.001)				(0.001)	(0.016)
Age		-0.001***			-0.002***	-0.002***
		(0.000)			(0.000)	(0.000)
Lists College			0.016***		0.010***	0.010***
			(0.000)		(0.000)	(0.001)
Dist. to HWG CZ				-0.016***	-0.001***	
				(0.000)	(0.000)	
2012 CZ FEs						✓
N	15,324,142	15,324,142	15,324,142	15,324,142	15,324,142	15,324,142
Mean	0.035	0.035	0.035	0.035	0.035	0.035
R^2	0.052	0.003	0.002	0.000	0.061	0.064

Notes: This table reports linear regression estimates based on the General Sample described in Section 2. In Panel (a), the dependent variable is an indicator for whether an individual lives in a different CZ in Q4 of 2025 than in Q1 of 2012. In Panel (b), the dependent variable is an indicator for whether an individual lives in a high-wage-growth CZ in Q4 of 2025. High-wage-growth CZs are defined as the ten CZs among the 50 largest CZs with the highest predicted wage growth between 2012 and 2024, based on a shift-share approach using local industry composition; see footnote 13 for details. Panel (b) is restricted to individuals who did not live in a high-wage-growth CZ in Q1 of 2012. "Prop. Friends Outs. 2012 CZ" denotes the share of an individual's friends living outside their 2012 CZ in Q1 of 2012. "Prop. Friends in HWG CZ" denotes the share of an individual's friends living in high-wage-growth CZs in Q1 of 2012. Friend locations are measured in Q1 of 2012, and the analysis restricts to friendships formed prior to 2012. "Grew Up in 2012 CZ" is an indicator for whether the CZ in which the individual lives in Q1 of 2012 is the same CZ as the one in which they attended high school. "Dist. to HWG CZ" measures the distance, in thousands of miles, from an individual's 2012 CZ to the nearest high-wage-growth CZ, based on CZ centroids. Columns (1)–(4) report univariate regressions. Column (5) includes all covariates jointly. Column (6) additionally includes 2012 CZ fixed effects. Standard errors are reported in parentheses. *** indicates significance at the 1% level.

Table 3: Network Effects on Out-Migration

	Has Left Home CZ by t^*					
	(1)	(2)	(3)	(4)	(5)	(6)
Network Effect (β_{-j})	0.335***	0.319***	0.326***	0.312***	0.325***	0.317***
	(0.007)	(0.007)	(0.013)	(0.007)	(0.013)	(0.009)
Baseline FE	✓					
... × Gender		✓				
... × Major Group			✓			
... + College-Cohort				✓		
... + College × Major Group					✓	
... + Abs. Time Friends' Moves						✓
N	1,310,479	1,273,066	410,556	1,310,439	458,472	1,310,479
Prop. Left Home CZ	0.125	0.123	0.126	0.125	0.133	0.125
R^2	0.151	0.173	0.235	0.160	0.201	0.151

Notes: This table reports estimates summarizing the event-time coefficients from equation 3 based on the College Graduation Sample described in Section 2. Specifically, the coefficient in the first row is a linear combination of the event-time coefficients: the average pre-period coefficient minus the average post-period coefficient, where the pre-period comprises quarters twelve to four before t^* and the post-period comprises quarters four to twelve after t^*. t^* is defined as Q4 of the year in which graduates are expected to graduate, when we measure their location. In all columns, the dependent variable is an indicator for whether, at t^*, the graduate lives outside the union of the CZs in which they attended high school or college. Column (1) reports the baseline specification, which includes fixed effects for the full interaction of high school CZ, college CZ, graduation cohort, and quintile of the initial share of friends living outside the home CZ, with the quintiles defined within high-school-CZ-by-college-CZ-by-graduation-cohort cells. Columns (2) and (3) additionally interact these baseline fixed effects with gender and major-group indicators, respectively. Columns (4)–(6) retain the baseline fixed effects and add college-by-cohort fixed effects, college-by-major-group fixed effects, and controls for friends' moves in absolute calendar time, respectively. Major groups classify majors into five groups based on the occupational distributions of recent graduates; see Appendix B for details. Standard errors, reported in parentheses, are computed for this linear combination using the full variance-covariance matrix of the event-time estimates. *** indicates significance at the 1% level.

Table 4: Network Effects on Directed Migration

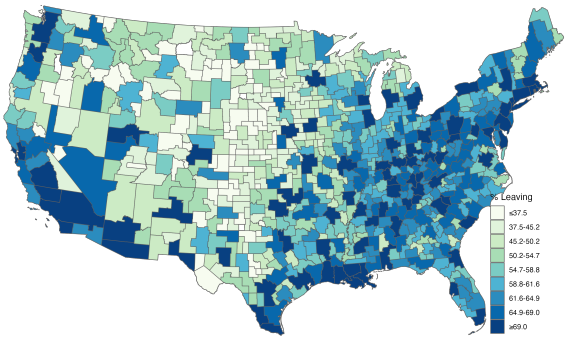
	Baseline	Alternative Controls				
		× Gender	× Major Group	+ College-Cohort	+ College × Major Group	+ Abs. Time Friends' Moves
Mean β_k	0.104	0.101	0.105	0.095	0.098	0.099
Mean β_k in % of Migration Rate	89.1	88.5	87.5	81.5	79.2	87.5
N ($\beta_k^{raw} > 0$)	50	50	47	50	47	50
N ($\beta_k^{raw} > 0$ & $p < 0.10$)	46	46	35	45	31	40
N ($\beta_k^{raw} > 0$ & $p < 0.05$)	45	45	29	44	29	39
N ($\beta_k^{raw} > 0$ & $p < 0.01$)	43	38	21	37	21	32

Notes: This table summarizes the destination-specific network effects reported in Panel (b) of Figure 4 and examines their robustness to richer controls, drawing on the College Graduation Sample described in Section 2. Each specification estimates effects separately for the 50 destination CZs and excludes graduates who attended high school or college in the corresponding CZ. For each destination, the network effect is given by the difference between the average pre- and post-period coefficients, where the pre-period comprises quarters twelve to four before t^* and the post-period comprises quarters four to twelve after t^*. At baseline, the regressions include fixed effects for the full interaction of high school CZ, college CZ, graduation cohort, and quintile of the initial share of friends living in the destination CZ, with the quintiles defined within high-school-CZ-by-college-CZ-by-graduation-cohort cells. Column 1 reports results from this baseline specification, summarizing the estimates shown in Panel (b) of Figure 4. Columns 2–6 re-estimate the regressions with alternative controls. These specifications respectively interact the baseline fixed effects with gender or major-group indicators, add college-by-cohort or college-by-major-group fixed effects, and control for friends' moves in absolute calendar time. The mean estimates are calculated using the empirical Bayes shrinkage procedure described in Appendix D. Mean β_k reports the unweighted average across destinations. The following row reports the unweighted average after scaling each destination-specific estimate to represent the percentage change in the baseline migration rate associated with a one-percentage-point increase in the share of friends living in that destination. The remaining rows report counts based on the raw, unshrunk estimates. The first reports the number of positive estimates; the subsequent rows report the number of positive estimates with p-values below the indicated thresholds.

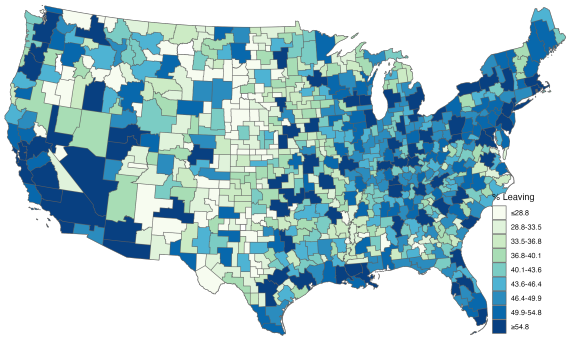
A Appendix Figures

Figure A1: Validation of Migration Data

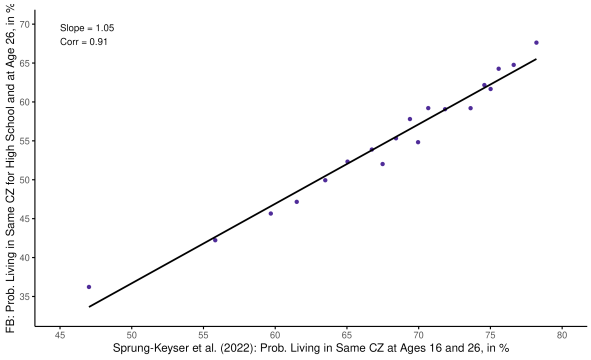
(a) Prob. of Leaving Childhood CZ (IRS)



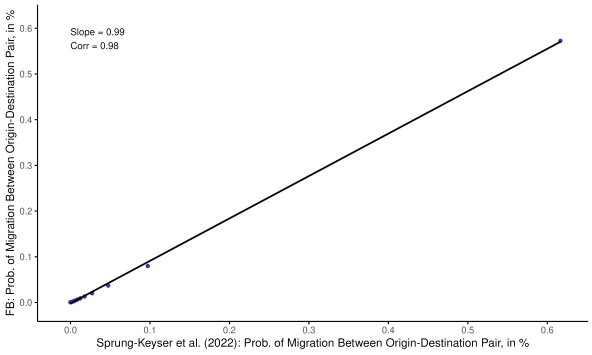
(b) Prob. of Leaving High School CZ (FB)



(c) Prob. of Living in Childhood / HS CZ

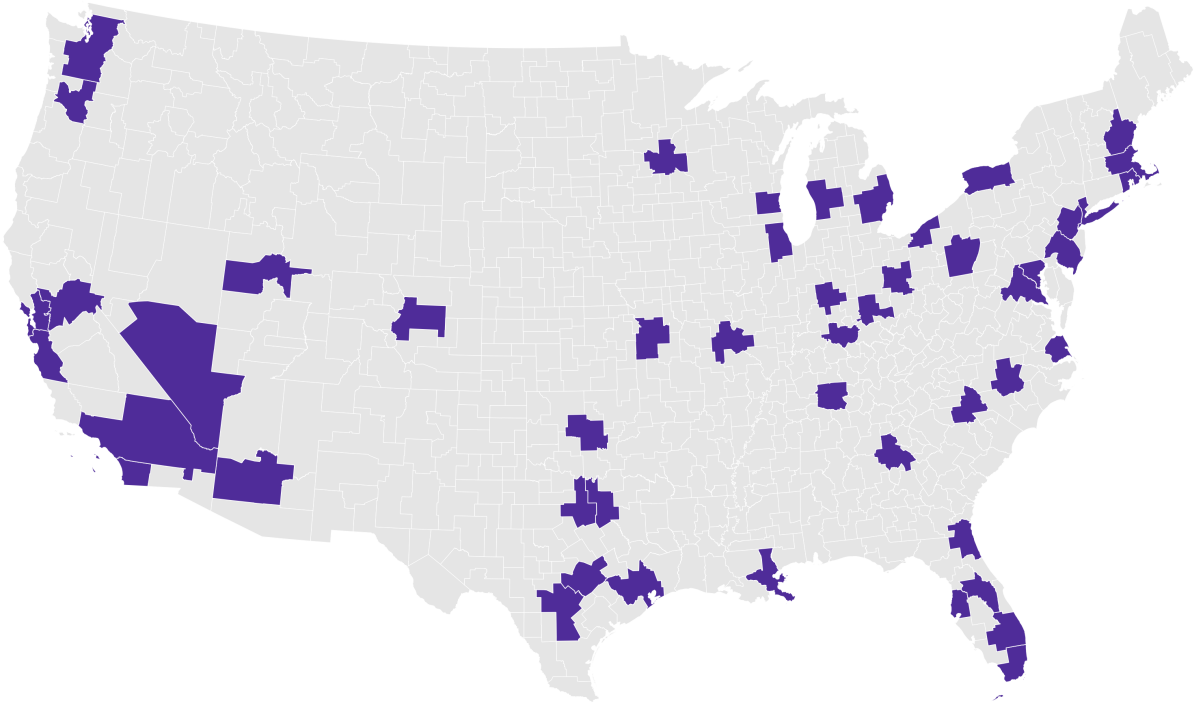


(d) Prob. of Moving to a Specific CZ



Notes: This figure presents validation tests of the migration patterns observed in the Facebook data. Panel (a) plots the percentage of individuals who lived in a given commuting zone at age 16 who no longer live in that CZ at age 26. The data are drawn from Sprung-Keyser et al. (2022), who observe the location of individuals born between 1984 and 1992 using linked data from the Census, American Community Survey, and Internal Revenue Service. In Panel (b), we produce a similar plot, which captures the percentage of Facebook users in our General Sample who were born between 1986 and 1992 and who, in Q4 of the year in which they turn 26, live in a CZ different than the one in which they attended high school. We use the 1986–1992 birth cohorts rather than the 1984–1992 cohorts because the Facebook location panel begins in 2012, so locations in the year individuals turn 26 are not observed for earlier cohorts. In Panel (c), we present a binned scatter plot of these two quantities. In Panel (d), we present a binned scatter plot comparing the percentage of individuals who move between each origin–destination CZ pair in the two data sets. We exclude CZ pairs in which the origin and destination are the same. The binned scatter plots in Panels (c) and (d), as well as the reported correlations and slopes, are weighted using the number of individuals in each origin in the data from Sprung-Keyser et al. (2022).

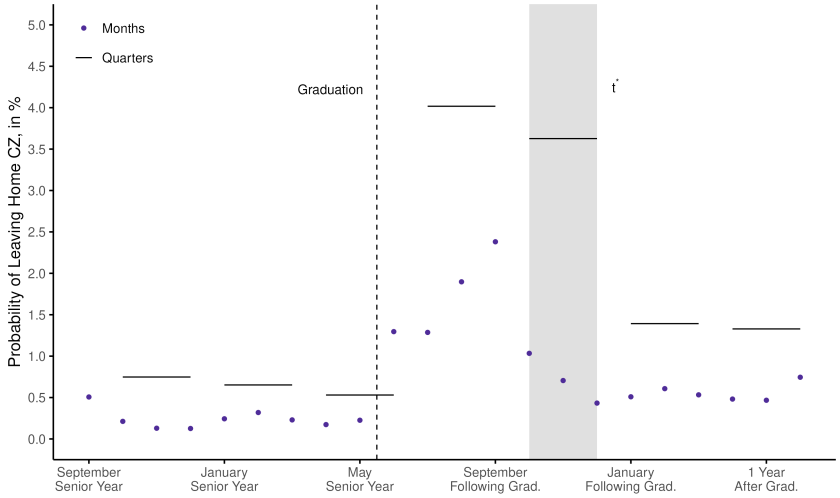
Figure A2: 50 Largest CZs by Facebook User Count



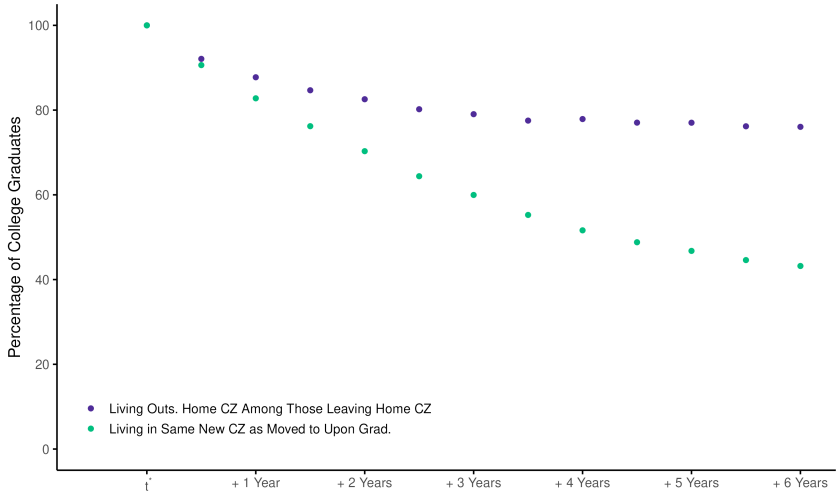
Notes: This figure presents a map of the 50 largest commuting zones (CZs), defined as the 50 CZs with the largest number of Facebook users residing there in Q1 of 2012. CZ boundaries follow the 1990 definition in Autor and Dorn (2013). The highlighted CZs and their corresponding identifiers are: Atlanta (9100), Austin (31201), Baltimore (11302), Boston (20500), Buffalo (18000), Charlotte (900), Chicago (24300), Cincinnati (12701), Cleveland (15200), Columbus (15900), Dallas (33100), Denver (28900), Detroit (11600), Fort Worth (33000), Grand Rapids (12200), Houston (32000), Indianapolis (14200), Jacksonville (7600), Kansas City (29502), Las Vegas (37901), Los Angeles (38300), Louisville (13101), Manchester (20600), Miami (7000), Milwaukee (24100), Minneapolis (21501), Nashville (5600), New Orleans (3300), New York (19400), Newark (19600), Oklahoma City (33803), Orlando (7400), Philadelphia (19700), Phoenix (35001), Pittsburgh (16300), Port St. Lucie (7100), Portland (38801), Providence (20401), Raleigh (1701), Sacramento (37400), Salt Lake City (36100), San Antonio (31301), San Diego (38000), San Francisco (37800), San Jose (37500), Seattle (39400), St. Louis (24701), Tampa (6700), Virginia Beach (2000), Washington, D.C. (11304).

Figure A3: College Graduation Event as "Transition Period" for Residential Choices

(a) Probability of Leaving Home CZ



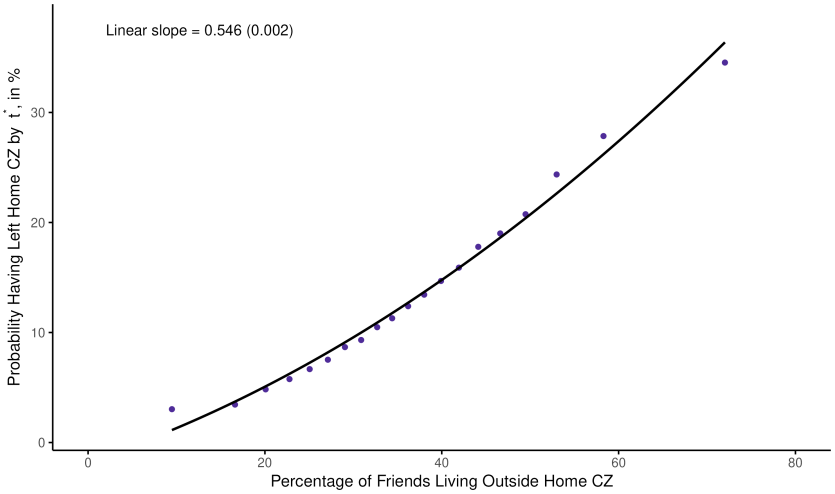
(b) Persistence of Location Choices



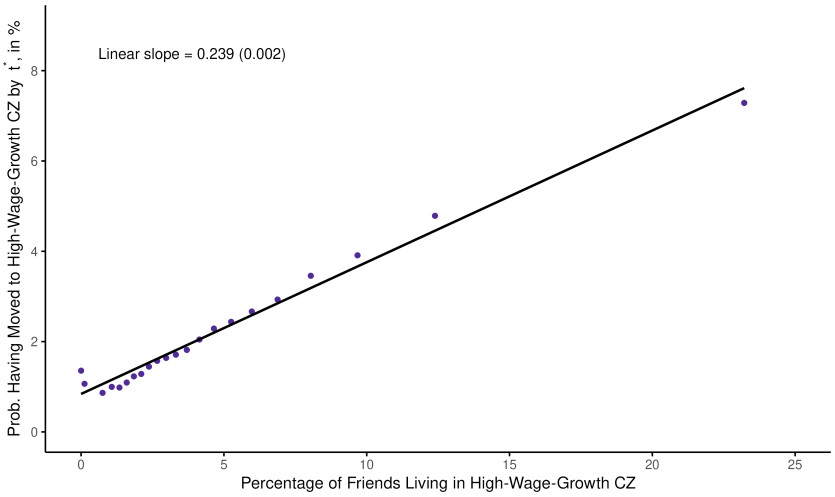
Notes: This figure presents descriptive evidence on the role of college graduation as a transition period for residential choices, drawing on the College Graduation Sample described in Section 2. Panel (a) shows the percentage of graduates leaving their home CZ over time, where the home CZ is defined as the union of the CZs in which they attended high school or college. Purple dots report monthly out-migration rates; horizontal black lines report quarterly values. Quarterly rates are larger because they aggregate moves over three months. Since quarterly locations are defined using an individual's modal location within the quarter, moves occurring late in a quarter tend to appear in the quarterly series in the following quarter, so the quarterly series responds with a short delay relative to the monthly series. The dashed vertical line indicates the expected time of graduation, i.e., late May of senior year. The shaded gray bar indicates t^*, i.e., Q4 of the graduation year. Panel (b) shows the persistence of the location choices made upon graduation over time among individuals who leave their home CZ. Purple dots report the percentage of these graduates who continue to live outside their home CZ in subsequent years. Green dots report the percentage who continue to live in the same non-home CZ to which they initially moved at graduation.

Figure A4: Migration Behavior and Location of Social Networks at College Graduation

(a) Probability of Leaving Home CZ



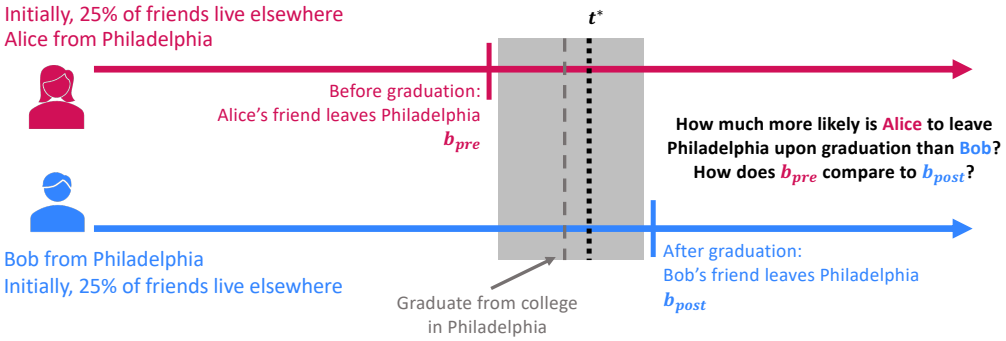
(b) Probability of Moving to CZ with High Wage Growth



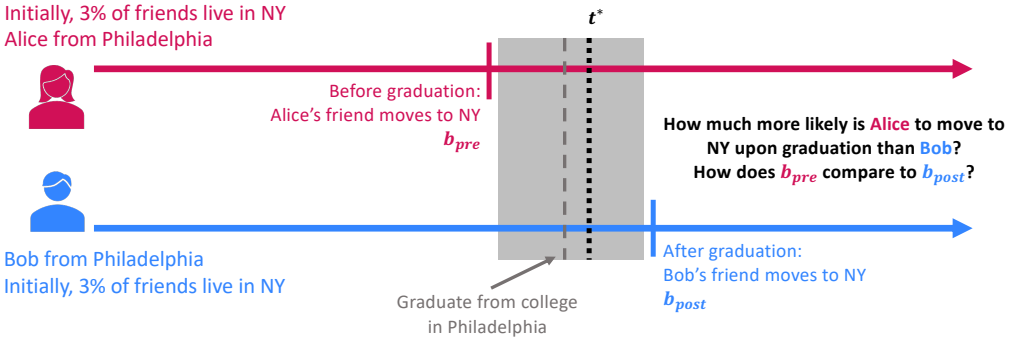
Notes: This figure presents descriptive evidence on the relationship between social networks and migration at the time of graduation, drawing on the College Graduation Sample described in Section 2. Panel (a) shows a binned scatter plot where the vertical axis corresponds to the probability (in %) that a graduate lives outside the home CZ at t^*, i.e., Q4 of the graduation year. The home CZ is defined as the union of the CZs in which they attended high school or college. The horizontal axis displays the percentage of friends living outside the home CZ. Panel (b) shows a binned scatter plot where the vertical axis corresponds to the probability (in %) that a graduate lives in a high-wage-growth CZ at t^*, and the horizontal axis displays the percentage of friends living in such CZs. This analysis is restricted to graduates who did not reside in a high-wage-growth CZ during high school or college. High-wage-growth CZs are defined analogously to Section 2.2, i.e., they correspond to the ten CZs among the 50 largest CZs with the highest predicted wage growth between 2012 and the individual's year of graduation. In both panels, friend locations are measured at t^*, and the analysis restricts to pre-graduation friendships defined in Appendix B. The black lines represent the best fit from a quadratic regression on the binned series in Panel (a) and from a linear regression on the binned series in Panel (b).

Figure A5: Illustrative Example of Identification Strategy

(a) Out-Migration

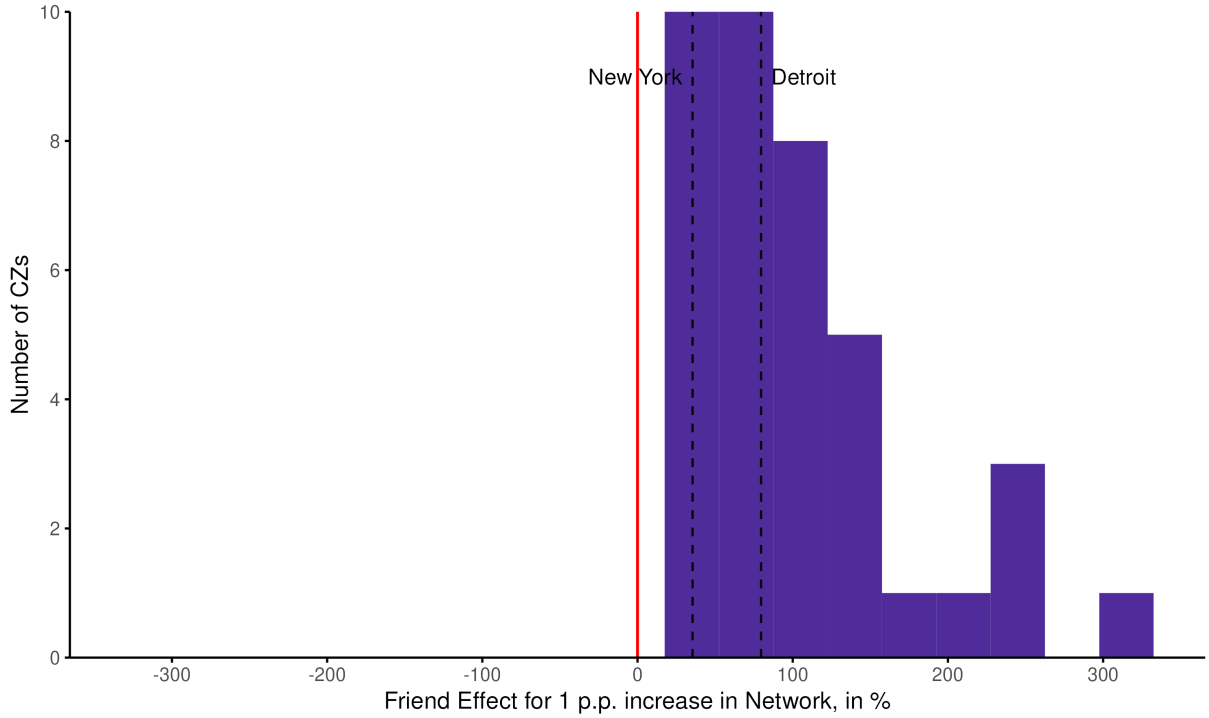


(b) Directed Migration



Notes: This figure presents a stylized example to illustrate the identification strategy described in Section 3. Panel (a) describes the out-migration case. In this case, there are two individuals, *Alice* and *Bob*, who both grow up and attend college in Philadelphia, graduate at the same time, and begin with the same share of friends living outside Philadelphia. While both *Alice* and *Bob* have one additional friend who leaves Philadelphia around the point in time at which they decide where to live after graduation, the timing of their friends' moves differs. *Alice*'s friend moves just before *Alice* and *Bob* decide whether to leave Philadelphia, while *Bob*'s friend moves slightly after. Panel (b) illustrates the same logic for directed migration to New York. *Alice* and *Bob* again grow up and attend college in Philadelphia, graduate at the same time, and begin with the same share of friends living in New York. Around graduation, *Alice* and *Bob* each have one additional friend who moves to New York. *Alice*'s friend moves before *Alice* and *Bob* decide where to live—in this case, whether they move to New York. In contrast, *Bob*'s friend moves slightly later. In both panels, the black dashed line denotes t^*, i.e., the point at which we measure post-graduation location, and the gray dashed line indicates the time of graduation. The shaded gray bar denotes the mid-period around graduation during which individuals are likely deciding where to live or may anticipate friends' moves occurring shortly thereafter.

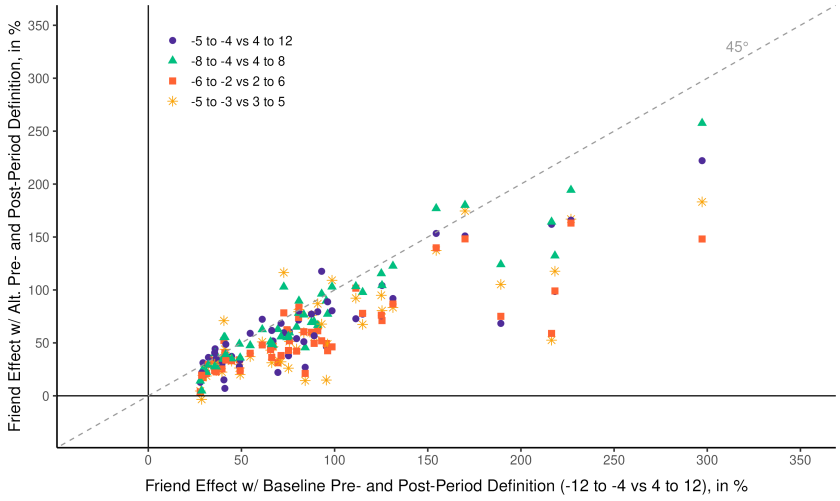
Figure A6: Distribution of CZ-Specific Network Effects — Raw Estimates



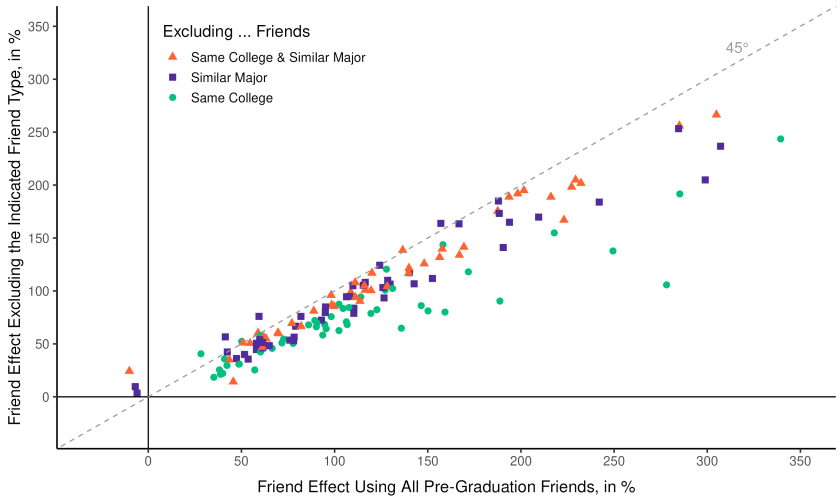
Notes: This figure shows the distribution of the destination-specific network effects, similar to Panel (b) of Figure 4, but using the raw, unshrunk estimates. Each effect is given by the difference between the average pre- and post-period coefficients and is scaled to represent the percentage change in the baseline migration rate to the corresponding destination associated with a one-percentage-point increase in the share of friends living there. For each destination-specific estimate, the analysis is restricted to graduates who did not attend high school or college in that CZ. The solid red line marks zero, and the dashed vertical lines mark the estimates for New York and Detroit, respectively.

Figure A7: Robustness of Network Effects on Directed Migration

(a) Alternative Definitions of Pre- and Post-Period

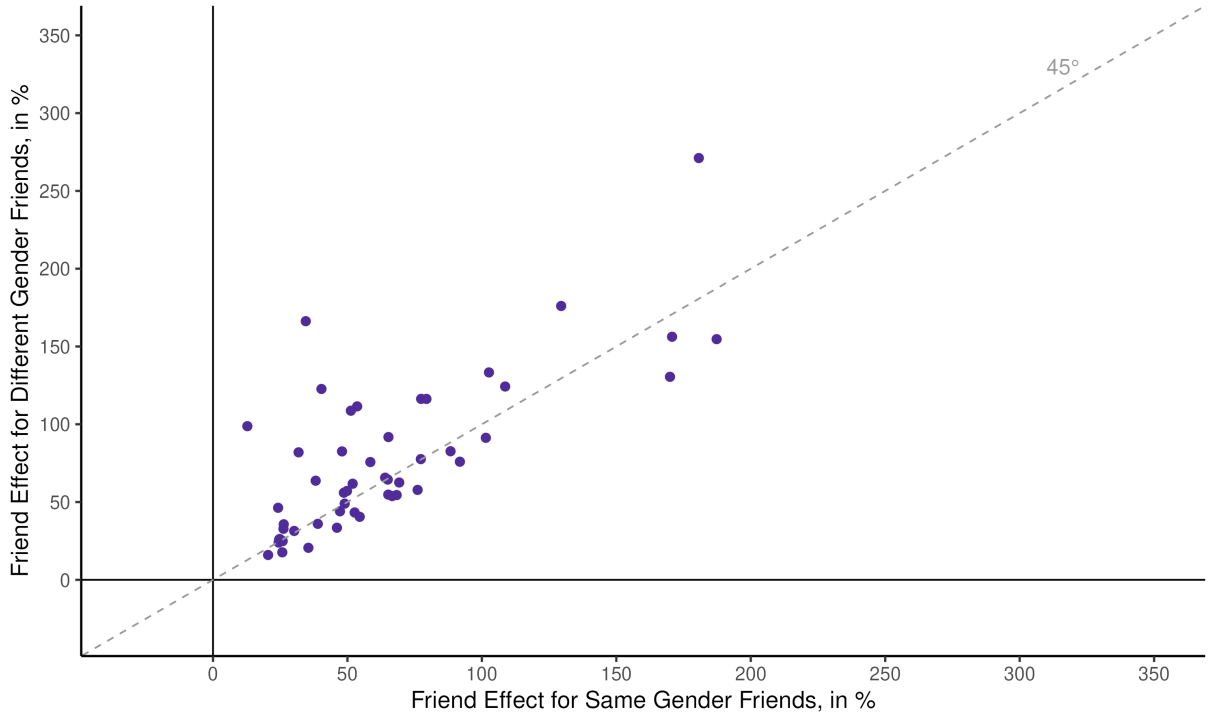


(b) Excluding Different Types of Friends



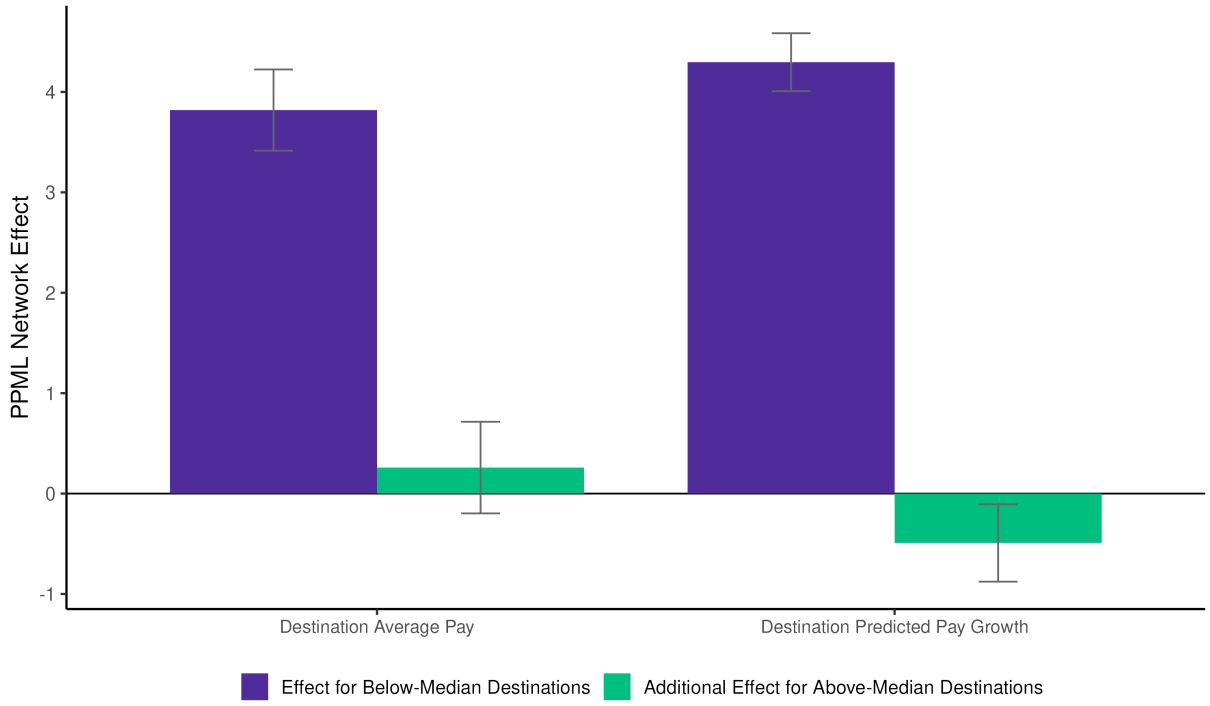
Notes: This figure presents robustness checks for the destination-specific network effects reported in Panel (b) of Figure 4. Each point represents one of the 50 destination CZs. All estimates correspond to the difference between the average pre- and post-period coefficients, are scaled to percent of the baseline migration rate to the corresponding destination, and use the empirical Bayes shrinkage procedure described in Appendix D. For each destination, the analysis excludes graduates who attended high school or college in that CZ. Panel (a) plots the baseline estimates, which define the pre- and post-periods as quarters twelve to four before and four to twelve after t^*, against estimates using the four alternative windows indicated in the legend. All four alternatives use the same baseline event-time regressions and estimated quarterly coefficients and differ only in the quarters averaged to construct the pre–post difference. The first alternative defines the pre-period using quarters five to four before t^* and retains quarters four to twelve after t^* as the post-period. Because the baseline network includes only friendships formed by the end of May of the year before expected graduation, all pre-period moves used in this specification occur after the corresponding friendships were formed. Panel (b) plots the baseline estimates based on all pre-graduation friends on the horizontal axis against estimates obtained after excluding, in turn, friends who attended the same college, had a similar major, or satisfied both conditions. We classify two majors as similar when they belong to the same major group, constructed based on the occupational distributions of recent graduates; see Appendix B for details. For each exclusion, the all-friends estimate is re-estimated on the same sample as the corresponding leave-out estimate. The leave-out specifications also include friends' moves among the excluded group and their initial share in the destination as controls. The dashed lines indicate the 45-degree line.

Figure A8: Network Effects on Directed Migration: Heterogeneity by Friend Gender



Notes: This figure compares destination-specific network effects for friends of the same gender as the graduate (horizontal axis) and friends of a different gender (vertical axis). Each point represents one of the 50 destination CZs. For each destination, the two effects are estimated jointly by including separate event-time friend-move variables for same- and different-gender friends in the same regression. Each estimate corresponds to the difference between the average pre- and post-period coefficients and is scaled to percent of the baseline migration rate to the corresponding destination. Estimates use the empirical Bayes shrinkage procedure described in Appendix D. For each destination, the analysis excludes graduates who attended high school or college in that CZ. The dashed line indicates the 45-degree line.

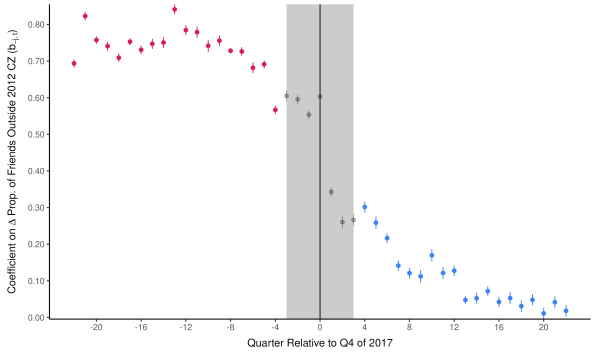
Figure A9: Interaction Effects



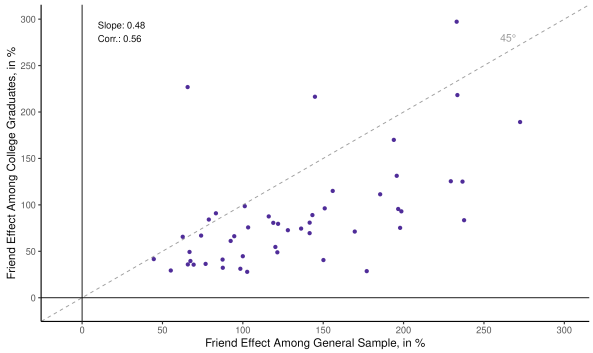
Notes: This figure examines whether average network effects differ across destinations with stronger versus weaker economic conditions. Estimates come from the stacked PPML specification described in Section 4.4 and used in Figure 7. In separate regressions, the event-time friend-move variables are interacted with indicators for whether the potential destination has above-median average pay or above-median predicted pay growth. Average pay is measured in the graduate's expected year of graduation. Predicted pay growth is measured between 2012 and that year using the same shift-share approach based on local industry composition that is used to define the high-wage-growth CZs in Figure 2 and Table 2; see footnote 13 for details. Each estimate corresponds to the difference between the average pre- and post-period coefficients. The first estimate gives the network effect for destinations below the median of the corresponding measure. The second gives the additional effect for above-median destinations relative to below-median destinations and may therefore be positive or negative. The total network effect for above-median destinations is given by the sum of the two estimates. Vertical lines denote 95% confidence intervals.

Figure A10: External Validity

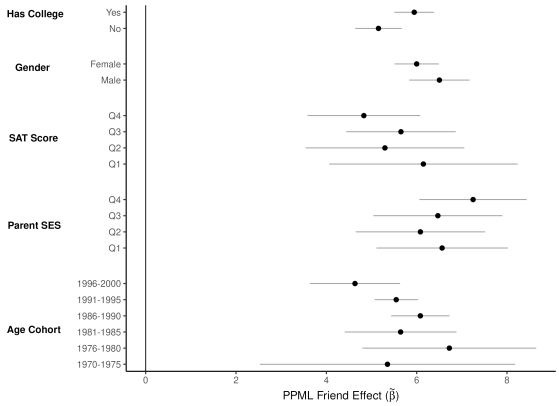
(a) Out-Migration



(b) CZ-Specific Network Effects



(c) Heterogeneity



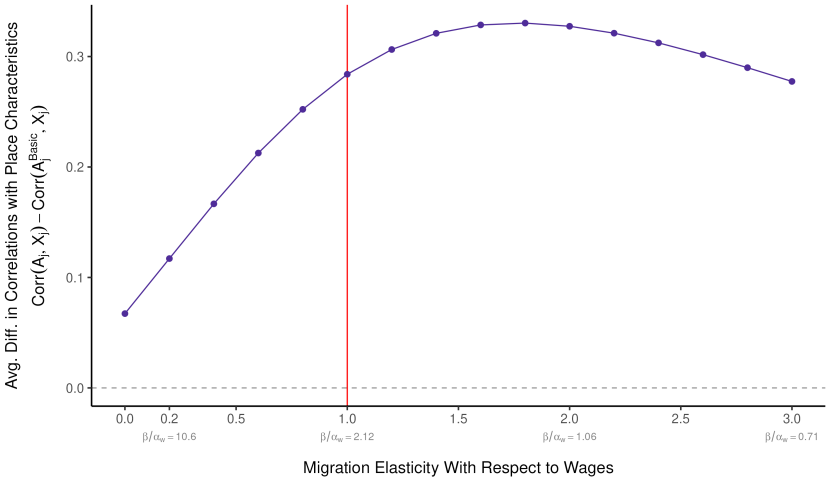
Notes: This figure presents estimation results for the external-validity exercise based on the General Sample described in Section 4.5. The empirical design follows equations 3 and 5, amended so that t = *early* corresponds to Q1 of 2012 and t^* corresponds to Q4 of 2017. Panel (a) reports event-time coefficient estimates for out-migration. The horizontal axis is measured in quarters relative to t^*. Red coefficient estimates correspond to friends' moves occurring between 22 and four quarters prior to t^*; blue coefficient estimates correspond to friends' moves occurring between four and 22 quarters after t^*; gray coefficient estimates correspond to the seven-quarter window around t^*, i.e., the three quarters before, quarter 0, and the three quarters after t^*. The shaded gray bar indicates this mid-period window. Vertical lines denote 95% confidence intervals. The outcome is an indicator for whether an individual lives, at t^*, outside the CZ in which they lived in Q1 of 2012. Panel (b) compares destination-specific network effects in the General Sample (horizontal axis) with the corresponding effects in the College Graduation Sample (vertical axis). Each point represents one of the 50 destination CZs. Each effect is given by the difference between the average pre- and post-period coefficients and is scaled by the baseline migration rate to that destination in the respective sample. For the General Sample, the analysis excludes individuals who lived in the destination CZ in Q1 of 2012; for the College Graduation Sample, it excludes graduates who attended high school or college there. Estimates use the empirical Bayes shrinkage procedure described in Appendix D. The College Graduation Sample estimates are therefore identical to those shown in Panel (b) of Figure 4. The dashed line indicates the 45-degree line. Panel (c) reports subgroup-specific estimates from a stacked PPML specification analogous to that in Section 4.4, estimated separately for each subgroup using the General Sample. Each estimate corresponds to the difference between the average pre- and post-period coefficients, and horizontal lines denote 95% confidence intervals. The SAT score measure corresponds to the 2013 average SAT score of the student body at an individual's college using data from Chetty et al. (2020); Q1–Q4 denote quartiles of SAT scores or predicted parental SES. The PPML analysis uses a 50% random sample for computational reasons.

Figure A11: Sensitivity: Role of Local Amenities A_j, A_j^{Basic}

(a) Reduction in Standard Deviation

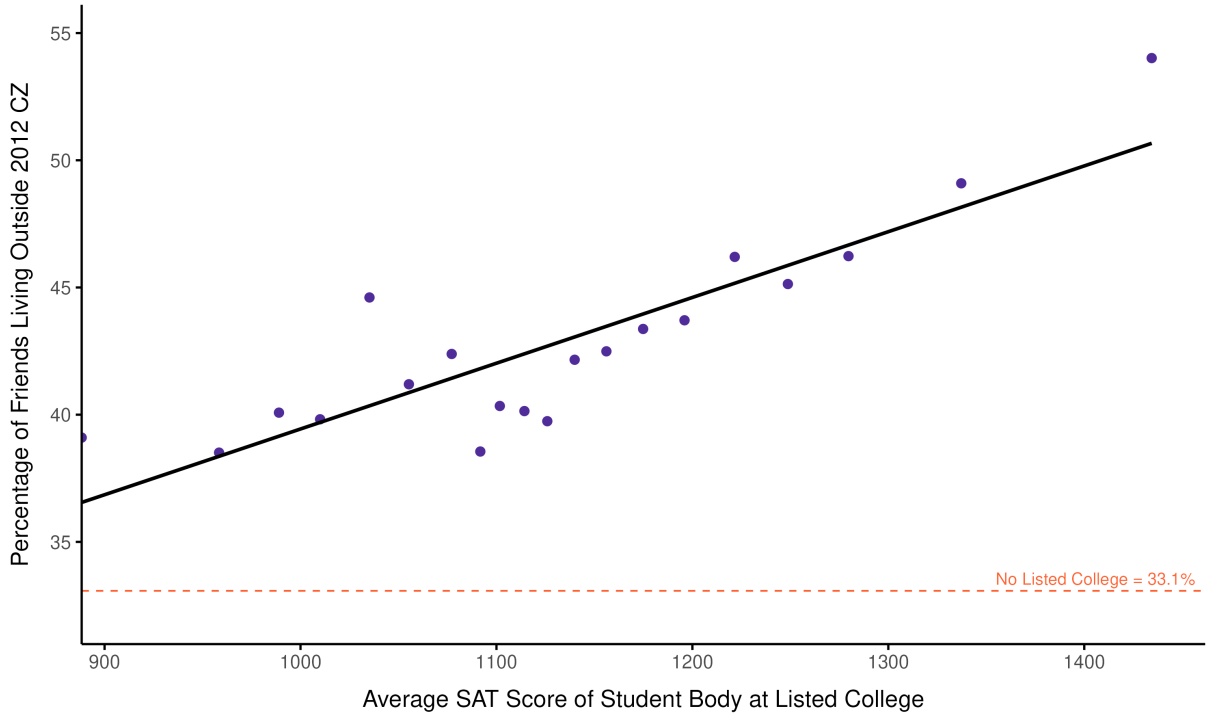


(b) Difference in Correlations with Place Characteristics



Notes: This figure examines the sensitivity of the model results in Figure 8 to alternative assumptions about the average migration elasticity with respect to wages. For each value shown on the horizontal axis, we separately recalibrate the network and basic models to match observed CZ population sizes and the indicated migration elasticity, while holding the network parameter $\widetilde{\beta}$ fixed. Panel (a) reports the ratio of the standard deviation of the residual location-specific components in the network model, A_j, to that in the basic model, A_j^{Basic}. Before computing the standard deviations, each vector is expressed in log-wage-equivalent percentage points by dividing by the corresponding model-specific α_w and demeaning. Values below one therefore indicate that the network model requires less dispersed residuals than the basic model. Panel (b) reports the average, across the six place characteristics shown in Panel (b) of Figure 8, of the difference between their population-weighted correlations with A_j and with A_j^{Basic}. Positive values indicate that the residuals from the network model are, on average, more positively correlated with these characteristics than those from the basic model. The red vertical line marks the baseline calibration, which targets a migration elasticity of one. The gray labels below the horizontal axis report the corresponding value of $\widetilde{\beta}/\alpha_w$ in the network model at selected elasticities.

Figure A12: Dispersion of Social Networks by Educational Background



Notes: This figure presents a binned scatter plot of the geographic dispersion of social networks by educational background using the Model Sample described in Section 2. The vertical axis reports the percentage of an individual's friends living outside their CZ of residence in Q1 of 2012. The horizontal axis reports the 2013 average SAT score of the student body at the college listed, using data from Chetty et al. (2020). Dots report means within 20 equally sized bins among individuals with observed college information, and the black line shows the linear best fit through the binned series. The dashed horizontal line marks the average percentage of friends living elsewhere among individuals without listed college information.

Appendix Tables

Table A1: Network Effects on Out-Migration: Alternative Definitions of Pre- and Post-Period

	Has Left Home CZ by t^*				
	Baseline	-5 to -4 vs 4 to 12	-8 to -4 vs 4 to 8	-6 to -2 vs 2 to 6	-5 to -3 vs 3 to 5
Network Effect (β_{-j})	0.335***	0.334***	0.303***	0.256***	0.275***
	(0.007)	(0.009)	(0.009)	(0.008)	(0.010)
N	1,310,479	1,310,479	1,310,479	1,310,479	1,310,479
Prop. Left Home CZ	0.125	0.125	0.125	0.125	0.125
R^2	0.151	0.151	0.151	0.151	0.151

Notes: This table reports the estimate from the baseline out-migration specification in Column 1 of Table 3 using alternative definitions of the pre- and post-periods. All columns use the same sample, event-time regression, fixed effects, and estimated quarterly coefficients as in the main table. They differ only in the quarters averaged to construct the pre–post coefficient difference. Column 1 uses the baseline windows, comprising quarters twelve to four before t^* and quarters four to twelve after t^*, and is identical to Column 1 of Table 3. Column 2 defines the pre-period using quarters five to four before t^* and retains quarters four to twelve after t^* as the post-period. Because the baseline network includes only friendships formed by the end of May of the year before expected graduation, all pre-period moves used in Column 2 occur after the corresponding friendships were formed. Columns 3–5 use the alternative windows shown in the column headings. Negative values denote quarters before t^* and positive values denote quarters after t^*. Standard errors are computed for each pre–post difference using the full variance-covariance matrix of the estimated quarterly coefficients and are reported in parentheses. *** indicates significance at the 1% level.

Table A2: Network Effects on Out-Migration: Alternative Clustering Schemes

	Has Left Home CZ by t^*			
	(1)	(2)	(3)	(4)
Network Effect (β_{-j})	0.335***	0.335***	0.335***	0.335***
	(0.007)	(0.010)	(0.009)	(0.010)
Baseline FE	✓	✓	✓	✓
Baseline SE clustering	✓			
Home CZ × Cohort clustering		✓		
College × Cohort clustering			✓	
Both clusterings				✓
N	1,310,479	1,310,479	1,310,479	1,310,479
Prop. Left Home CZ	0.125	0.125	0.125	0.125
R^2	0.151	0.151	0.151	0.151

Notes: This table reports the estimate from the baseline out-migration specification in Column 1 of Table 3 under alternative standard-error specifications. All columns use the same sample, regression specification, fixed effects, and pre–post coefficient difference as in the main table, and therefore report the same point estimate. Column 1 uses the baseline individual-level clustering and is identical to Column 1 of Table 3. Column 2 clusters standard errors by high-school-CZ-by-college-CZ-by-graduation-cohort. Column 3 clusters by college and graduation cohort. Column 4 uses two-way clustering along both dimensions. Standard errors are reported in parentheses. *** indicates significance at the 1% level.

Table A3: Network Effects on Out-Migration: Alternative Home CZ Definitions

	HS CZ = College CZ	HS CZ ≠ College CZ		
	(1)	(2)	(3)	(4)
	Left Home CZ	Left Both CZs	Left HS CZ	Left College CZ
Network Effect (β_{-j})	0.287***	0.380***	0.609***	1.297***
	(0.009)	(0.011)	(0.017)	(0.025)
Baseline FE	✓	✓	✓	✓
N	606,276	704,203	704,203	704,203
Prop. Left CZ	0.089	0.170	0.600	0.570
R^2	0.064	0.183	0.203	0.199

Notes: This table reports variants of the baseline out-migration specification in Column 1 of Table 3 using alternative definitions of the home CZ. All columns use the baseline event-time specification, fixed effects, pre- and post-periods, and standard-error clustering from the main table. Each coefficient corresponds to the difference between the average pre-period coefficient and the average post-period coefficient. Column 1 restricts the sample to graduates whose high school and college CZs coincide and defines the home CZ as this common CZ. Columns 2–4 restrict the sample to graduates whose high school and college CZs differ. Column 2 retains the baseline definition of the home CZ as the union of the high school and college CZs, so that the outcome indicates whether the graduate leaves both CZs. Columns 3 and 4 instead define the relevant home CZ as the high school CZ and college CZ, respectively. In these columns, both the outcome and the event-time friend-move variables are defined relative to the corresponding CZ. Standard errors are reported in parentheses. *** indicates significance at the 1% level.

Table A4: Network Effects on Directed Migration: Alternative Definitions of Pre- and Post-Period

	Baseline	Alternative Defs. of Pre- and Post-Period			
		-5 to -4 vs 4 to 12	-8 to -4 vs 4 to 8	-6 to -2 vs 2 to 6	-5 to -3 vs 3 to 5
Mean β_k	0.104	0.084	0.088	0.069	0.070
Mean β_k in % of Migration Rate	89.1	66.3	76.9	56.5	60.7
N ($\beta_k^{raw} > 0$)	50	48	49	49	46
N ($\beta_k^{raw} > 0$ & $p < 0.10$)	46	35	41	35	32
N ($\beta_k^{raw} > 0$ & $p < 0.05$)	45	33	39	30	27
N ($\beta_k^{raw} > 0$ & $p < 0.01$)	43	22	32	21	22

Notes: This table examines the sensitivity of the destination-specific network effects reported in Panel (b) of Figure 4 to alternative definitions of the pre- and post-periods. Each specification estimates effects separately for the 50 destination CZs and excludes graduates who attended high school or college in the corresponding CZ. For each destination, the network effect is given by the difference between the average pre- and post-period coefficients. Column 1 uses the baseline windows, comprising quarters twelve to four before t^* and quarters four to twelve after t^*, and summarizes the estimates shown in Panel (b) of Figure 4. Columns 2–5 use the same baseline event-time regressions and estimated quarterly coefficients but calculate the pre–post difference using the alternative windows shown in the column headings. Column 2 defines the pre-period using quarters five to four before t^* and retains quarters four to twelve after t^* as the post-period. Because the baseline network includes only friendships formed by the end of May of the year before expected graduation, all pre-period moves used in Column 2 occur after the corresponding friendships were formed. The specifications in Columns 2–5 correspond to Panel (a) of Figure A7. The mean estimates are calculated using the empirical Bayes shrinkage procedure described in Appendix D. Mean β_k reports the unweighted average across destinations. The following row reports the unweighted average after scaling each destination-specific estimate to represent the percentage change in the baseline migration rate associated with a one-percentage-point increase in the share of friends living in that destination. The remaining rows report counts based on the raw, unshrunk estimates. The first reports the number of positive estimates; the subsequent rows report the number of positive estimates with p-values below the indicated thresholds.

Table A5: Network Effects on Directed Migration: Alternative Clustering Schemes

Clustering scheme	N ($\beta_k^{raw} > 0$ & $p < 0.10$)	N ($\beta_k^{raw} > 0$ & $p < 0.05$)	N ($\beta_k^{raw} > 0$ & $p < 0.01$)
Baseline clustering	46	45	43
Home CZ × cohort	45	43	38
College × cohort	45	44	38
Two-way clustering	45	43	38

Notes: This table reports inference for the 50 destination-specific network effects underlying Panel (b) of Figure 4 under alternative standard-error specifications. All rows use the same baseline regressions, samples, and point estimates as in the main analysis. They differ only in the calculation of standard errors and p-values. Baseline standard errors are clustered at the individual level. The second row clusters by high-school-CZ-by-college-CZ-by-graduation-cohort. The third row clusters by college and graduation cohort. The final row uses two-way clustering along both dimensions. Entries report the number of positive destination-specific estimates, among the 50 estimates, that are statistically significant at the indicated level. Counts are based on the original, unshrunk regression estimates, whose distribution is shown in Appendix Figure A6.

Table A6: Network Effects on Out-Migration: Excluding Friends with Shared College or Similar Major

	Has Left Home CZ by t^*						
	Baseline	Excl. Same College & Similar Major		Excl. Similar Major		Excl. Same College	
		All Friends	Excl.	All Friends	Excl.	All Friends	Excl.
Network Effect (β_{-j})	0.335***	0.582***	0.482***	0.496***	0.385***	0.403***	0.232***
	(0.007)	(0.019)	(0.019)	(0.016)	(0.015)	(0.008)	(0.007)
N	1,310,479	373,500	373,456	429,397	429,337	1,273,037	1,272,642
Prop. Left Home CZ	0.125	0.138	0.138	0.133	0.133	0.127	0.127
R^2	0.151	0.188	0.190	0.182	0.183	0.152	0.156

Notes: This table reports variants of the baseline out-migration specification in Column 1 of Table 3 that exclude friends who share college- or field-specific migration pathways with the graduate. Each coefficient corresponds to the difference between the average pre-period coefficient and the average post-period coefficient. Column 1 reproduces the baseline estimate from the main table. Columns 2–3 examine the exclusion of friends who attended the same college and had a similar major to the graduate. Columns 4–5 examine the exclusion of friends with a similar major, and Columns 6–7 examine the exclusion of friends who attended the same college. We classify two majors as similar when they belong to the same major group, constructed based on the occupational distributions of recent graduates; see Appendix B for details. Within each pair, the first column re-estimates the effect for all pre-graduation friends on the sample used for the corresponding exclusion specification. The second column reports the effect of friends outside the excluded group. Moves among the excluded friends remain included in the regression as separate controls, together with their initial share of the graduate's network outside the home CZ. The paired all-friends and exclusion specifications use the same fixed-effect structure. Small differences in the number of observations within each pair arise because of the slight differences in the included friend-move and control variables. Standard errors are reported in parentheses. *** indicates significance at the 1% level.

Table A7: Network Effects on Directed Migration: Excluding Friends with Shared College or Similar Major

	Baseline	Excl. Same College & Similar Major		Excl. Similar Major		Excl. Same College	
		All Friends	Excl.	All Friends	Excl.	All Friends	Excl.
Mean β_k	0.104	0.192	0.168	0.158	0.130	0.139	0.092
Mean β_k in % of Migration Rate	89.1	147.7	128.9	126.3	106.1	118.2	81.0
N ($\beta_k^{raw} > 0$)	50	47	46	47	48	50	50
N ($\beta_k^{raw} > 0$ & $p < 0.10$)	46	37	33	37	35	47	43
N ($\beta_k^{raw} > 0$ & $p < 0.05$)	45	32	30	34	27	45	40
N ($\beta_k^{raw} > 0$ & $p < 0.01$)	43	25	22	27	25	42	36

Notes: This table reports variants of the destination-specific network effects summarized in Column 1 of Table 4 that exclude friends who share college- or field-specific migration pathways with the graduate, drawing on the College Graduation Sample described in Section 2. Each specification estimates effects separately for the 50 destination CZs and excludes graduates who attended high school or college in the corresponding CZ. Column 1 reproduces Column 1 of Table 4 exactly. Columns 2–3 examine the exclusion of friends who attended the same college and had a similar major to the graduate. Columns 4–5 examine the exclusion of friends with a similar major, and Columns 6–7 examine the exclusion of friends who attended the same college. We classify two majors as similar when they belong to the same major group, constructed based on the occupational distributions of recent graduates; see Appendix B for details. Within each pair, the first column re-estimates the effects for all pre-graduation friends on the sample used for the corresponding exclusion specification. The second column reports the effects of friends outside the excluded group. The exclusion specifications jointly estimate separate effects for friends outside the excluded group and for excluded friends and also control for the excluded group's initial share of the graduate's network in each destination. The paired all-friends and exclusion specifications use the same fixed-effect structure. The mean estimates are calculated using the empirical Bayes shrinkage procedure described in Appendix D. Mean β_k reports the unweighted average across destinations. The following row reports the unweighted average after scaling each destination-specific estimate to represent the percentage change in the baseline migration rate associated with a one-percentage-point increase in the share of friends living in that destination. The remaining rows report counts based on the raw, unshrunk estimates. The first reports the number of positive estimates; the subsequent rows report the number of positive estimates with p-values below the indicated thresholds.

Table A8: Model Parameters

Parameter	Description	Value	Source
Network Model			
α_w	Weight on Log Wages in Utility	1.89	Section 5.1
α_h	Weight on Log Rents in Utility	0.63	Set to $\frac{1}{3}\alpha_w$; U.S. Bureau of Labor Statistics (2025)
$\tilde{\beta}$	Weight on Friend Share in Utility	4.0	Section 4.4
Basic Model			
α_w^{Basic}	Weight on Log Wages in Utility	1.03	Section 5.1
α_h^{Basic}	Weight on Log Rents in Utility	0.34	Set to $\frac{1}{3}\alpha_w^{\text{Basic}}$; U.S. Bureau of Labor Statistics (2025)
Joint Parameters			
η^H	Housing Supply Elasticity	2	Saiz (2010)
α^Y	Capital Share in Production	0.33	Standard calibration
η^Y	Elasticity of Substitution Across CZ Varieties	4	Feenstra et al. (2018)
ρ	Rental Rate of Capital	0.025	Standard calibration

Notes: This table reports the baseline parameter values for the network and basic models described in Section 5. In the network model, $\widetilde{\beta}$ is calibrated to the main PPML estimate from Section 4.4. The wage coefficient α_w and the vector of residual location-specific components A_j are jointly chosen so that the model matches observed CZ population sizes and generates an average migration elasticity with respect to wages of one. For the basic model, α_w^{Basic} and A_j^{Basic} are recalibrated to match the same targets after excluding the network term. In both models, the rent coefficient is set equal to one-third of the corresponding wage coefficient, implying a housing expenditure share of one-third. The remaining parameters are common to both models and govern housing supply and production in the productivity-shock simulations.

B Sample Construction

This appendix provides detailed information on the construction of the three main samples used in the paper: the General Sample, the College Graduation Sample, and the Model Sample. All three samples are derived from the same underlying spine of de-identified Facebook data used in prior work on social capital and economic outcomes (Chetty et al., 2022a,b). This spine combines individual-level panel data on locations and social connections for a large sample of U.S. Facebook users.

The three samples serve different purposes and are not all nested. The General Sample is used for the descriptive analyses in Section 2.2 and the external-validity exercises in Section 4.5. The College Graduation Sample is constructed separately for the graduation-based research design in Section 4. The Model Sample is a subset of the General Sample restricted to individuals in the model's CZ choice set. Unless otherwise noted, the restrictions listed below are cumulative. All friend-related measures are based on U.S.-based friends.

B.1 General Sample

The General Sample forms the basis for the descriptive analyses in Section 2.2 and the external-validity exercises in Section 4.5. To construct this sample, we begin with the set of users in the Facebook spine who have friendship connections on the platform and impose the following restrictions:

Birth cohort. Users born between 1970 and 2000, inclusive.

Friend count. Users have at least 50 friends on the platform as of the date on which the analysis is conducted.

Activity. Users are active on the platform at three points in time: Q1 of 2012, Q4 of 2017, and Q4 of 2025. A user is considered active in Q1 of 2012 if they were 1-day active on January 1, 2012, or 30-day active on February 1 or March 1, 2012, or 30-day active on April 1, 2012 (if not 1-day active on January 1). Analogous definitions apply for Q4 of 2017 (using October 1, November 1, December 1, 2017, and January 1, 2018) and Q4 of 2025 (using October 1, November 1, December 1, 2025, and January 1, 2026).

Early friends. Users have at least 50 friendships formed before December 31, 2011.

Continuous network observation. Users have non-missing data on the geographic distribution of their friends for every quarter from Q1 of 2012 through Q4 of 2025.

Location data. Users have non-missing commuting zone (CZ) locations in Q1 of 2012, Q4 of 2017, and Q4 of 2025. High school CZ is non-missing.

B.2 College Graduation Sample

The College Graduation Sample serves as the primary analysis sample for the causal estimates in Section 4. We begin with the same spine of Facebook users used to construct the

General Sample and impose the same friend-count restrictions. We then apply the following additional restrictions:

College attendance. Users attended a four-year college in the United States. Colleges are identified by matching self-reported college names to the Integrated Postsecondary Education Data System (IPEDS), keeping only institutions with `iclevel = 1` (four-year colleges) and excluding online colleges. The matching procedure follows that described in the data appendix of Chetty et al. (2022b).

Graduation cohort. Users are expected to graduate in the spring of 2017, 2018, or 2019, assuming they follow a "standard" educational timeline. We define cohorts by expected high school graduation year, so these college graduation cohorts correspond to high school cohorts 2013, 2014, and 2015.[B1]

High school location. High school CZ is non-missing.

Activity during college. Users are active in Q4 of their graduation year (defined analogously to the General Sample activity restrictions). Users have at least three months of 30-day activity during the fall semester of their senior year (August through December of the year prior to graduation) and at least nine months of 30-day activity during their graduation year.

Time in college location. Users spent at least 25 percent of their time in their college CZ during their college years. The college period is defined as beginning in Q4 of the year four years prior to graduation and ending in Q1 of the graduation year.

B.3 Model Sample

The Model Sample is used for the quantitative model in Section 5.

The Model Sample begins with all users who satisfy the restrictions of the General Sample and further imposes that, in Q1 of 2012, users lived in one of the 50 largest CZs, defined in more detail below.

50 largest CZs. We define the 50 largest CZs as the 50 CZs with the largest number of Facebook users residing there in Q1 of 2012. Appendix Figure A2 displays these CZs on a map, and the figure notes list them by name. For comparison, 45 of the 50 CZs selected by Facebook user count also rank among the 50 largest CZs by 2012 ACS population, and the remaining five all fall within the top 62 by ACS population. Table 1 shows that 54 percent of individuals in the General Sample lived in one of these 50 CZs in Q1 of 2012, while 58 percent of individuals in the College Graduation Sample did so upon graduation. By construction, all individuals in the Model Sample live in one of these CZs in Q1 of 2012.

[B1] Equivalently, these are the cohorts expected to graduate from college four years after high school graduation. In the network definitions below, c denotes the user's expected college graduation year.

B.4 Network Definitions

We define several categories of friends based on the timing of friendship formation and shared characteristics. These definitions are used across our analyses to isolate different types of social ties.

General Sample and Model Sample networks. For both the General Sample and the Model Sample, we measure each individual's social network as of December 31, 2011, including all friends befriended on or before that date.

College Graduation Sample networks. For the College Graduation Sample, we define the following categories of friends:

- *Pre-graduation friends*: Friends befriended no later than May of the year prior to the user's expected college graduation. Formally, for a user who is expected to graduate in year c, a friend counts as pre-graduation if the friendship was formed in year $\leq c-2$, or in year $c-1$ with month < 6.
- *Post-graduation friends*: Friends befriended no earlier than June of the year four years after the user's expected college graduation. Formally, friendships formed in year $\geq c+5$, or in year $c+4$ with month ≥ 6, and where the friend did not attend the same high school or college as the user (or the cohort difference is at least 5 years).
- *Same-college friends*: Within pre-graduation friends, we define same-college friends as friends who attended the same college as the user. Both the user and friend must have non-missing college information.
- *Similar-major friends*: Within pre-graduation friends, we define similar-major friends as friends whose listed majors fall in the same occupation-based major group as the user's listed major. We construct five such groups by clustering majors according to the occupational distributions of recent graduates. Thus, two majors are classified as similar when they are associated with similar labor-market paths. To construct them, we use ACS data on individuals with at least a bachelor's degree who are ages 21–23 in 2017, 2018, or 2019, so that the sample roughly corresponds to recent graduates in the cohorts used in our main analysis. For each major, we estimate the distribution of graduates across broad occupation groups, using the ten largest occupation groups and grouping all remaining occupations into a residual category. We then group majors with similar occupational distributions into five clusters using k-means clustering. Therefore, similar-major friends are friends whose fields of study are associated with similar labor-market paths. Both the user and friend must have non-missing major information.
- *Same-college and similar-major friends*: Within pre-graduation friends, we define same-college and similar-major friends as friends who both attended the same college as the user and belong to the same broad major cluster described above. College and major information must be non-missing for both the user and friend.

C Reverse Causality in the College Graduation Research Design

One caveat to using post-period friends' moves to quantify the bias term is the possibility of reverse causality. In principle, some friends' relocations after t^* may be affected by the graduate's own location choice at t^*. To the extent this occurs, the post-period coefficients capture not only the extent to which friends' later moves proxy for unobserved determinants of the graduate's location choice, but also the extent to which the graduate's location choice affects friends' subsequent moves.

Importantly, this concern does not generate a spurious positive estimate of the network effect in our pre–post design. If a graduate's move causes friends to move in the same direction after t^*, then this channel raises the post-period coefficients. Since our estimate subtracts the average post-period coefficient from the average pre-period coefficient, such reverse causality would lead us to subtract too much. As a result, it would attenuate the estimated network effect rather than inflate it.

To see this more formally, consider the coefficient on friends' moves in a given event-time quarter t. Let $W_{i,t}$ denote the controls and the other event-time friend-move variables included in equation 3. Let $\widetilde{Y}_{i-j,t^*}$ denote the residual from projecting Y_{i-j,t^*} on $W_{i,t}$, and let $\widetilde{\Delta n}_{i-j,t}$ denote the residual from projecting $\Delta n_{i-j,t}$ on $W_{i,t}$. By the Frisch-Waugh-Lovell theorem, the event-time coefficient can be written as

$$b_{-j,t} = \frac{\mathrm{Cov}\left(\widetilde{\Delta n}_{i-j,t}, \widetilde{Y}_{i-j,t^*}\right)}{\mathrm{Var}\left(\widetilde{\Delta n}_{i-j,t}\right)}.$$

In the pre-period, this coefficient combines the causal effect of having more friends outside j at the time of the graduate's location choice and the bias from unobserved factors that also affect friends' moves:

$$b^{pre}_{-j,t} = \beta_{-j,t} + \delta^{pre}_{-j,t}.$$

The term $\delta^{pre}_{-j,t}$ captures the extent to which friends' pre-period moves are correlated with unobserved determinants of the graduate's location choice, for example because the graduate and their friends are exposed to common shocks.

In the post-period, absent reverse causality, friends' moves cannot causally affect the graduate's earlier location choice at t^*. The post-period coefficient would therefore only identify the corresponding bias term,

$$b^{post}_{-j,t} = \delta^{post}_{-j,t}.$$

This is the logic underlying the pre–post comparison. If friends' moves before and after graduation are equally informative about unobserved determinants of the graduate's location choice, then

$$\delta^{pre}_{-j,t} = \delta^{post}_{-j,t},$$

so that

$$b^{pre}_{-j,t} - b^{post}_{-j,t} = \beta_{-j,t}.$$

Reverse causality modifies only the post-period expression. To see this, decompose residualized post-period friend moves into two components:

$$\widetilde{\Delta n}_{i-j,t} = \widetilde{s}_{i-j,t} + \widetilde{r}_{i-j,t}.$$

Here, $\widetilde{s}_{i-j,t}$ is the residualized non-reflection component of friends' post-period moves: the part of friends' moves after t^* that is not caused by the graduate's own location choice. This includes friends' own idiosyncratic migration decisions, common shocks, and other determinants of friends' moves, after residualizing with respect to $W_{i,t}$. In contrast, $\widetilde{r}_{i-j,t}$ is the residualized reflection component: the part of friends' post-period moves caused by the graduate's location choice at t^*.

Substituting this decomposition into the post-period coefficient gives

$$b^{post}_{-j,t} = \frac{\text{Cov}\left(\widetilde{s}_{i-j,t}, \widetilde{Y}_{i-j,t^*}\right)}{\text{Var}\left(\widetilde{\Delta n}_{i-j,t}\right)} + \frac{\text{Cov}\left(\widetilde{r}_{i-j,t}, \widetilde{Y}_{i-j,t^*}\right)}{\text{Var}\left(\widetilde{\Delta n}_{i-j,t}\right)}.$$

The first term is the post-period analogue of the bias term δ: it captures the extent to which friends' post-period moves proxy for unobserved determinants of the graduate's location choice. The second term is the component due to reverse causality. We can therefore write

$$b^{post}_{-j,t} = \delta^{post}_{-j,t} + \kappa_{-j,t},$$

where

$$\kappa_{-j,t} = \frac{\text{Cov}\left(\widetilde{r}_{i-j,t}, \widetilde{Y}_{i-j,t^*}\right)}{\text{Var}\left(\widetilde{\Delta n}_{i-j,t}\right)}.$$

To interpret $\kappa_{-j,t}$, write the linear projection of the reflection component on the graduate's residualized location choice as

$$\widetilde{r}_{i-j,t} = \rho_t \widetilde{Y}_{i-j,t^*} + \zeta_{i-j,t}, \quad \text{Cov}\left(\zeta_{i-j,t}, \widetilde{Y}_{i-j,t^*}\right) = 0.$$

Then

$$\kappa_{-j,t} = \frac{\rho_t \,\text{Var}\left(\widetilde{Y}_{i-j,t^*}\right)}{\text{Var}\left(\widetilde{\Delta n}_{i-j,t}\right)}.$$

This expression clarifies when reverse causality should matter for the post-period coefficients. The key object is not the overall amount of residual variation in post-period friends' moves, but the extent to which that residual variation is induced by the graduate's own location

choice. Put differently, reverse causality shows up in the post-period coefficients only to the extent that friends' post-period moves contain additional information about the residual component of the graduate's location choice.

There are three reasons why this component may be limited. First, the graduate's location choice may have only a small effect on friends' subsequent moves. In the notation above, this corresponds to a small ρ_t. In that case, even if friends' post-period moves are correlated with the graduate's choice for other reasons, the part of that correlation arising from graduate-to-friend influence is small.

Second, only residual variation in the graduate's location choice can generate reverse-causality bias in the event-time coefficient. Any component of Y_{i-j,t^*} that is already predicted by the controls and by the other event-time friend-move variables has been partialed out. Thus, reverse causality is not governed by the unconditional relationship between graduates' choices and friends' later moves. It is governed by whether friends' later moves are related to the part of the graduate's choice that remains unexplained after conditioning on the full specification. If the graduate's location choice is already largely captured by the baseline controls, the initial network composition, and the sequence of friends' moves at other event times, then post-period moves have little additional variation through which to reflect the graduate's choice.

Third, even if ρ_t is positive, the post-period coefficient depends on the strength of the graduate-induced component relative to the total residual variation in post-period friends' moves. This is the role of the denominator in the expression above. Importantly, this is not the same as saying that a large $\mathrm{Var}(\widetilde{\Delta n}_{i-j,t})$ mechanically makes reverse causality unimportant. If post-period friend moves vary a lot precisely because graduates strongly cause their friends to move, then the numerator will also be large. Instead, the relevant distinction is between graduate-induced variation and other residual variation in friends' moves. If most residual post-period friend moves reflect friends' own migration decisions, common shocks, measurement error, family reasons, job moves, or other factors not caused by the graduate's location choice, then the graduate-induced component is diluted and $\kappa_{-j,t}$ is small.

The same expression also indicates when reverse causality should show up strongly. It should appear as larger positive post-period coefficients when three conditions hold: graduates' choices substantially affect friends' later moves; the affected friend moves occur in the post-period used to estimate the bias term rather than only in the mid-period around graduation; and these friend moves are related to the residual component of the graduate's location choice, after conditioning on the controls and the other event-time friend-move variables. In such a case, friends' later moves would contain additional information about where the graduate chose to live, and the post-period coefficients would be pushed upward.

This timing distinction is important in our setting. Some coordinated moves or responses to the graduate's choice may occur very close to graduation. These are precisely the quarters we exclude from both the pre- and post-period averages, since they fall in the mid-period during which graduates may be making location decisions and may anticipate friends' moves. Reverse causality that operates mainly through such immediate coordination would therefore

not appear strongly in the post-period coefficients used to form the pre–post difference. By contrast, reverse causality that operates through friends following the graduate with a longer lag should raise the post-period coefficients.

Combining the common-shock and reverse-causality components, the observed post-period coefficient can be written as

$$b^{post}_{-j,t} = \delta^{post}_{-j,t} + \kappa_{-j,t}.$$

Thus, the post-period coefficients are informative about both sources of concern. If reverse causality were quantitatively important, and if friends tend to follow graduates rather than systematically move away from them, it should contribute positively to $b^{post}_{-j,t}$. In the data, however, the post-period coefficients are much smaller than the pre-period coefficients and are often close to zero. This implies that the sum of the post-period bias term and the reverse-causality component is small:

$$\delta^{post}_{-j,t} + \kappa_{-j,t} \approx 0.$$

As a result, unless a positive reverse-causality component is being offset by a negative post-period bias term of similar magnitude, any attenuation of the pre–post estimate from reverse causality must be limited.

Under the identifying assumption that the common-shock bias is symmetric across the pre- and post-period,

$$\delta^{pre}_{-j,t} = \delta^{post}_{-j,t},$$

the pre–post difference becomes

$$b^{pre}_{-j,t} - b^{post}_{-j,t} = \beta_{-j,t} - \kappa_{-j,t}.$$

Thus, graduate-to-friend reverse causality does not inflate the estimated network effect. Under the natural case in which friends are more likely to follow graduates than to move away from them, $\kappa_{-j,t} \geq 0$, so reverse causality makes the pre–post difference conservative. The small post-period coefficients suggest that this conservative bias is unlikely to be large.

The same logic applies to directed migration after replacing friends outside j with friends in destination k. If graduates' moves to k caused friends to move to k after t^*, this would raise the post-period coefficient for destination k. Since the directed-migration estimates also subtract post-period coefficients from pre-period coefficients, such reverse causality would again attenuate the estimated network effect rather than generate a spurious positive effect.

D Empirical Bayes Shrinkage of Destination-Specific Estimates

Several figures in the paper report destination-specific network effects across CZs. Since some CZ-specific estimates are noisier than others, especially for destinations with lower baseline migration rates, we apply an empirical Bayes shrinkage procedure following Chetty and Hendren (2018b). The goal of the shrinkage is to reduce noise in the displayed distribution of CZ-specific point estimates.

Let $\hat{\beta}_k$ denote the raw estimate of the network effect for destination CZ k, and let $\hat{\sigma}_k$ denote its standard error. In our main directed-migration application, $\hat{\beta}_k$ is the difference between the average pre-period and post-period coefficients from equation 5. We write

$$\hat{\beta}_k = \beta_k + e_k,$$

where β_k is the destination-specific network effect and e_k is sampling error with variance $\hat{\sigma}_k^2$.

To estimate the signal variance across destinations, we randomly split the sample into two equally sized subsamples and estimate the destination-specific effects separately in each subsample. Let $\hat{\beta}_k^{(1)}$ and $\hat{\beta}_k^{(2)}$ denote the estimates from the two split samples. We estimate the signal variance as the weighted covariance of the two split-sample estimates:

$$\hat{\tau}^2 = \max\left\{0, \frac{\sum_k w_k \left(\hat{\beta}_k^{(1)} - \bar{\beta}^{(1)}\right)\left(\hat{\beta}_k^{(2)} - \bar{\beta}^{(2)}\right)}{\sum_k w_k}\right\},$$

where

$$\bar{\beta}^{(s)} = \frac{\sum_k w_k \hat{\beta}_k^{(s)}}{\sum_k w_k}, \qquad s \in \{1, 2\}.$$

The weights are given by the average precision of the two split-sample estimates,

$$w_k = \frac{1}{2}\left(\frac{1}{\left(\hat{\sigma}_k^{(1)}\right)^2} + \frac{1}{\left(\hat{\sigma}_k^{(2)}\right)^2}\right).$$

We truncate $\hat{\tau}^2$ at zero.

We then apply shrinkage to the full-sample estimates. Let

$$\bar{\beta} = \frac{\sum_k \hat{\sigma}_k^{-2} \hat{\beta}_k}{\sum_k \hat{\sigma}_k^{-2}}$$

denote the precision-weighted mean of the full-sample estimates. The shrinkage factor for

destination k is

$$\lambda_k = \frac{\hat{\tau}^2}{\hat{\tau}^2 + \hat{\sigma}_k^2},$$

and the shrunk estimate is

$$\hat{\beta}_k^{EB} = \bar{\beta} + \lambda_k \left(\hat{\beta}_k - \bar{\beta} \right).$$

We use $\hat{\beta}_k^{EB}$ when displaying the distribution of CZ-specific point estimates in figures such as Figure 4. Since the shrinkage procedure is intended only to summarize the distribution of effects more cleanly, we conduct inference using the raw estimates $\hat{\beta}_k$ in Appendix Table A5.

E Construction of Sister CZs

For the sister CZ exercise in Section 4.2, we construct for each CZ in our directed-migration analysis a comparison CZ with similar industrial composition. The goal is to identify, for each destination CZ, another CZ that is likely to be exposed to similar industry-level shocks, but that is distinct from the destination itself.

We use annual data from the Quarterly Census of Employment and Wages (QCEW). We construct CZ-level industry employment shares in 2012 by mapping county-level QCEW data to commuting zones and restrict attention to the 50 largest CZs used in the directed-migration and model analyses shown in Appendix Figure A2. For each CZ j, let $s_{j\ell}$ denote the share of CZ employment in industry ℓ in 2012.

We then measure the distance in industrial composition between CZ j and each other CZ k as

$$d(j,k) = \left[\sum_{\ell} \left(s_{j\ell} - s_{k\ell}\right)^2\right]^{1/2}.$$

The sister CZ of j is defined as the CZ $k \neq j$ with the smallest value of $d(j,k)$:

$$\text{Sister}(j) = \arg\min_{k \neq j} d(j,k).$$

Thus, sister CZs are selected solely on the basis of baseline industrial composition, using information from before the outcome period. Across focal–sister pairs, the correlation in percentage employment (wage) growth between 2012 and 2024 is 0.61 (0.58).

F Survey Evidence on Mechanisms

This appendix describes the off-platform survey used in Section 4.3 to provide suggestive evidence on why individuals value having friends and family nearby. The survey was fielded online through Qualtrics between April 2 and April 18, 2024. Respondents were recruited through Prolific and restricted to U.S. residents between the ages of 19 and 45. The full survey sample includes 517 respondents.

Our main analyses focus on the 298 respondents who report that they currently live within 100 miles of the city where they grew up. We focus on this group because the mechanisms we seek to understand are especially relevant for individuals who continue to live close to home, despite potentially having access to economic opportunities elsewhere. The restriction does not require respondents to have always lived near their home city; some respondents may have lived elsewhere at earlier points in their lives. Because the survey is self-reported and fielded on a nonrepresentative online sample, we interpret the results as suggestive evidence on mechanisms.

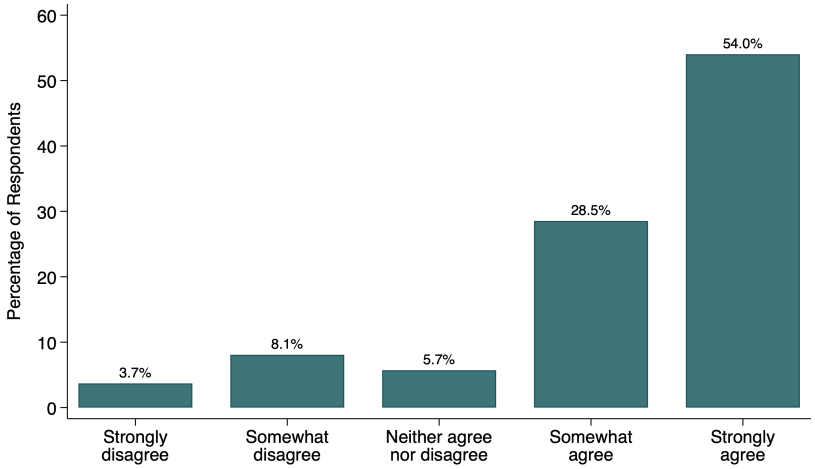
Appendix Figure F1 shows that social networks are an important self-reported reason for residential choice among these respondents. Panel (a) reports responses to the statement that friends and family are one of the main reasons why the respondent lives in their current city. More than 80% of respondents agree with this statement. Panel (b) reports responses to a hypothetical question asking what pay raise respondents would require to consider moving to a city where they did not know anyone. Around half report that they would require a pay raise of at least 50% of their current income, or would not consider moving at all. These responses are consistent with the main findings in the paper: social networks appear to be an important component of location choice, and respondents report a high willingness to forgo economic gains in order to remain near their social ties.

Appendix Figure F2 explores what respondents value about these ties. Panel (a) shows how often respondents see the people who are most important for their decision of where to live for different reasons. Leisure is the dominant reason: more than 70% of respondents report seeing these people always or most of the time for leisure. Panel (b) shows that respondents report receiving less practical help from local friends and family with jobs or housing, although help with child care is more common among respondents with children. Panel (c) provides a direct comparison between companionship and practical assistance: nearly 85% of respondents agree that they value the company of these individuals far more than any practical help they provide.

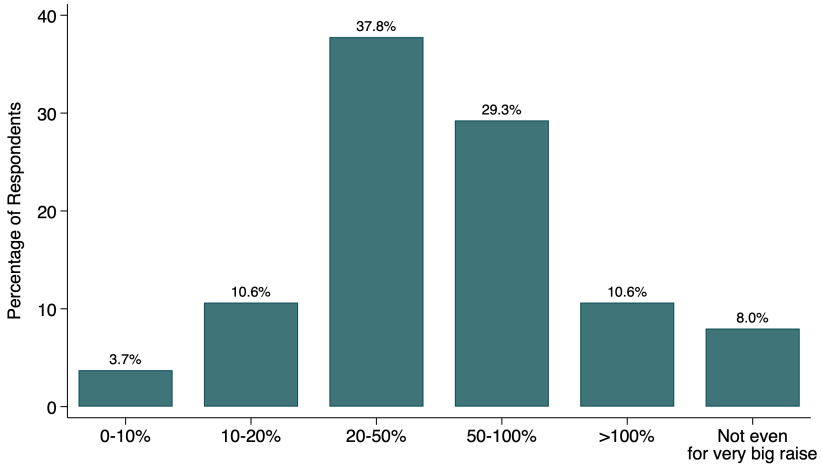
Taken together, the survey evidence points toward the social-amenity value of friends and family as an important mechanism behind the network effects documented in the Facebook data. Respondents report that friends and family are central to their residential choices, and that the value of these ties comes primarily from companionship and leisure rather than from practical assistance with jobs, housing, or other needs. While these results are only suggestive, they support the interpretation that contemporaneous social presence is itself an important component of the value of local social networks.

Figure F1: Survey Evidence on Importance of Friends for Residential Choice

(a) Friends + Family one of Main Reasons to Live in City



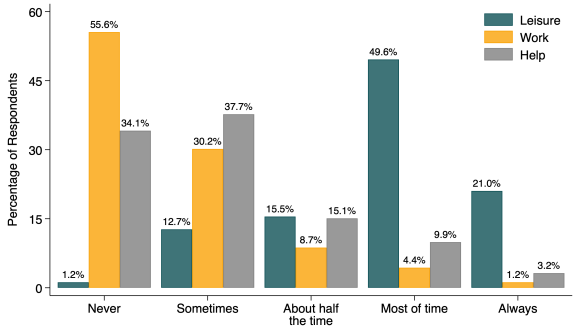
(b) Pay Raise to Move to City Without Friends



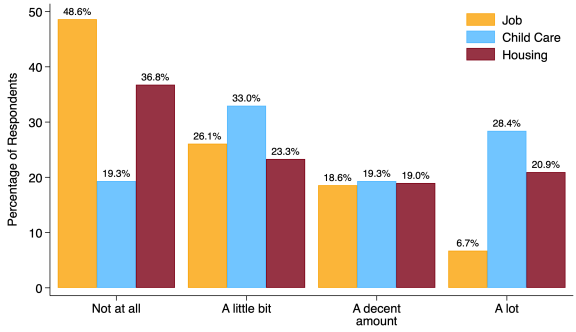
Notes: This figure presents survey evidence on the importance of social networks for residential choice. Panel (a) shows the distribution of respondents agreeing with the statement that friends and family are one of the main reasons they live in their current city. Panel (b) shows the distribution of pay raises respondents say they would have to receive in order to consider moving to a city without any prior connections. The sample for Panel (b) is restricted to respondents who have never lived in a city other than the one they grew up in due to the design of the survey.

Figure F2: Survey Evidence on Mechanisms of Network Effects

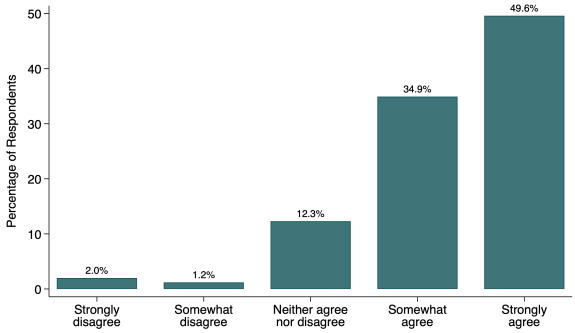
(a) Frequency of Seeing Friends, by Purpose



(b) Help Received from Friends in Recent Years



(c) Enjoy Friends' Company More than Needing Help



Notes: This figure presents survey evidence regarding the role played by social networks for residential choice. Panel (a) shows the frequency with which respondents see the people that are most influential for their decision of where to live for reasons related to leisure (in turquoise), work (in yellow), and other forms of help (in gray). Panel (b) shows the distribution of the extent to which respondents recently received help from their local friends and family with respect to their job (in yellow), child care (in light blue), and housing (in dark red). For the values corresponding to child care, we subset to individuals with children. Panel (c) shows the distribution of respondents agreeing with the statement that they enjoy the company of their local social networks much more than they need their help.

G Model Details

This appendix provides additional details on the model described in Section 5. The main text introduces the residential-choice component of the model and, for the static exercises, takes wages and rents as given. For the shock simulations in Section 5.2, we augment this residential-choice block with standard local production and housing-market blocks. The production block follows Zabek (2024), and the housing block uses a closely related upward-sloping housing supply structure. This appendix describes all three blocks and the equilibrium conditions used in the simulations.

Throughout these exercises, n_{ij} denotes the observed share of individual i's friends living in CZ j. We hold this observed network distribution fixed when solving the model and simulating local shocks. This fixed-network approach is consistent with the accounting exercise in the main text: we ask how much the observed geography of social networks, combined with the estimated network effect, can account for residential choices and responses to local shocks.

Residential Choice and Labor Supply

Individual i's indirect utility from living in CZ j is

$$U_{ij} = \alpha_w \ln(w_j) - \alpha_h \ln(r_j) + A_j + \tilde{\beta} n_{ij} + \epsilon_{ij}, \tag{8}$$

where w_j denotes local wages, r_j denotes local rents, A_j is a residual location-specific component, and n_{ij} is the observed share of i's friends living in CZ j. We assume that ϵ_{ij} follows a Type I extreme value distribution. The probability that individual i chooses CZ j is therefore

$$\psi_{ij} = \frac{\exp\left(\alpha_w \ln(w_j) - \alpha_h \ln(r_j) + A_j + \tilde{\beta} n_{ij}\right)}{\sum_{k \in J} \exp\left(\alpha_w \ln(w_k) - \alpha_h \ln(r_k) + A_k + \tilde{\beta} n_{ik}\right)}. \tag{9}$$

Aggregate labor supply in CZ j is then

$$N_j = \sum_{i \in I} \psi_{ij}. \tag{10}$$

In the simulations, n_{ij} remains fixed when shocks are applied. Thus, equation 9 captures only the direct effect of changing wages and rents on location choices.

Housing

We model local housing supply as

$$H_j^S = \nu_j r_j^{\eta^H}, \tag{11}$$

where η^H is the housing supply elasticity and ν_j is a location-specific housing supply shifter.

The indirect utility in equation 8 can be motivated by an underlying Cobb-Douglas utility over consumption and housing. Under this interpretation, households spend a constant share α_h/α_w of income on housing. Aggregate housing demand in CZ j is therefore

$$H_j^D = \frac{\alpha_h}{\alpha_w}\frac{w_j N_j}{r_j}. \tag{12}$$

Equating housing demand and supply gives

$$r_j = \left[\frac{\alpha_h}{\alpha_w}\frac{w_j N_j}{\nu_j}\right]^{\frac{1}{1+\eta^H}} \tag{13}$$

We use this condition to recover the housing supply shifter that rationalizes observed rents in the baseline equilibrium:

$$\nu_j = \frac{\alpha_h}{\alpha_w}\frac{w_j N_j}{r_j^{1+\eta^H}} \tag{14}$$

Production and Labor Demand

Each CZ j has a representative, perfectly competitive firm producing a local good variety Y_j using capital and labor:

$$Y_j = \theta_j K_j^{\alpha^Y} N_j^{1-\alpha^Y}. \tag{15}$$

Capital is supplied elastically at rental rate ρ, and labor is paid wage w_j. The local good is sold at price p_j. Cost minimization implies

$$w_j = (1-\alpha^Y)(p_j\theta_j)^{\frac{1}{1-\alpha^Y}}\left(\frac{\alpha^Y}{\rho}\right)^{\frac{\alpha^Y}{1-\alpha^Y}}. \tag{16}$$

At the national level, a final-good producer combines local varieties using a CES aggregator,

$$Y = \left[\sum_{j\in J} Y_j^{\frac{\eta^Y-1}{\eta^Y}}\right]^{\frac{\eta^Y}{\eta^Y-1}}, \tag{17}$$

where η^Y is the elasticity of substitution across local varieties. The first-order condition for the final-good producer implies that relative prices are proportional to relative marginal products:

$$\frac{p_j}{p_{\bar{j}}} = \left(\frac{Y_{\bar{j}}}{Y_j}\right)^{\frac{1}{\eta^Y}}, \tag{18}$$

where $\bar{j}$ denotes the reference CZ. We normalize $p_{\bar{j}} = 1$.

Using the local firm's first-order conditions, output can be written as

$$Y_j = \theta_j N_j \left(\frac{p_j \theta_j \alpha^Y}{\rho} \right)^{\frac{\alpha^Y}{1-\alpha^Y}} . \tag{19}$$

Combining equations 16–19 yields the wage schedule used in the simulations. In the baseline equilibrium, we choose the local productivity terms θ_j so that the model matches the observed distribution of wages, given observed population sizes and the remaining production parameters. In the shock simulations, we perturb θ_j and solve for the new equilibrium values of wages, rents, and population levels.

Equilibrium

Given parameters

$$\left\{ \alpha_w, \alpha_h, \tilde{\beta}, \alpha^Y, \eta^Y, \eta^H, \rho \right\}$$

and location-specific primitives

$$\{A_j, \theta_j, \nu_j\}_{j \in J} ,$$

an equilibrium consists of choice probabilities $\{\psi_{ij}\}_{i,j}$, population levels $\{N_j\}_j$, wages $\{w_j\}_j$, rents $\{r_j\}_j$, prices $\{p_j\}_j$, and outputs $\{Y_j\}_j$ such that:

1. Individuals choose locations according to the logit probabilities in equation 9.
2. Labor supply in each CZ is given by equation 10.
3. Housing markets clear, so rents satisfy equation 13.
4. Local firms satisfy the production first-order conditions summarized by equation 16.
5. Relative prices across local varieties satisfy the CES final-good producer's first-order conditions.

Because location choices determine local population sizes, and local population sizes in turn affect wages and rents, the equilibrium is characterized by a fixed point. We solve this fixed point numerically. Starting from baseline wages, rents, and population levels, we update choice probabilities using equation 9, update populations using equation 10, update rents from housing-market clearing, and update wages and prices using the production block. We iterate until populations, wages, and rents converge.

In the baseline calibration, we use observed wages, rents, and populations to recover the residual location components A_j and the local primitives θ_j and ν_j. In the shock simulations, we perturb local productivity and solve the full equilibrium system described above, holding the observed network distribution fixed.